

Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

Worked design

90 to 264 Vac, 47 to 63 Hz
25 V / 26.3 A = 657.5 W

Single-Stage PFC LLC Converter Design

Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. None of its numbers is measured: each is calculated, or read from a component datasheet or parameter listing. The safety-insulation values come from the seminars and certification reports listed under References, not from the standards themselves.

The figures were drawn for this note, except the power-stage, controller and compensator drawings of Chapter 6, which come from the ST L6790A design spreadsheet; their captions say so. Where a figure follows the argument of a published application note, the caption says so too.

Revision history

Version	Date	Change
V1.0	August 2026	First issue. Worked design: 25 V / 26.3 A, 657.5 W.
V1.1	September 2026	Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$, the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer.
V1.2	September 2026	Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.
V1.3	September 2026	Design example on one transformer, a TDK PQ 50/50 core on its catalogue coil former with a partition added; the winding, its leakage and the pins drawn, and the tank set by that leakage: L_r 25 μ H, L_m 45 μ H, C_r 180 nF. Compensator chapter rewritten with the loop equations, the op-amp equivalent of the TL431 network and a worked loop design. V_{CC} from the auxiliary winding through a Zener regulator. Safety insulation of the transformer for household AV equipment in the US, EU, Japan, Korea and China. Text shortened throughout. Parts fitted: STO60N045DM9 primary switches, two L6498LD gate drivers with external bootstrap diodes and a TEA2095TE SR controller, with the V_{CC} start-up re-sized for them.

Conventions

- Frequencies are switching frequencies unless written f_l , the line frequency.
- $f_n = f_{sw}/f_r$, the switching frequency normalised to the series resonance; gain curves are drawn against it. Other documents may write f_s for f_{sw} .
- M is the gain of the resonant network: the output referred to the primary over the fundamental of the drive. **$M = 1$ at the series resonance** at any load.
- $Z_0 = \sqrt{L_r/C_r}$ and $Q = Z_0/R_{ac}$: $Q = 0$ is no load; larger Q is more power and a flatter curve.

Single-Stage PFC LLC Converter Design

- $V_{o,eff} = V_{out} + N_{rect} V_f$, with N_{rect} the number of rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- n is the turns ratio of the gain model; $n_T = N_p/N_s$ is the ratio actually wound.
- $\lambda = L_r/L_m$; older literature uses $m = 1 + 1/\lambda$.
- θ is the line phase angle, zero at the mains zero crossing. “Equivalent input” is what the tank sees after topology morphing, not the mains voltage.

Single-Stage PFC LLC Converter Design

Table of contents

1 Introduction	8
2 The two converters this one is made of	8
2.1 What an LLC converter is	8
2.2 Why an LLC, and not something simpler	9
2.3 What resonance is for	9
2.4 How the circuit becomes $M(f_n, Q)$	10
2.5 The two resonances	11
2.6 The gain function and the three regions	18
2.7 Capacitive and inductive, and why those words	19
2.8 Why a diode has to recover on one side and not on the other	20
2.9 The two boundaries are not the same boundary	20
2.10 ZVS and ZCS are not the same thing	21
2.11 Sizing the dead time so that ZVS actually happens	22
2.12 Not a flyback, and why that changes the core	23
2.13 How an LLC is normally designed	26
2.14 What power factor correction is	27
2.15 Power factor, and why it is not the same as distortion	27
2.16 How a corrector fixes it	28
2.17 Why they are normally two stages	29
3 Why single stage, and what it costs	29
3.1 The energy a unity power factor cannot deliver	29
3.2 Moving the buffer from 400 V to the output	30
3.3 What the load must tolerate	30
4 Operating principle	30
4.1 Gain is a boundary condition, not a control variable	30
4.2 Two divergences that cancel	30
4.3 Frequency modulation is the power factor correction	31
4.4 Solving for the profile without a root search	32
4.5 Reading the gain chart of a single-stage converter	33
4.6 The feedback pin is a power command	34
4.7 Which side of resonance this converter runs on	34
4.8 Why λ must be about 0.5	34
4.9 $f_{sw,max}$ does not check the tank — it computes it	35
4.10 The other bound on λ , and where it has no solution	35

Single-Stage PFC LLC Converter Design

5 Topology morphing	35
5.1 Full bridge	36
5.2 Half bridge	36
5.3 How the controller does it	37
5.4 The hysteresis band, and the range it really implies	37
6 Design procedure	39
6.1 The specification	39
6.2 The equivalent input range	39
6.3 Turns ratio and reflected voltage	40
6.4 The resonant tank	40
6.5 ZVS verification	40
6.6 Which side of resonance the converter runs on	41
6.7 Currents over the line cycle	41
6.8 The transformer	41
6.9 Two ratios, two inductances	42
6.10 What the transformer specification must say	43
6.11 Peak flux is set by the secondary, not the primary	44
6.12 Choosing the core: two areas, and the one that usually decides	44
6.13 The winding arrangement is the leakage	44
6.14 The gap, and why the drawing must not name it	45
6.15 The window: wire, current density and what actually fits	45
6.16 Skin depth, and why the wire is not a wire	46
6.17 Safety insulation: what the winding has to carry	47
6.18 Loss, and the temperature the drawing has to survive	47
6.19 The saturation test is not the peak winding current	48
6.20 The specification, and what it is not allowed to leave out	50
6.21 If one transformer is not practical	50
6.22 The input capacitor	50
6.23 The output capacitor bank	50
6.24 Semiconductor requirements	51
6.25 Driving the bridge: external gate drivers	51
6.26 Controller network	53
6.27 Supplying V_{CC} from the auxiliary winding	54
6.28 Voltage loop and compensation	55
6.29 What the loop must achieve, and what goes wrong when it does not	56
6.30 Why the crossover must be low: the $2f_l$ ripple	56

Single-Stage PFC LLC Converter Design

6.31 The compensator: TL431, optocoupler and the FB pin	57
6.32 Placing the zero and the pole: the K-factor method	59
6.33 Checking the loop: crossover, phase margin and gain margin	60
6.34 Gain margin, and why it needs checking	61
6.35 Feedback ripple against the burst threshold	62
7 Design example	62
7.1 The specification, sorted by what kind of number it is	62
7.2 What the design came out as	64
7.3 What a verification margin is	64
7.4 Following the numbers through	65
7.5 ZVS over the whole operating space	69
7.6 The gain chart of this design	69
7.7 Which side of resonance this design runs on	71
7.8 The currents this design has to carry	73
7.9 What the transformer must provide	74
7.10 Choosing the core	75
7.11 The winding and the pins	79
7.12 Safety insulation of this transformer	82
7.13 Checking the flux, and the specification	84
7.14 The output bank, as sized	86
7.15 Where the power goes	88
7.16 What the semiconductors have to be	89
7.17 The primary switches: STO60N045DM9	89
7.18 The gate drivers: two L6498LD	90
7.19 The SR MOSFETs: STL160N10F8	93
7.20 Synchronous rectification: TEA2095TE	93
7.21 The voltage loop, as built	95
7.22 The parts around the controller	101
7.23 The oscillator: C_T first, then R_T	102
7.24 The current-sense resistor, which sets three things at once	103
7.25 Burst mode: one resistor, read once at power-up	104
7.26 Brown-out and bridge configuration: the CFG pin	104
7.27 Output sensing and over-voltage: the ZCD divider	105
7.28 V_{CC} from the auxiliary winding, as built	106
7.29 Controller pin rules	108
7.30 Protections, and where each threshold is set	108

Single-Stage PFC LLC Converter Design

- 7.31 System design rules 109
- 7.32 What to measure first on hardware 109
- 7.33 The whole circuit, with this design's values 110
- 8 List of symbols 112**
- 9 References 119**
- 10 Errata and open items 120**
 - 10.1 Datasheet: the draft is not self-consistent 121
 - 10.2 Traps in the obvious way of building a design sheet 121
 - 10.3 Constants used here without a derivation 122
 - 10.4 Open items in this design 123

Single-Stage PFC LLC Converter Design

1 Introduction

A conventional universal-input LLC supply has two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps the bus down. The bus capacitor stores the energy a unity-power-factor input cannot deliver evenly, and it gives the LLC an almost constant input, so the tank covers only a narrow gain range.

A **single-stage PFC LLC** removes the boost stage and the bus capacitor. The rectified mains feeds the resonant tank directly. The input is then a 100/120 Hz half sine, the gain the tank must provide follows the input through the line cycle, and the output capacitor must absorb what the bus capacitor used to. Most two-stage design habits do not apply.

This note explains how the converter works and how to design one with the STMicroelectronics **L6790A**. One design runs through it: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out.**

Scope. Power stage, resonant tank, transformer, output bank, controller network, V_{CC} supply, voltage loop and the insulation the transformer must carry. Not covered: EMI filter, layout, and the safety approval itself.

2 The two converters this one is made of

This chapter covers an ordinary LLC and ordinary power factor correction. Readers who know both can skip to Section 3.

2.1 What an LLC converter is

An LLC converter is a square-wave generator driving a resonant network, which feeds a transformer and a rectifier. The switches only set the frequency. There is no duty cycle and no programmed inductor current.

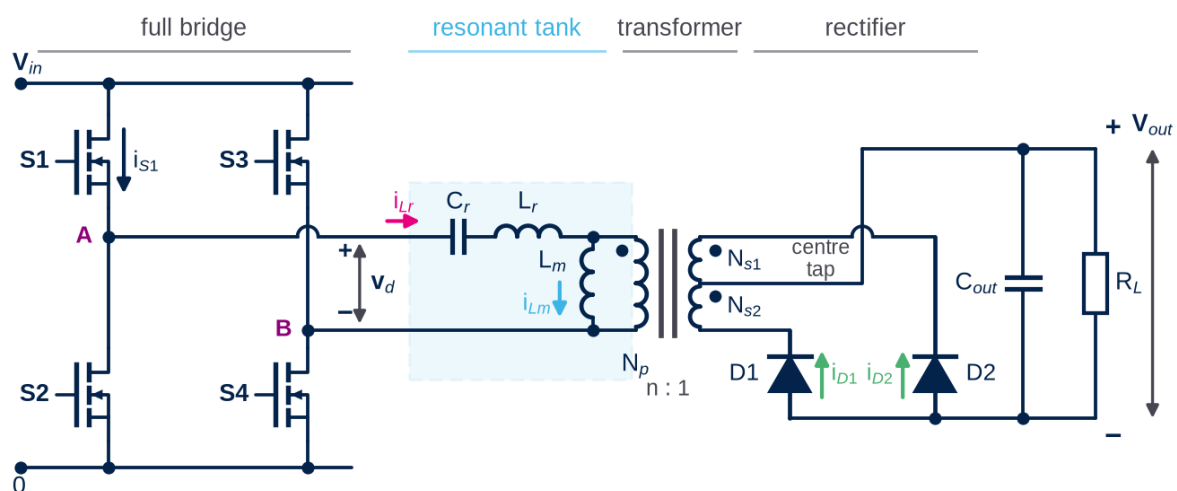


Figure 1: An LLC stage. The switches set the frequency; the tank sets the power; the transformer sets the voltage. The bridge output $v_d = v_A - v_B$ drives the tank. Each arrow is the positive direction of a current plotted in Figure 2.

Single-Stage PFC LLC Converter Design

The three tank elements are the series inductance L_r , the magnetising inductance L_m of the transformer, and the series capacitance C_r . L_m is the transformer's own inductance and L_r is usually its leakage inductance, so only C_r is always an added part.

2.2 Why an LLC, and not something simpler

Compared with a forward or flyback converter, an LLC gives:

- **Soft switching at every load.** The primary switches turn on at zero voltage, and below resonance the rectifiers turn off at zero current. The magnetising current that discharges the mid-point flows at any load, so ZVS is not lost at light load.
- **Higher frequency**, because switching loss no longer limits it, so smaller magnetics and filters.
- **The leakage inductance is used as L_r** instead of needing a snubber, and there is no output inductor.

The costs: the frequency moves with line and load, the magnetising current circulates at any load so light-load efficiency is modest, and the design comes from a gain curve, not from a duty-cycle expression.

2.3 What resonance is for

In a hard-switched converter the energy in each switch's output capacitance is lost at every turn-on, and that loss limits the frequency.

An LLC running **in its inductive region** presents an inductive load to the bridge, so the tank current lags the drive voltage. In the dead time that lagging current discharges the mid-point by itself, and the next switch turns on at zero volts. **That is zero-voltage switching**; Section 2.10 gives the condition.

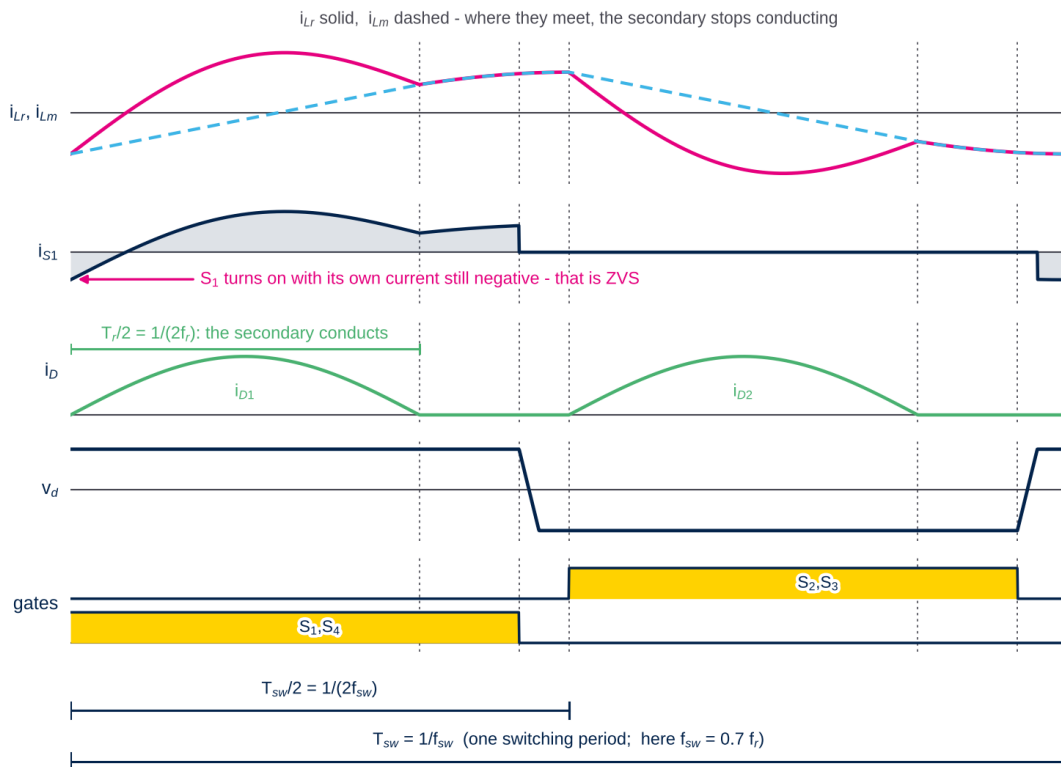


Figure 2: One switching period below resonance. The difference between i_{Lr} and i_{Lm} crosses to the secondary. The switch turns on while its current is still negative; that current has already discharged the mid-point. The secondary conducts for the resonant half period $T_r/2 = 1/(2f_r)$; the bridge changes over every $T_{sw}/2 = 1/(2f_{sw})$, which below resonance is the longer of the two.

Single-Stage PFC LLC Converter Design

The condition is *inductive operation*, not a fixed frequency. Below the capacitive boundary the current leads, the bridge hard-switches into a low impedance, and parts are destroyed rather than heated. The boundary moves with load (Section 2.9).

2.4 How the circuit becomes $M(f_n, Q)$

The gain curve comes from three simplifications, each of which discards something.

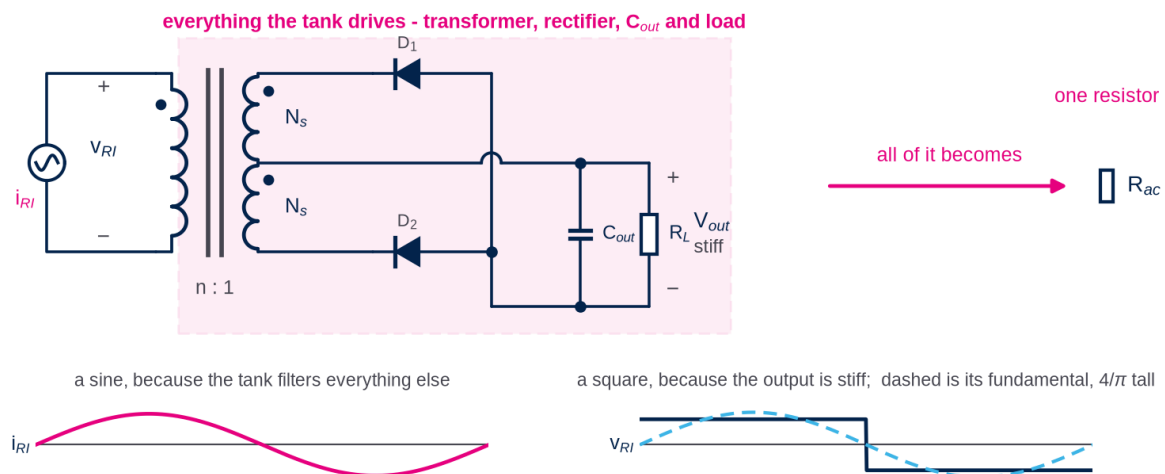


Figure 3: Why the rectifier and load can be one resistance. The output is stiff, so the tank sees a square voltage v_{RI} ; the tank passes only the fundamental, so the current i_{RI} is a sine. At the fundamental the pair takes the same power as a resistor.

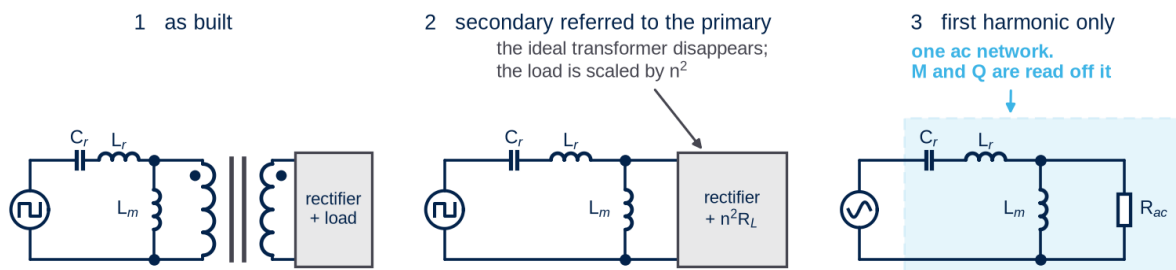


Figure 4: The reduction in three steps: as built; secondary referred to the primary; fundamental only. The result is one ac network with gain M and damping Q .

- **Refer the secondary to the primary.** The load resistance is multiplied by n^2 .
- **Replace the rectifier and output filter by one resistor.** At the fundamental they look like one.
- **Keep only the fundamental of the drive.** This is the *first harmonic approximation*. It is why an LLC design ends with a simulation or a measurement, not with the equations.

Two numbers describe the remaining network:

- **The gain M :** the voltage across R_{ac} over the fundamental that drives it. C_r and L_r cancel at the series resonance f_r , so **$M = 1$ there at any load**. Below f_r the tank can step up; above it, only down.
- **The quality factor $Q = Z_0/R_{ac}$,** with $Z_0 = \sqrt{L_r/C_r}$. **Q rises with output power.** At $Q = 0$ the curve is tall and peaked; load flattens it.

Single-Stage PFC LLC Converter Design

With f_n normalised to the series resonance and λ the ratio of the two inductances:

Equation 1

$$f_n = \frac{f_{sw}}{f_r}, \quad Z_0 = \sqrt{\frac{L_r}{C_r}}, \quad Q = \frac{Z_0}{R_{ac}}, \quad \lambda = \frac{L_r}{L_m}$$

$M(f_n, Q)$ always means the gain of this reduced circuit.

2.5 The two resonances

While the secondary conducts, the reflected output voltage clamps L_m , leaving the series resonance

Equation 2

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

When the secondary is off, both inductances resonate:

Equation 3

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

At f_r the gain is 1 for any load. At f_o the no-load gain is unlimited. The tank can boost between the two; above f_r it cannot.

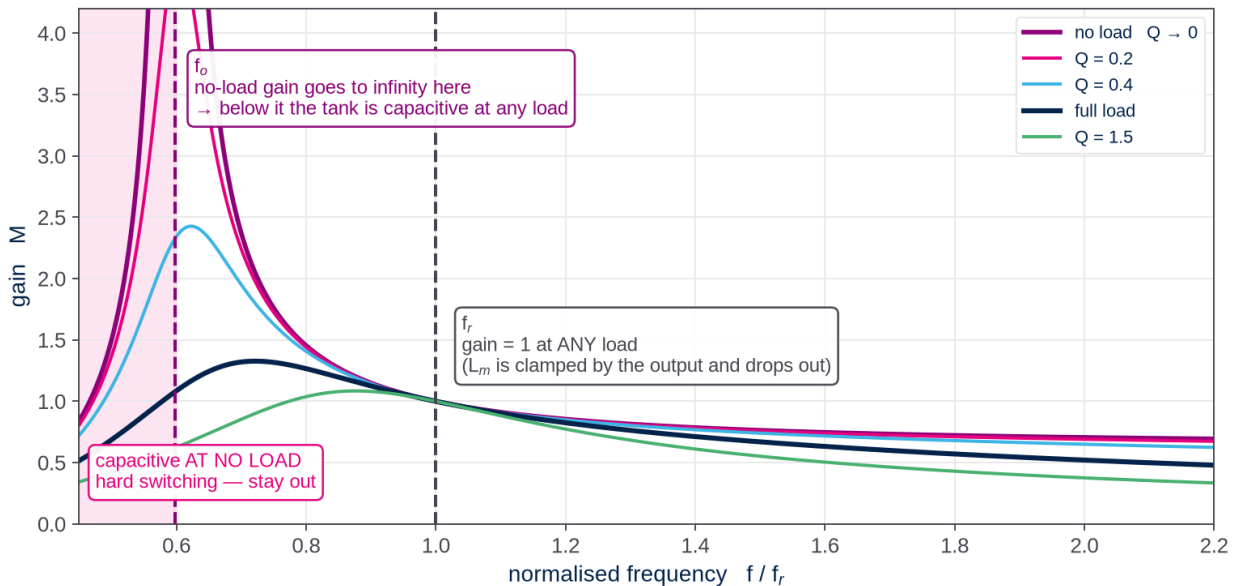


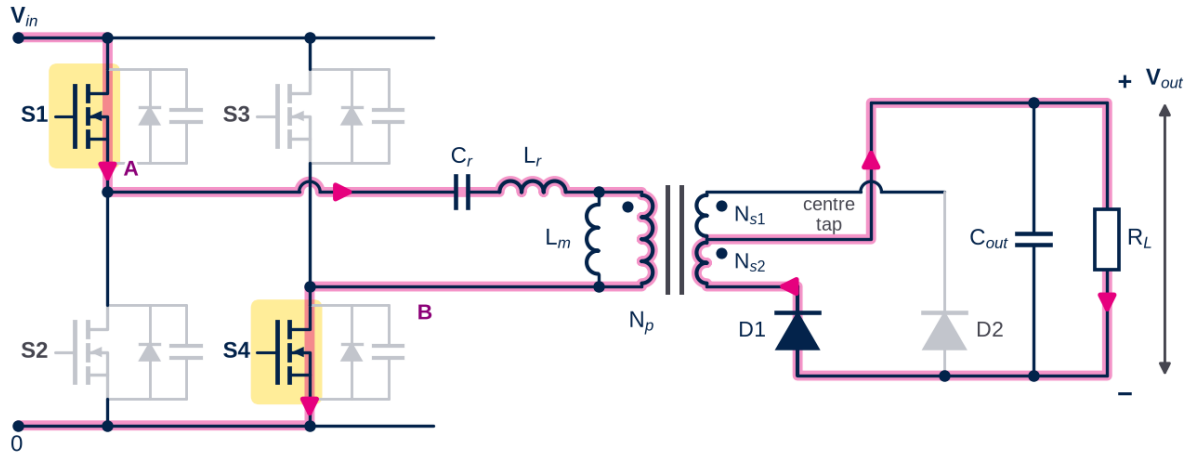
Figure 5: The two resonances and the band between them in which the tank can boost. The shaded edge is the no-load capacitive boundary; it moves up under load.

Below resonance each half cycle has two parts, with the same pair of switches on throughout: power delivery, then freewheeling. At resonance the half cycle is power delivery only; above resonance power delivery is cut short.

Single-Stage PFC LLC Converter Design

1 POWER DELIVERY

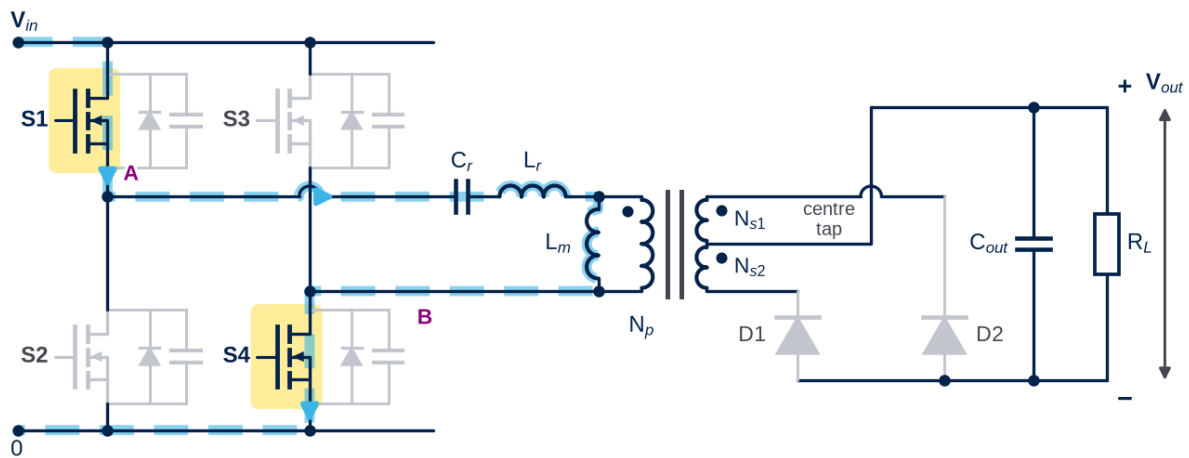
S1, S4 on · D1 conducts · lower half of N_s



L_m is clamped, so i_{Lr} is a straight ramp and $i_{Lr} - i_{Lm}$ is the half sine that crosses.
What is traced is the load component; i_{Lm} flows in L_m too, but it reverses inside this interval.

2 FREEWHEELING

S1, S4 STILL on · secondary off



i_{Lr} has fallen to i_{Lm} . Every rectifier is off, L_m is free again, and the tank rings at f_o .

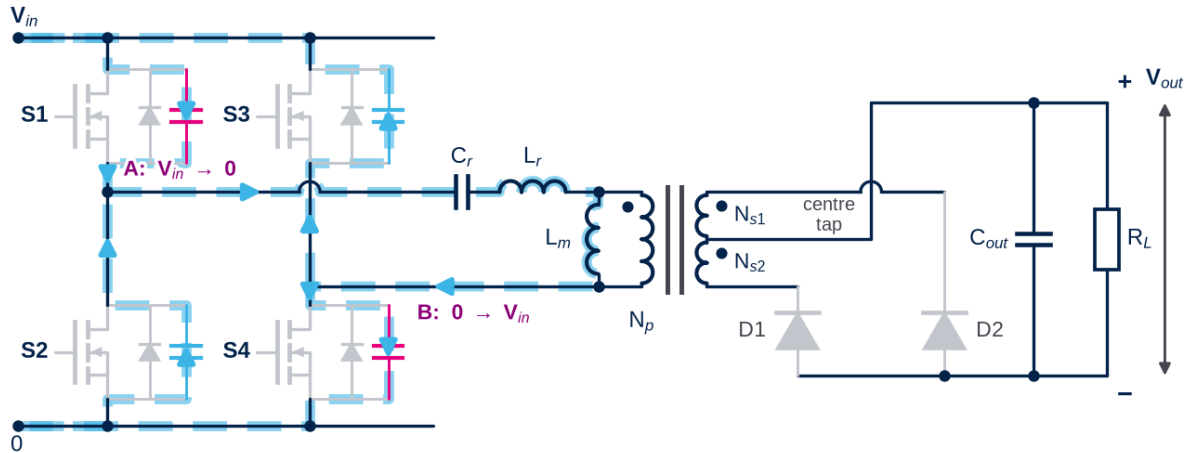
— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 6: First half, steps 1 and 2. 1 Power delivery: S_1, S_4 on, the resonant current exceeds the magnetising current and the difference reaches D_1 . L_m is clamped, so this interval resonates at f_r . 2 Freewheeling: the resonant current has fallen to the magnetising current, the rectifiers are off, and L_m joins the resonance at f_o . The same switches are still on. Greyed body diodes and C_{oss} do nothing in that step.

Single-Stage PFC LLC Converter Design

3 DEAD TIME (a)

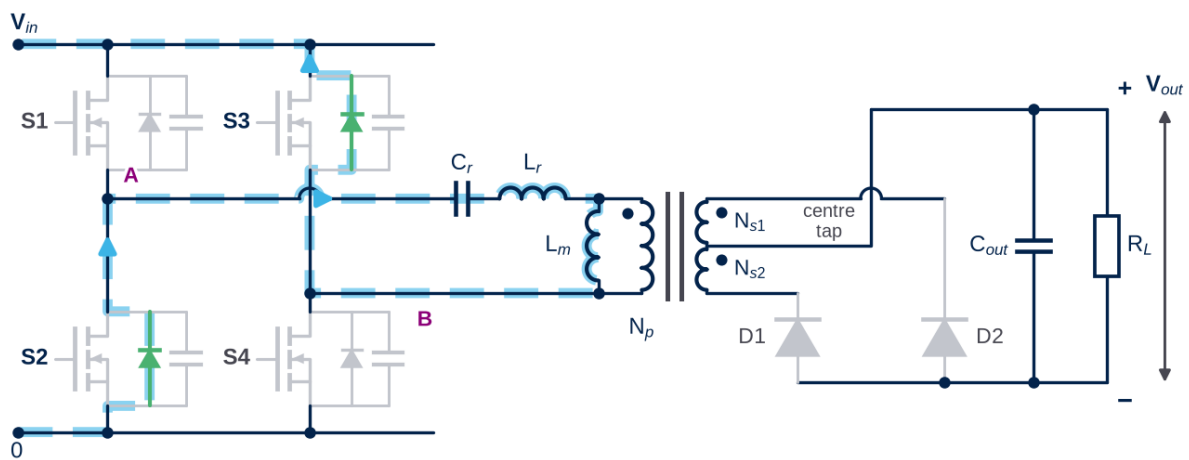
all four off · C_{oss} swaps the midpoints over



All four channels are off, so the ONE tank current can reach the rails only THROUGH the four C_{oss} . It arrives at A from both, so A is pulled down; it leaves B into both, so B is pushed up. That swap IS the transition.

4 DEAD TIME (b) — ZVS

body diodes of S2 and S3 clamp



$V_{ds} = 0$ on S2 and S3 now, because their own body diodes are clamping them. Gate them on HERE.

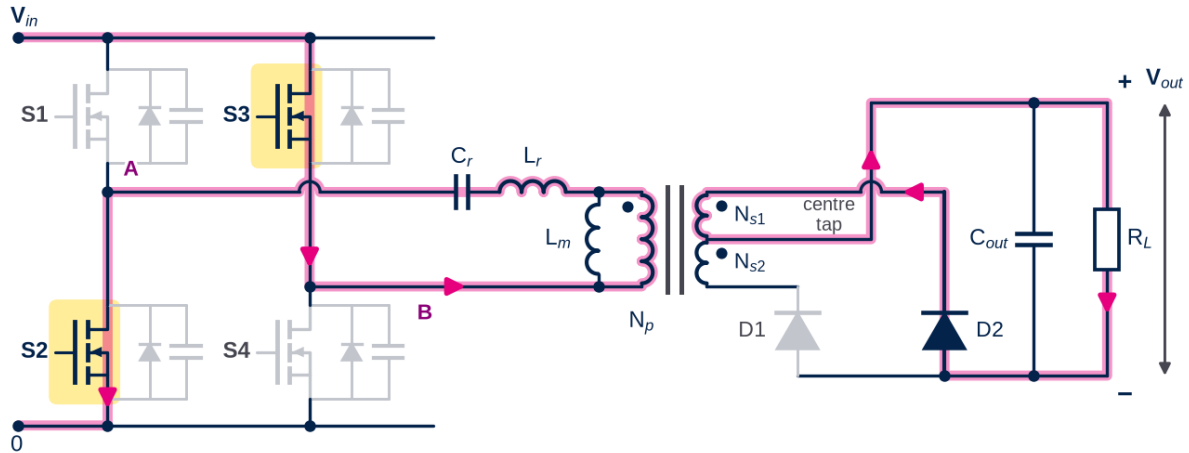
— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 7: Steps 3 and 4, the dead time. 3 All switches off; the magnetising current can only flow through the four C_{oss} , pulling A down and pushing B up. 4 The swing is complete and the body diodes of S_2, S_3 clamp V_{ds} at zero. The next gate must arrive inside this window.

Single-Stage PFC LLC Converter Design

5 POWER DELIVERY

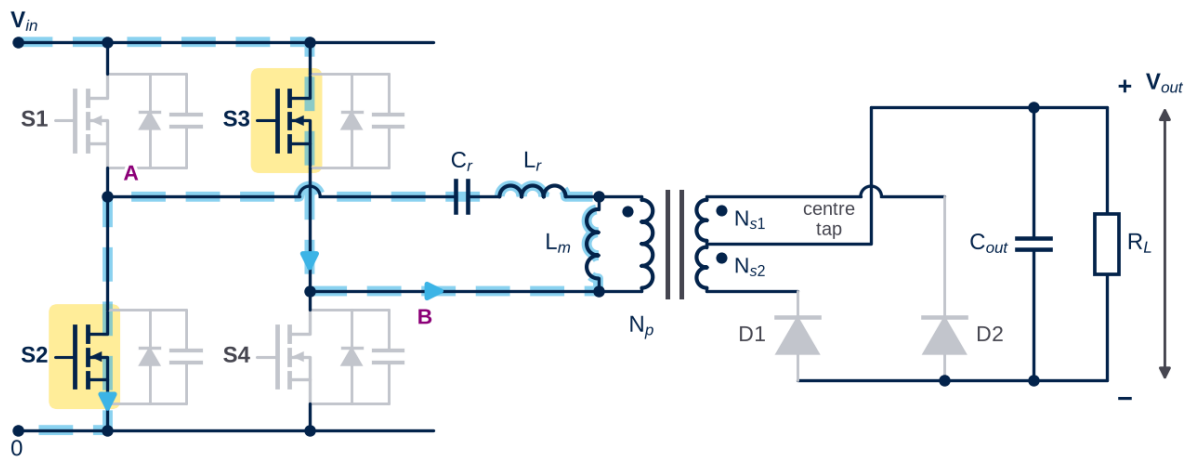
S2, S3 on · D2 conducts · upper half of N_s



The mirror of 1. Just after turn-on the old current still runs back through the two channels. Again only the load component is traced - i_{Lm} reverses inside this interval as well.

6 FREEWHEELING

S2, S3 STILL on · secondary off



The mirror of 2, and the easy one to miss: it ends at the right-hand edge of a waveform plot.

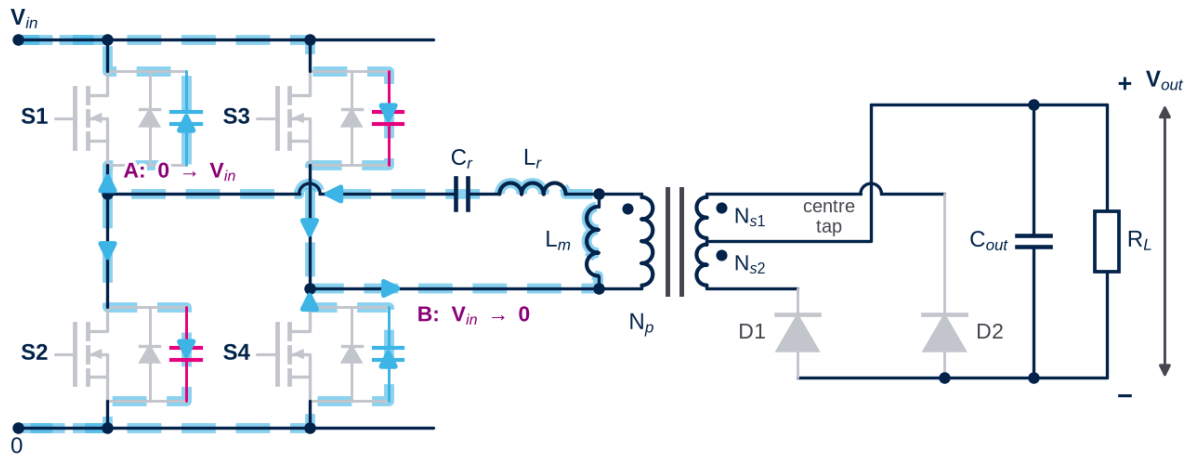
— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 8: Second half, steps 5 and 6: the mirror image of 1 and 2 with S_2, S_3 and D_2 .

Single-Stage PFC LLC Converter Design

7 DEAD TIME (a)

all four off · C_{oss} swaps the midpoints back

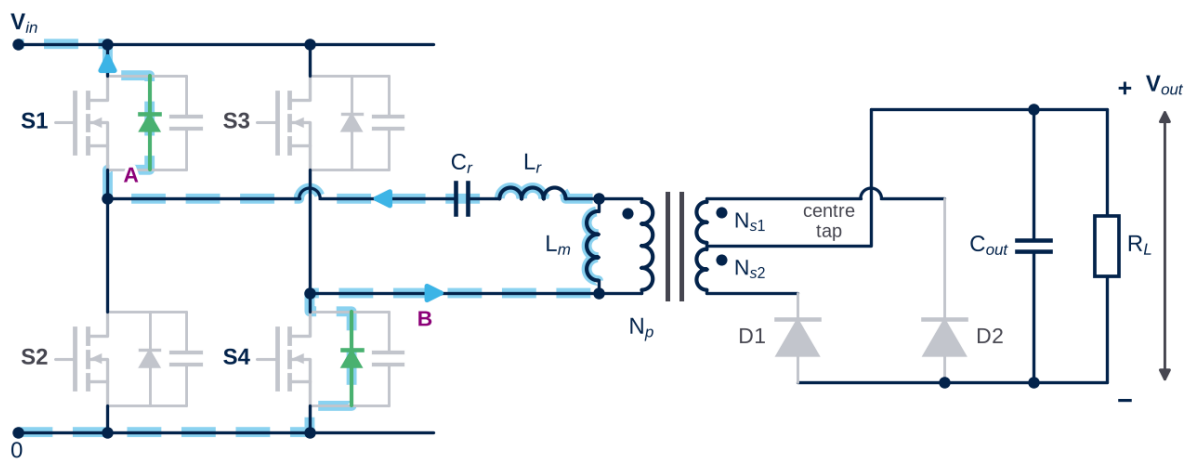


The mirror of 3: the current now arrives at B and leaves A, so B falls and A rises.

Through the capacitances again, never a channel. Magenta: C_{oss} charging, V_{ds} rising; cyan: discharging.

8 DEAD TIME (b) — ZVS

body diodes of S1 and S4 clamp



The same for S1 and S4, and the period closes. ZVS is in 4 and 8 - and in none of the other six.

— load current — magnetising current only gated on
 body diode conducting = the ZVS window — C_{oss} charging, V_{ds} rising — C_{oss} discharging

Figure 9: Steps 7 and 8, the second dead time. The mid-points swing back (7), the body diodes of S_1, S_4 clamp (8), and step 1 begins at zero volts. ZVS happens in steps 4 and 8 only.

Single-Stage PFC LLC Converter Design

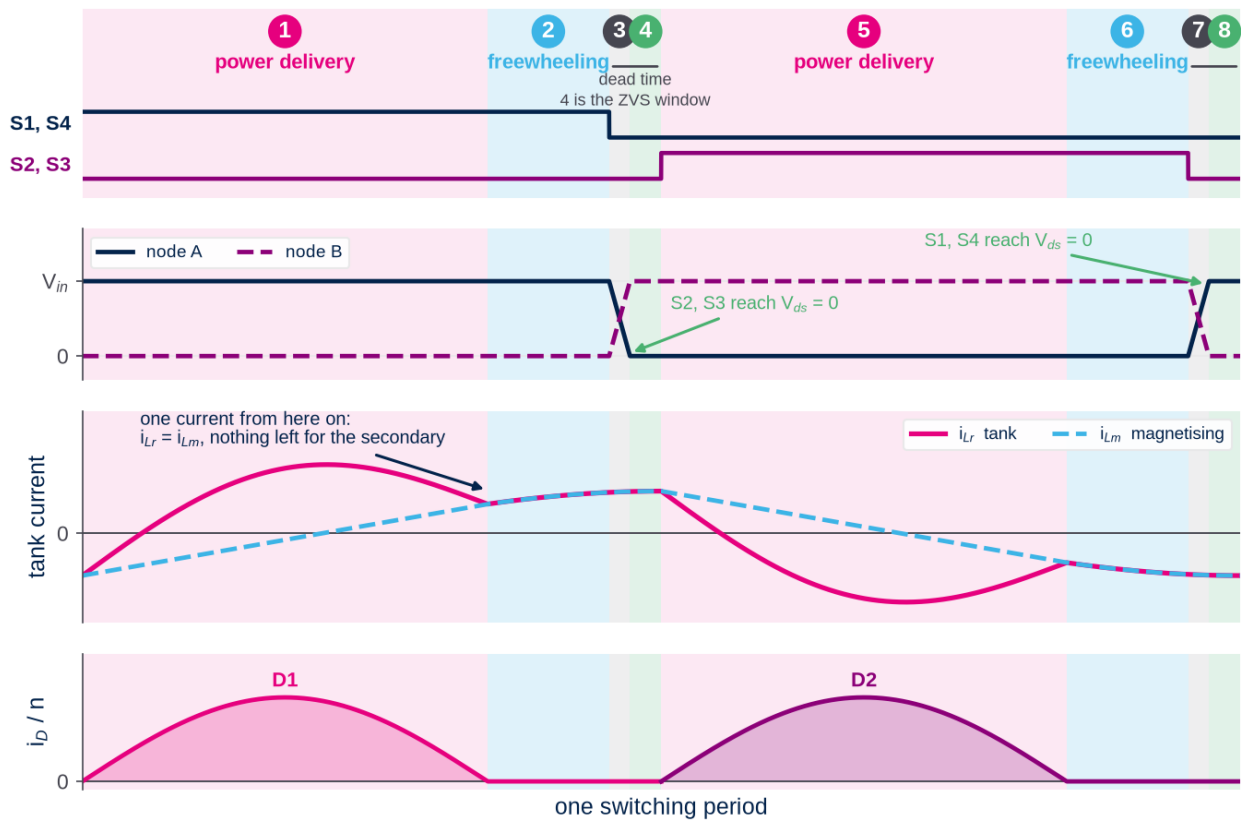


Figure 10: Where each step sits on the waveforms. The numbered bands are the eight panels above in time order. Drawn below resonance at $f_{sw}/f_r = 0.70$ with the dead time widened so that steps 3, 4, 7 and 8 are visible; the current amplitudes are this design's $I_{Lr,pk}$ and $I_{Lm,pk}$ at the lowest voltage the tank sees. The rectifier current is drawn referred to the primary, $i_D/n = i_{Lr} - i_{Lm}$.

“Freewheeling” means two different intervals in the literature. Here it is steps 2 and 6: rectifiers off, L_m in the resonance, same switches on. Toshiba’s *Resonant Circuits and Soft Switching* (2019) uses it for the dead time, when the body diodes conduct. The dead time exists at every frequency; steps 2 and 6 exist only below resonance.

Single-Stage PFC LLC Converter Design

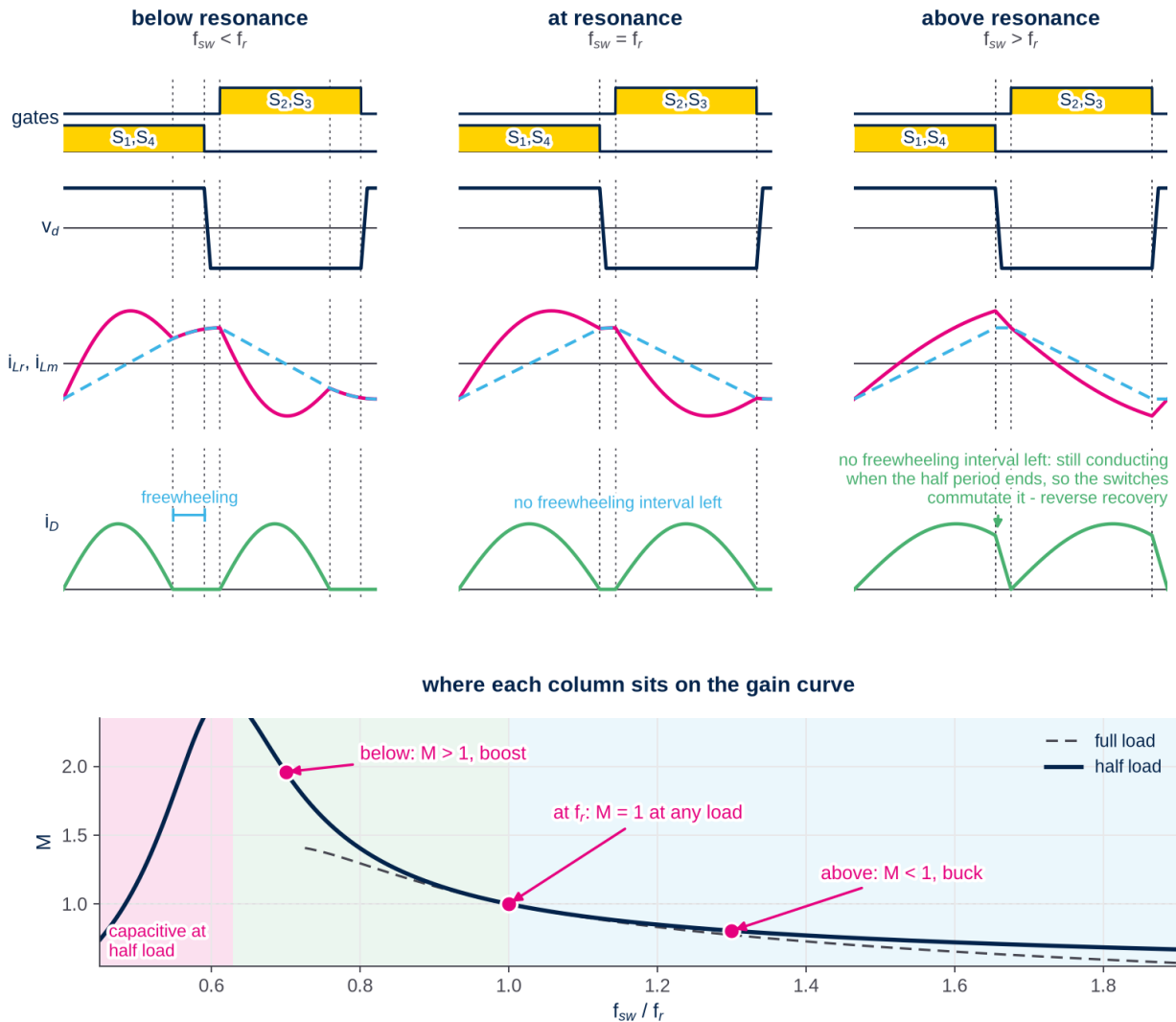


Figure 11: Below, at and above resonance. Top: gates, bridge voltage, resonant current with the magnetising current dashed, and rectifier current. Bottom: the same three switching frequencies as points on the half-load gain curve. At full load $f_{sw}/f_r = 0.70$ is already capacitive, so the full-load curve is drawn only from its ZVS edge up.

The bottom panel is the gain M of Section 2.6. Which side of f_r the converter runs on decides whether the tank boosts or bucks:

- **Below f_r : $M > 1$, boost.** The resonant half sine ends early and i_{Lr} **lands on** i_{Lm} . From then on no power crosses to the secondary: freewheeling.
- **At f_r : $M = 1$ at any load.** i_{Lr} meets i_{Lm} as the bridge switches, and the rectifier current reaches zero at the same instant. This is the most efficient point.
- **Above f_r : $M < 1$, buck.** The switching half period ends first and the resonant half sine is **cut off**. The switches turn off more current and the rectifier is interrupted while conducting.

The freewheeling interval rings at f_o . Nothing clamps L_m , so $L_r + L_m$ rings with C_r . It looks nearly flat because it is short against the period of f_o and sits near the crest of that ring. The lower f_{sw} , the larger its share of the half period. Everything useful happens between the two frequencies.

Single-Stage PFC LLC Converter Design

Symbols are not standard. At least one widely used guide calls the *series* resonance f_o , the opposite of the meaning here. Check before comparing.

A conventional LLC runs close to f_r at nominal input, where the circulating current is smallest, and goes below it only in hold-up. This converter runs on both sides, depending on the input voltage and the line phase (Section 4.7).

2.6 The gain function and the three regions

With the first-harmonic approximation the tank gain is

Equation 4

$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2\left(f_n - \frac{1}{f_n}\right)^2}}$$

Two checks by eye: at $f_n = 1$ the Q term vanishes and **$M = 1$ at any Q** , so every curve passes through that point. At no load ($Q = 0$) raising f_n only brings M down to $1/(1+\lambda)$; that floor matters in Section 4.10.

The curve has three regions. Below the capacitive/inductive boundary the bridge hard-switches: not allowed. Between the boundary and f_r the tank is inductive and boosts; above f_r it is inductive and bucks. The boundary is drawn for the full-load curve; at no load it sits at f_o and it moves up with load.

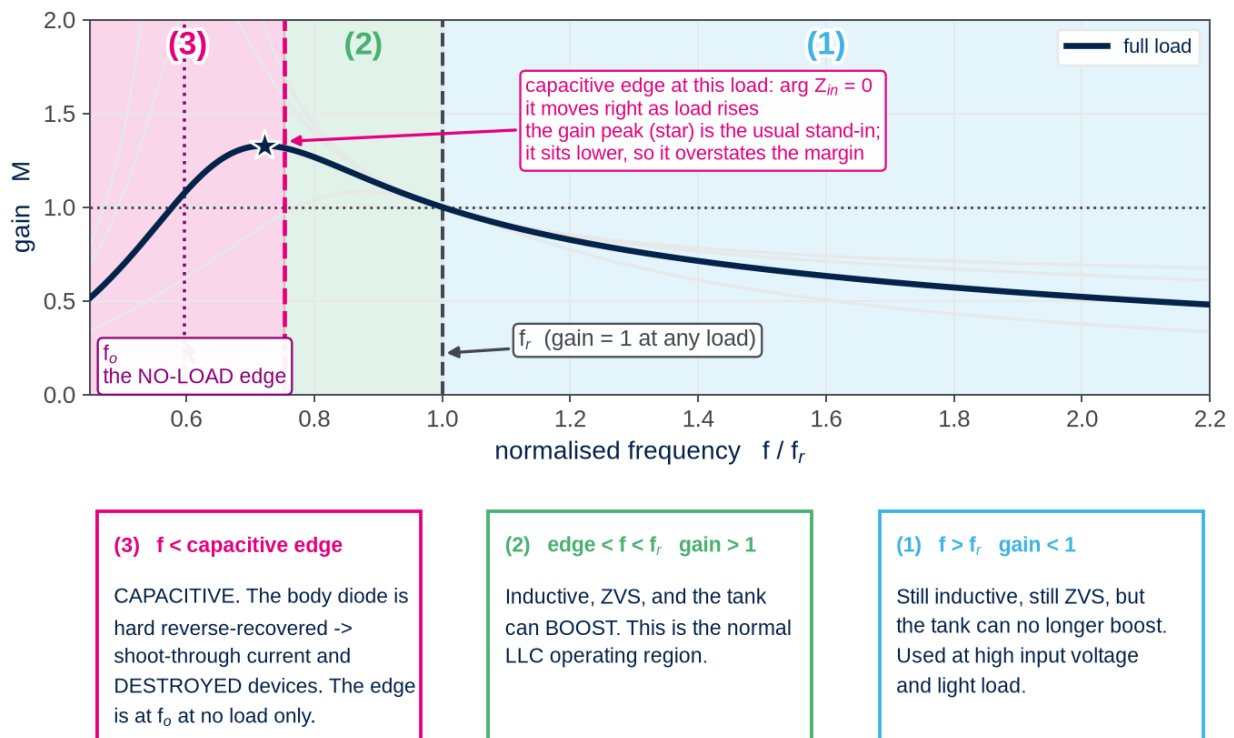


Figure 12: The three regions, shaded for the full-load curve. Only the inductive ones are usable. Framing after ROHM TechWeb.

Single-Stage PFC LLC Converter Design

The side can be read from the chart without a waveform. Draw the required gain as a horizontal line and find where it crosses the curve for the present load. Above 1 the crossing is left of $f_n = 1$ (below resonance, the left column of Figure 11); at 1 it is f_r at any load; below 1 it is to the right (above resonance).

So the input voltage decides the side, not the load: the required gain is the reflected output voltage over the input the tank sees, and the load only picks the curve. A higher input lowers the line and moves the crossing right. The load decides whether the crossing is still to the right of the capacitive boundary (Section 2.9).

2.7 Capacitive and inductive, and why those words

The names describe **what the bridge sees**: the phase of the tank current against the square wave that drives it. At low frequency C_r dominates and the current *leads*; at high frequency the inductances dominate and it *lags*. That phase decides whether the switching is soft or hard.

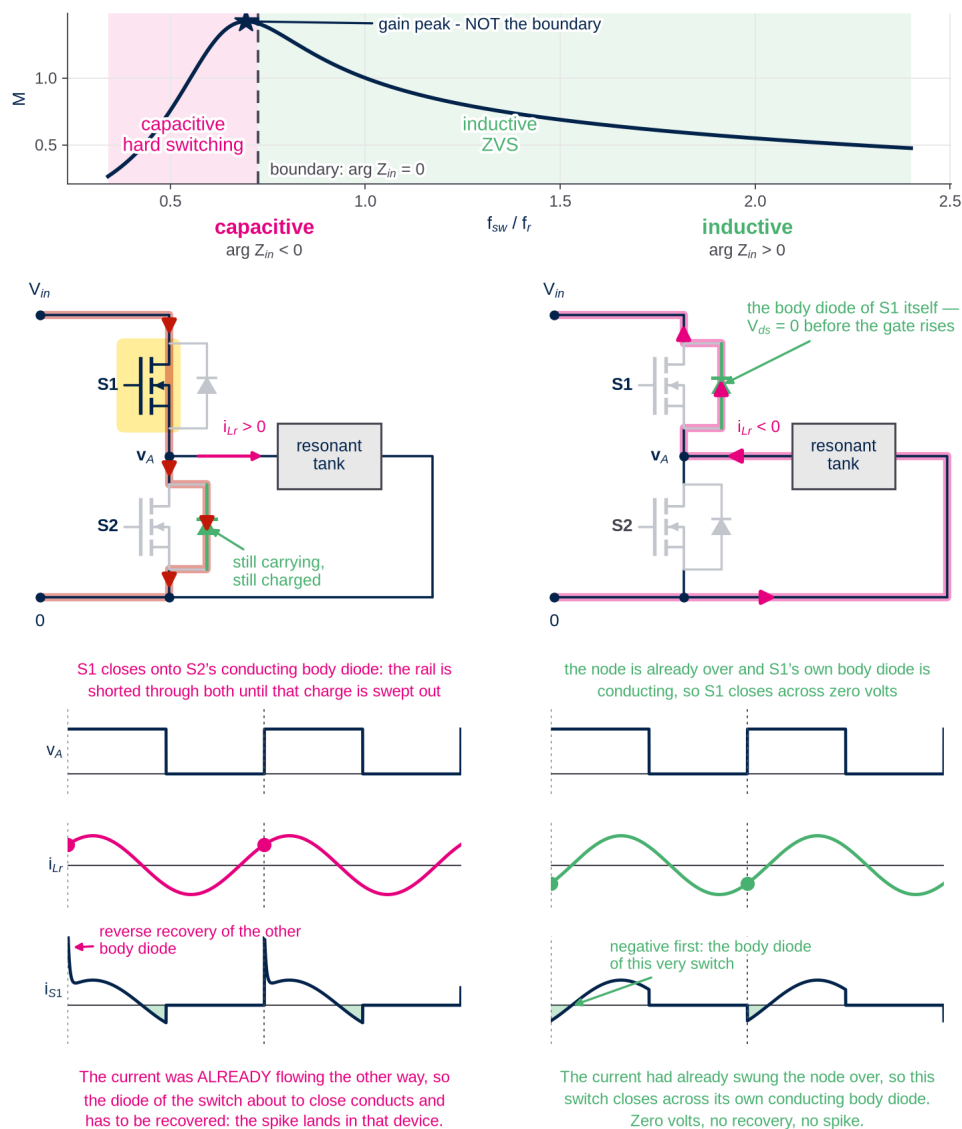


Figure 13: Either side of the boundary. Top: the boundary sits a little above the gain peak. Middle: the leg as S_1 is gated on. Left, the tank current is already positive, flowing in S_2 's body diode, so S_1 closes onto a conducting diode. Right, it is negative, has carried the mid-point up during the dead time, and S_1 finds its own body diode conducting at zero volts. Bottom: the mid-point voltage v_A (to 0), the tank current and the drain current i_{S1} of the closing switch, to one scale.

Single-Stage PFC LLC Converter Design

- **Inductive: the current lags.** At turn-off it still pushes the bridge node towards the other rail; the dead time lets it, and the next device turns on at zero volts. **ZVS.**
- **Capacitive: the current leads.** It has already reversed, so it pushes the node back. The next device turns on into the full rail, and the other body diode is forced off with reverse recovery through a low impedance. **Hard switching, and the failure that destroys parts.** The design stays out of this region; the controller has a protection in case it does not.

2.8 Why a diode has to recover on one side and not on the other

Take one leg: a high-side switch, a low-side switch and the tank on the mid-point. When either device turns off, **the tank current does not stop**. What carries it during the dead time decides everything. The middle row of Figure 13 shows the two cases, after onsemi AN-4151.

- **Inductive.** The current carries the mid-point to the *other* rail through the two C_{oss} , the incoming body diode picks it up, and that device is gated on at zero volts. Its own body diode then hands over to the channel with no reverse voltage: **nothing to recover.**
- **Capacitive.** The current has already reversed. It holds the mid-point on the rail it came from through the body diode of the device that just turned off. The incoming device is gated on across the full rail, onto a conducting diode.

That diode cannot block until its stored charge is swept out, so for that time both devices conduct and the rail is shorted through them. The current is a spike limited only by loop inductance, dissipated in the incoming device at full rail voltage; when the diode snaps off, the di/dt in the loop inductance appears as overshoot across it.

This is why a fast-recovery body diode is required on the primary (Section 7.16): any excursion into the capacitive region turns a device on onto the conducting body diode of the other. It is also why f_{Min} is kept above f_o , and why the anti-capacitive protection exists.

2.9 The two boundaries are not the same boundary

Both pairs of words describe positions on the same frequency axis, but they are different lines on it.

	capacitive / inductive	below / above resonance
the boundary	where $\arg Z_{in} = 0$	f_r , the series resonance
does it move?	yes — with load	no, it is fixed by L_r and C_r
what it decides	ZVS or hard switching, on the primary	boost or buck, and ZCS or not, on the secondary
which side is allowed	inductive only, always	both sides are legitimate

Table 1: Two classifications, two boundaries, two consequences.

The capacitive boundary **rises with load**, from f_o at no load towards f_r as the load grows, so a frequency that is inductive at light load can be capacitive at overload.

The boundary is not the gain peak. It is where $\arg Z_{in} = 0$, a little *above* the peak. Using the peak is optimistic in the dangerous direction. Section 7.5 gives both frequencies for this design; the verdict comes from the ZVS check of Section 6.5.

Single-Stage PFC LLC Converter Design

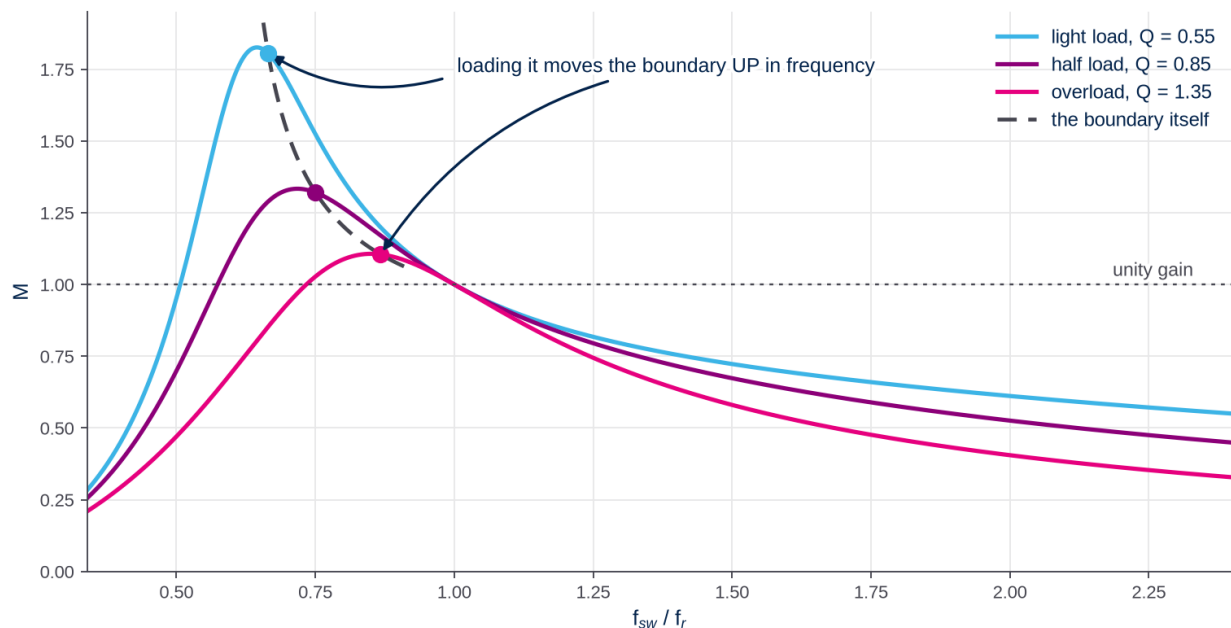


Figure 14: The boundary moves up with load. The dashed curve is the boundary itself, traced as Q varies.

- Below f_r and inductive is the normal boosting region.
- Below f_r and capacitive is the fault. Going below f_r is not the danger; going below the boundary is.
- Above f_r is inductive at every load.

Design rule. Keep f_{Min} above f_o . That is only the no-load boundary, so it is necessary but not sufficient; the real check is the ZVS sweep of Section 6.5, run under load at every line phase.

2.10 ZVS and ZCS are not the same thing

Both switch a device while its voltage or its current is zero. They apply to different devices, and only one is guaranteed.

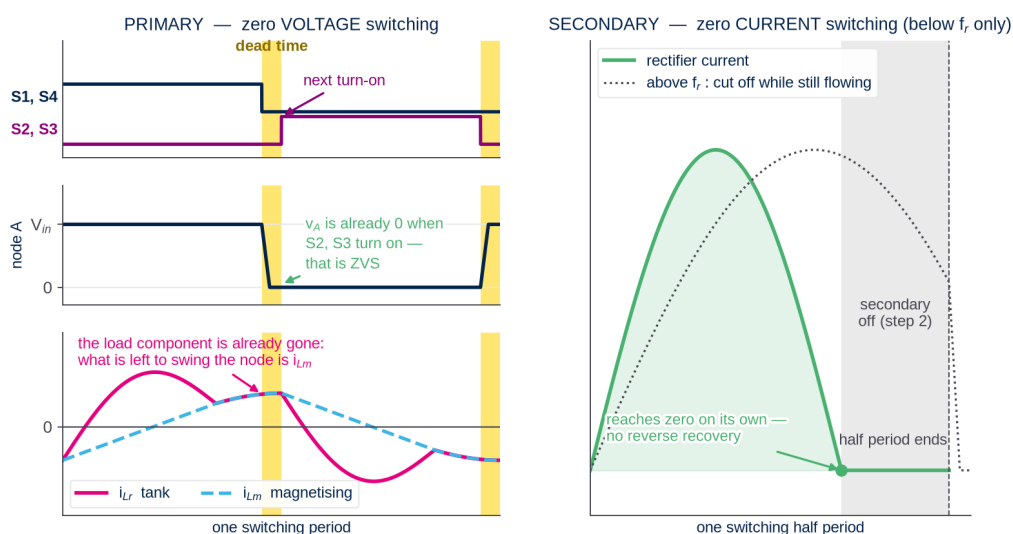


Figure 15: ZVS on the primary, ZCS on the secondary: different devices, different mechanisms, different conditions.

Single-Stage PFC LLC Converter Design

- **ZVS on the primary switches** needs inductive operation, which is possible on both sides of f_r . It is required everywhere; Section 6.5 gives the margin.
- **ZCS on the secondary rectifiers** happens when the resonant current reaches zero by itself, which is only **below** f_r . Above f_r the rectifier is cut off while conducting and reverse recovery returns.

A converter whose input crosses f_r must be checked on both sides: its rectifier body diode and synchronous-rectifier dead time matter whenever it runs above f_r .

2.11 Sizing the dead time so that ZVS actually happens

ZVS is a charge problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail. The design quantity is T_{ZC} , the time the tank current takes to reach zero after the gates turn off, and it must exceed the dead time:

Equation 5

$$T_{ZC} > t_D$$

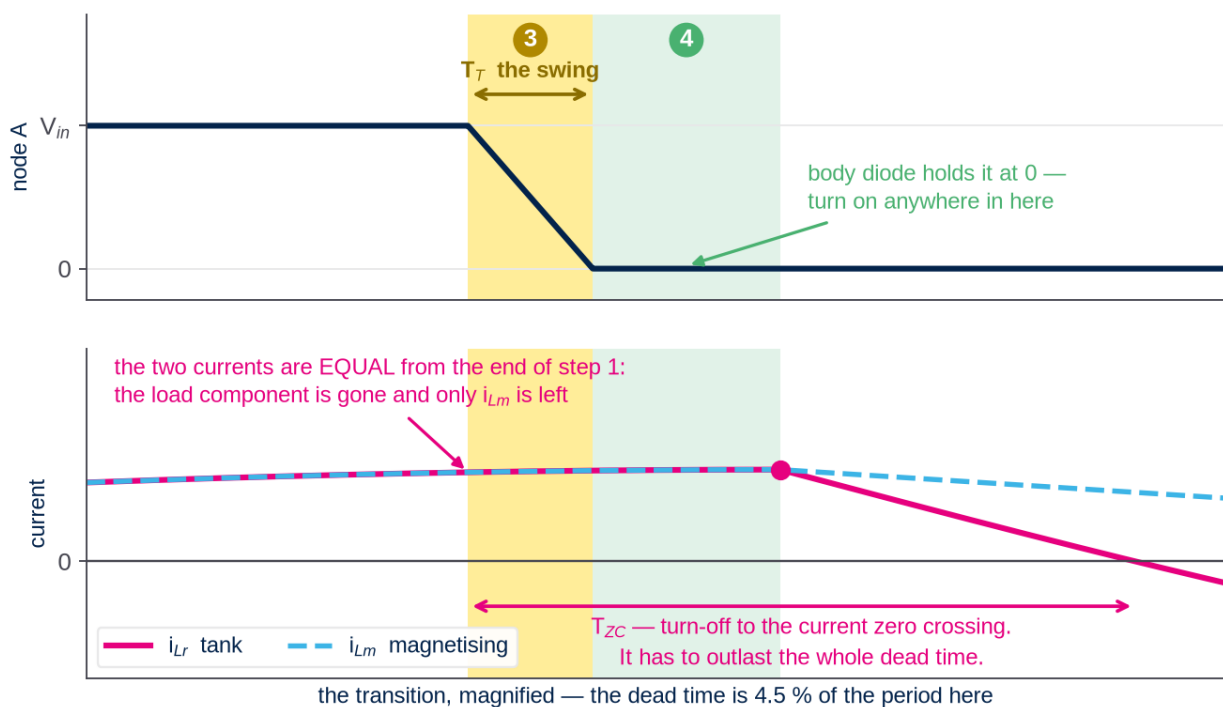


Figure 16: The magnetising current discharges the bridge node during the dead time. Two conditions: the current must not reach zero before the dead time ends ($T_{ZC} > t_D$), and the swing T_T must finish inside the dead time.

Use $C_{o(tr)}$ (equivalently Q_{oss}), not the headline C_{oss} : the datasheet values differ by three to five times, and ZVS is a charge question.

Single-Stage PFC LLC Converter Design

2.12 Not a flyback, and why that changes the core

Two flyback habits are wrong here and cost money: sizing the core from the current, and asking for saturation at the peak winding current. Both come from when the two windings conduct.

In a flyback they never conduct together, so at every instant the whole winding current is magnetising current. Every joule delivered was first stored in the core, the flux follows the *current*, and A_e is sized from the peak current.

In an LLC they conduct together. Their ampere-turns oppose, and only the difference magnetises the core:

Equation 6

$$N_p i_p - N_s i_s = N_p i_\mu$$

The load current passes through by transformer action and never enters the core. What the core holds is the magnetising energy $\frac{1}{2}L_\mu i_\mu^2$, with L_μ the magnetising inductance of the physical transformer (it is not quite the tank's L_m ; Section 6.9). That energy is not a by-product: i_μ is what swings the bridge node in the dead time. The gap exists to set L_μ , and through it L_m , not to store load energy.

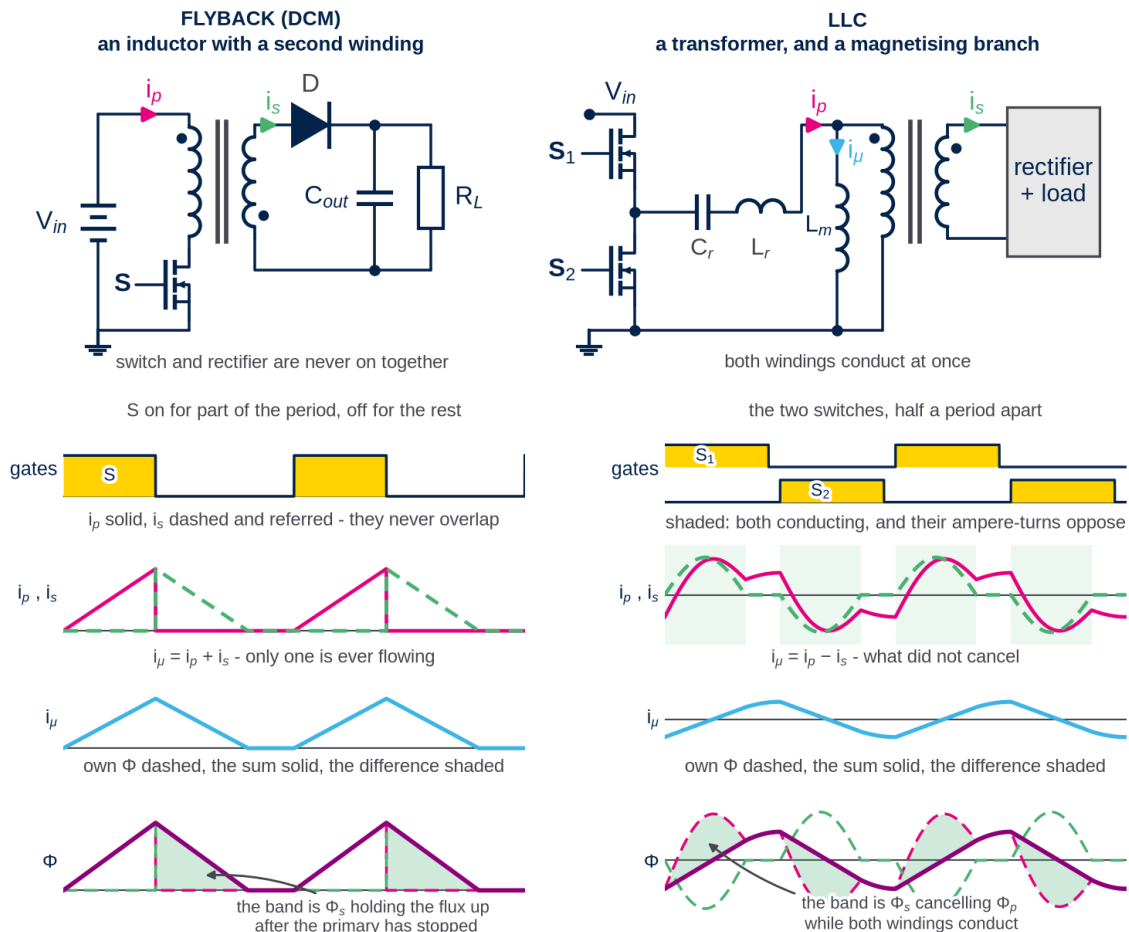


Figure 17: Left: a flyback in discontinuous conduction, as in Figure 18. The switch and the rectifier are never on together, so the conducting winding carries all the magnetising current and the flux ramps with it. Right: both conduct at once and only $i_\mu = i_p - i_s$ magnetises the core. The secondary current is drawn referred to the primary.

The arrows are the reference directions: i_s is taken into the dot in the flyback and out of it in the LLC, so the flyback's ampere-turns add. Bottom row: dashed, what each winding would drive alone; solid, their sum; shaded, the difference the secondary makes.

Single-Stage PFC LLC Converter Design

Figure 18 follows one period of each converter interval by interval: which winding conducts, what voltage it holds, and so which way the flux moves and what stops it.

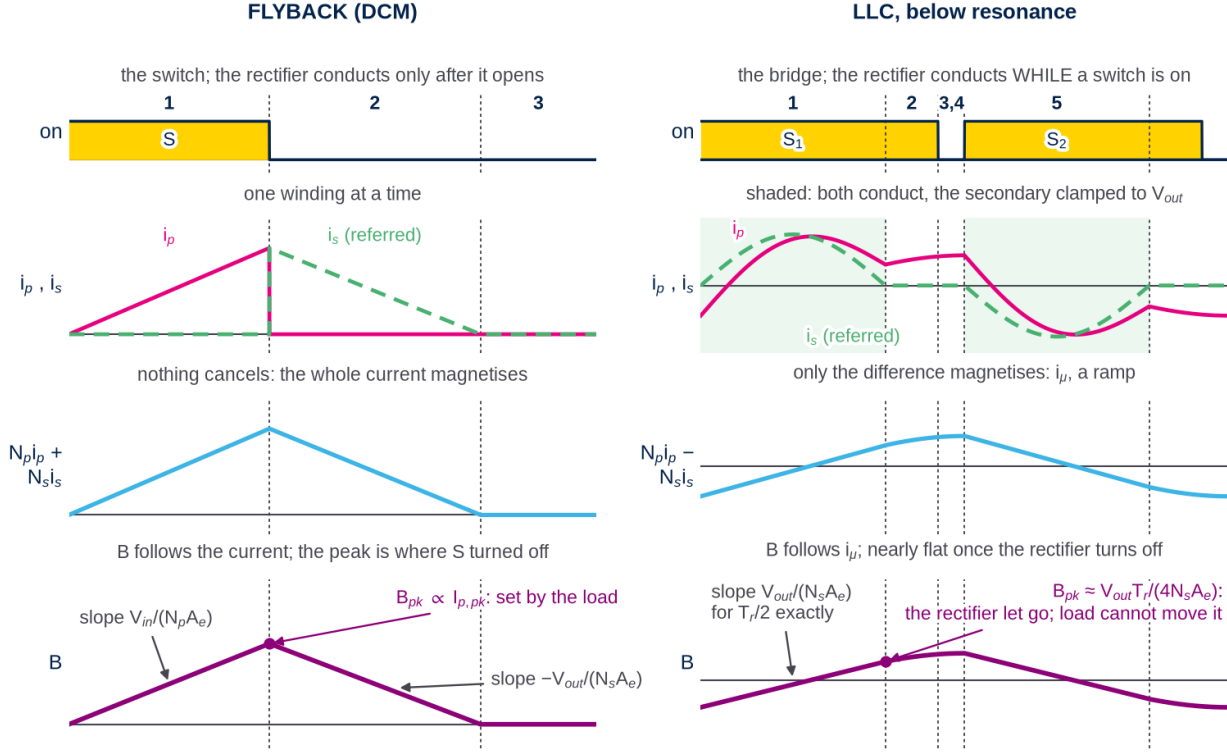


Figure 18: The flux built step by step. Left, a flyback in discontinuous conduction; right, an LLC below resonance, drawn from the eight-interval model with this design's currents and numbered with its steps (Figure 10). Compare the bottom row, the flux.

- **Flyback, 1.** The switch closes. V_{in} sits on the primary, i_p ramps at V_{in}/L_p , and because no other winding conducts every ampere-turn magnetises: B rises at $V_{in}/(N_p A_e)$. The energy goes into the gap.
- **Flyback, 2.** The switch opens. The flux cannot jump, so the ampere-turns pass to the secondary at once, $N_s i_s = N_p i_{p,pk}$. The rectifier conducts, V_{out} sits on the secondary, and B falls at $V_{out}/(N_s A_e)$ while the gap energy leaves for the output.
- **Flyback, 3.** The secondary current reaches zero; both windings are off and the flux rests near zero until the next cycle. The peak flux was fixed at the turn-off instant by $i_{p,pk}$, which is whatever the load and the input asked for.
- **LLC, 1.** A bridge switch closes. As soon as the secondary voltage reaches V_{out} the rectifier conducts, so the secondary is clamped to V_{out} and the magnetising branch sees nV_{out} . The load current flows in both windings at once, their ampere-turns cancel, and only i_μ ramps: B rises at $V_{out}/(N_s A_e)$.
- **LLC, 2.** The resonant half period ends: i_s reaches zero, the rectifier turns off and the clamp is gone. L_m joins the resonance with C_r , i_μ is nearly flat, and B holds at its peak. That peak was set by V_{out} and $T_r/2$ and by nothing else.
- **LLC, 3 and 4.** Dead time. i_μ swings the bridge node to the other rail; the flux hardly moves.
- **LLC, 5.** The other switch closes and the other rectifier clamps the secondary to $-V_{out}$; B ramps back down at the same slope to $-B_{pk}$.

Single-Stage PFC LLC Converter Design

	Flyback	LLC
Which winding sets the flux	whichever one conducts; B follows its current	the clamped secondary; B follows its volt-seconds
What the peak flux depends on	$I_{p,pk}$: load, input voltage and conduction mode	V_{out} , T_r and N_s only
Flux against load	rises with load	does not move
Flux against switching frequency	falls as the frequency rises (shorter on-time)	constant below resonance, falls above it
What the gap does	stores the load energy; gap and A_e together set the power	sets L_m , i.e. the magnetising current and ZVS; holds only the magnetising energy
What threatens the core	overload, over-current	over-voltage on the output
Which protection guards the core	the current limit	the output OVP; the current limit only where there is no OVP (Section 6.19)
Saturation test current	the peak winding current, with margin	$i_{\mu,pk}$ scaled to the OVP ceiling; below the winding peak
What sizes A_e	peak current and L_p : energy	output volt-seconds at f_r : Faraday

Table 2: The same two cores, compared.

“An LLC transformer stores no energy” is a slogan. It stores the magnetising energy twice per period; it does not store the load energy.

A_e still matters because Faraday’s law does not ask whether energy is stored:

Equation 7

$$B(t) = \frac{1}{NA_e} \int v dt$$

with v the winding voltage, N its turns and A_e the core area. **Only the input to the calculation differs.** In a flyback the volt-seconds are the input voltage times the on-time. In an LLC the conducting secondary clamps the winding to the output for a resonant half period, so the flux is set by the **output voltage alone** (Section 6.11).

Three consequences:

- **The flux does not rise with load.** More load raises i_s and the reflected part of i_p together; their difference i_μ does not move. Overload threatens the copper and the semiconductors, not the core.
- **The flux rises with output voltage**, so the saturation specification is set at the over-voltage threshold, not the nominal output (Section 6.19).
- **The flux does not fall with switching frequency** below resonance: the rectifier clamps the winding for $T_r/2$ whatever f_{sw} is, which is why f_r appears in the flux equation.

What to keep in mind when designing.

- **Size A_e from volt-seconds, not from current.** The input is V_{out} , N_s and f_r (Section 6.11); the winding current does not enter.

Single-Stage PFC LLC Converter Design

- **Do not add gap for margin.** In a flyback more gap means more stored energy. Here more gap means a smaller L_m : more magnetising current, more circulating loss, a different λ , and no change in the flux at all. The gap is set by L_m and checked by L_{open} ; the flux margin is set by the OVP ceiling and checked by the DC-overlap test.
- **The test current follows L_μ ; the flux does not.** A smaller gap raises L_μ and lowers $i_{\mu,pk}$ at the same flux. Recompute I_{eq} from the actual L_μ whenever the gap changes; never carry it over.
- **B_{pk} is fixed, so core loss follows frequency.** The flux is the same at every point of the line cycle below resonance, and the core loss at that flux rises with f_{sw} . Above resonance the flux falls as $1/f_{sw}$ and the loss falls with it, so the worst core loss is near f_r .
- **The flyback habit that costs the most** is asking the supplier for saturation at the tank peak current. That asks for a larger core than the converter needs, for a flux the converter can never produce.

A DC-overlap test is made with every other winding open, which removes the cancellation: all of the test current is magnetising current. Section 6.19 takes that to a number.

2.13 How an LLC is normally designed

The standard sequence. Section 6 follows it and departs from it in two places.

- 1 **Turns ratio** so that the converter runs near $M = 1$ at nominal input, where circulating current is lowest.
- 2 **Gain range** $M_{min} \dots M_{max}$ from the input range and output tolerance.
- 3 **Load resistance referred to the primary, R_{ac} .**
- 4 **$m = (L_r + L_m)/L_r$ (or λ) and Q together** from the peak-gain chart. For each m there is a curve of attainable peak gain against Q , and the design must sit under it with margin. Larger m : less circulating current, less peak gain.
- 5 **Component values.** Q and R_{ac} give Z_0 ; Z_0 and f_r give C_r and L_r ; m gives L_m .
- 6 **Verify** gain margin at the worst corner, ZVS at the worst point, and currents and flux against real parts.

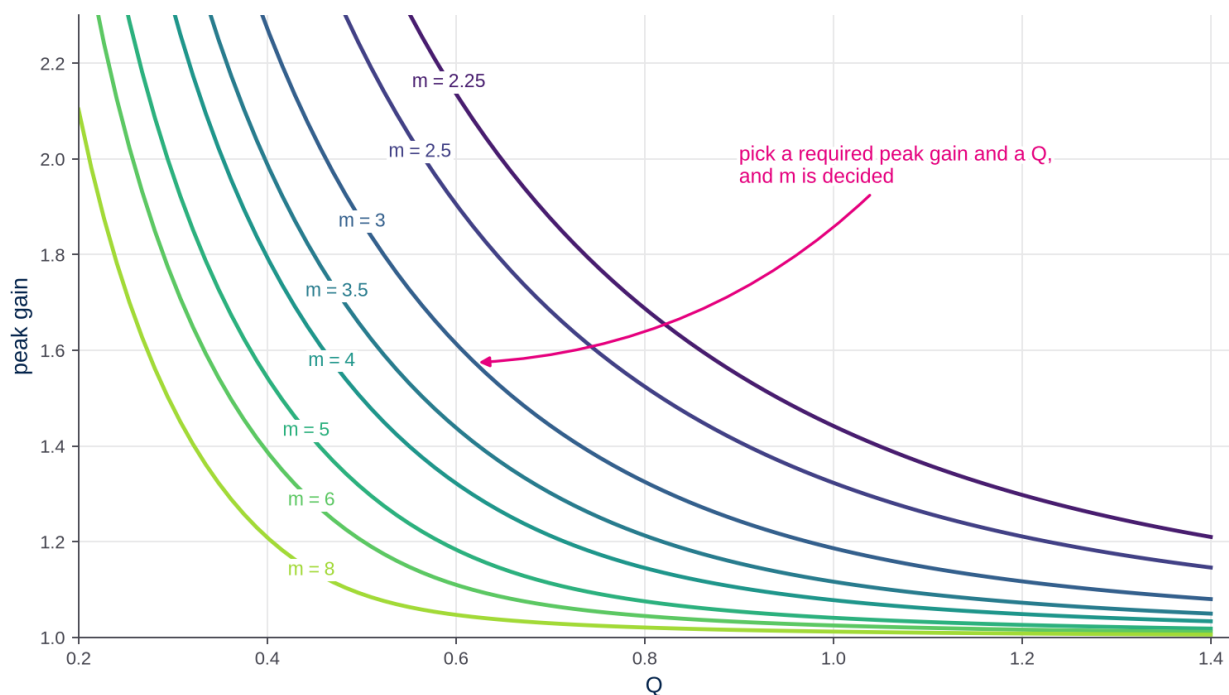


Figure 19: Step 4: the peak gain depends on both m and Q , so a required gain and a Q leave only a band of usable m .

Single-Stage PFC LLC Converter Design

2.14 What power factor correction is

A bridge rectifier feeding a capacitor holds the bus near the mains peak, so the diodes conduct only in the short window when the mains is above the capacitor voltage. All the charge for the whole cycle arrives in that window.

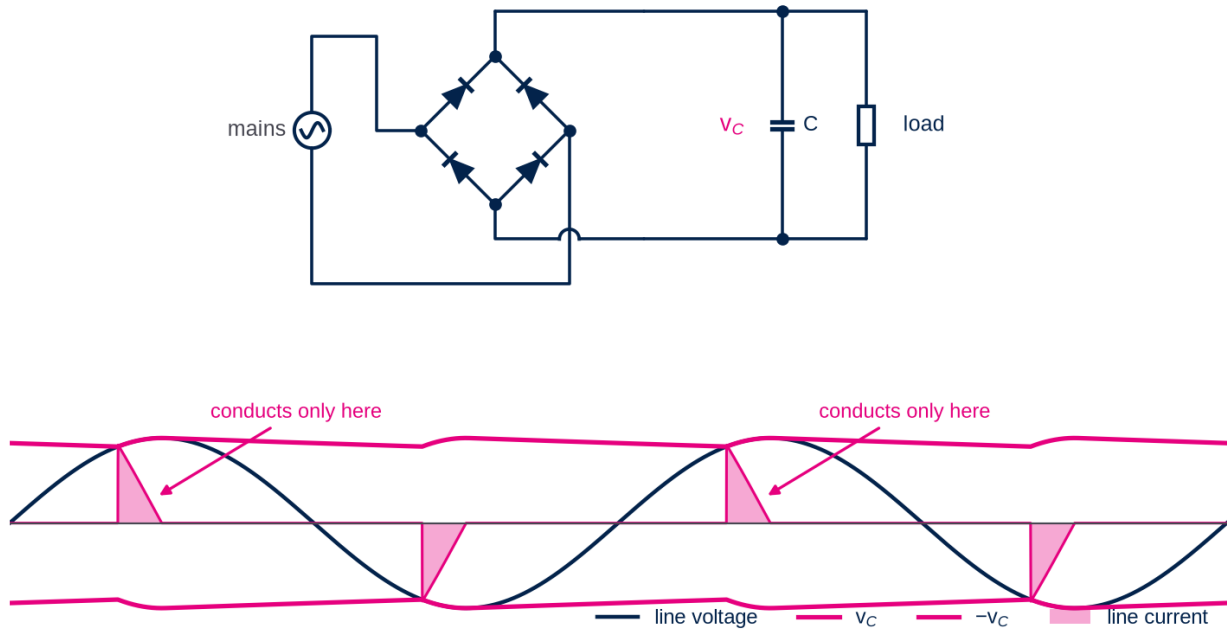


Figure 20: A capacitor-input rectifier: v_C stays near the crest, the diodes conduct in narrow windows, and the current inside them is tall.

The peak current is several times what a resistor of the same power would draw, and a pulse train that narrow is mostly harmonics.

2.15 Power factor, and why it is not the same as distortion

Power factor is the ratio of real power to apparent power, with V_{ac} and I_{ac} the rms input voltage and current:

Equation 8

$$PF = \frac{P}{V_{ac} I_{ac}} = \cos \varphi_1 \times \frac{I_{1,rms}}{I_{ac}}$$

with P the average power, $I_{1,rms}$ the rms of the fundamental alone, and φ_1 its phase against the voltage. **Displacement** (the phase) is what a motor gets wrong; **distortion** (the harmonic share of the rms) is what a rectifier gets wrong. The current above is roughly in phase and still has a power factor of only about 0.6.

Total harmonic distortion, THD, is the harmonic content as a fraction of the fundamental:

Equation 9

$$THD = \frac{\sqrt{I_{ac}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{ac}} = \frac{1}{\sqrt{1 + THD^2}}$$

Single-Stage PFC LLC Converter Design

A converter can hold PF above 0.99 and still fail IEC 61000-3-2, which limits *individual* harmonics in amperes. Class D covers only the product types the standard lists, up to 600 W; above that the absolute Class A limits apply. Settle the class first.

2.16 How a corrector fixes it

Force the input current to follow the input voltage and the converter looks like a resistor to the mains. The standard implementation is a **boost converter after the bridge**, switching fast enough that the mains is constant within one switching cycle.

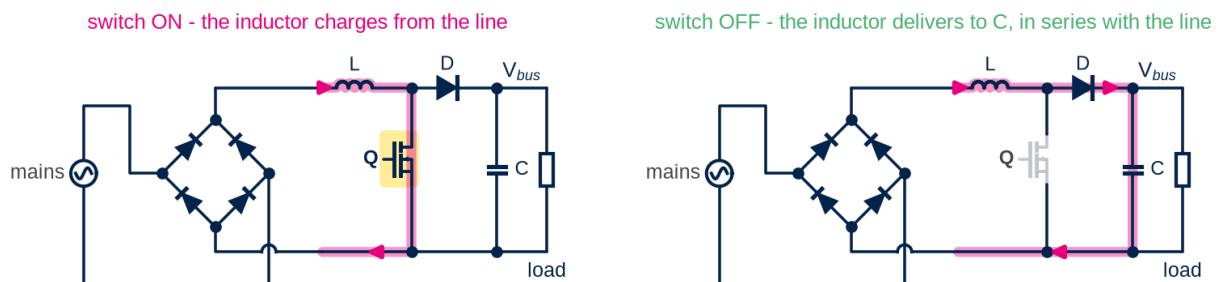


Figure 21: A boost corrector: the inductor charges from the line while the switch is on and delivers to the bus in series with the line while it is off.

- **Why a boost.** Near the zero crossing the stage must step up by an unlimited ratio. A boost can, and its input current is continuous, which makes shaping possible.
- **The inner loop** regulates the inductor current, fast.
- **The outer loop** regulates the bus voltage and is made **slower than $2f_l$** . A fast voltage loop would fight the bus ripple and flatten the current shape the corrector exists to make.
- **The multiplier** joins them: shape from the mains, level from the output.

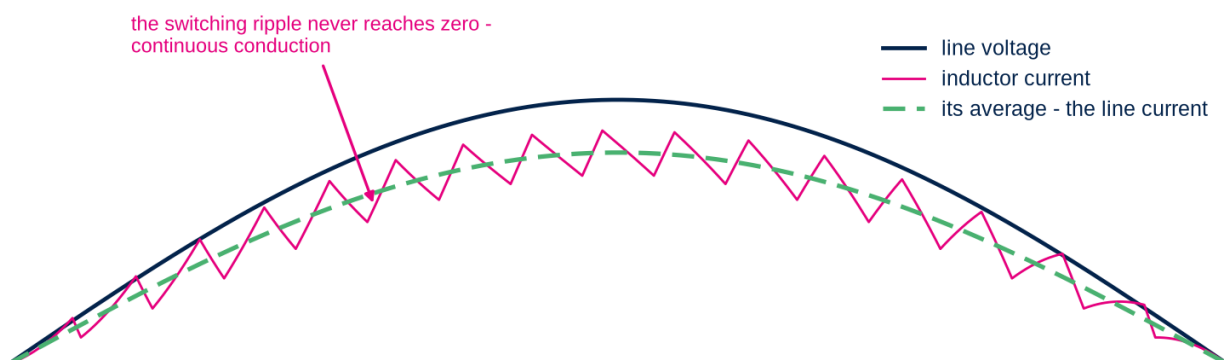


Figure 22: The result in continuous conduction: switching ripple rides on the inductor current, and its average follows the input voltage.

A current that follows the voltage means an input power that follows $\sin^2\theta$: zero twice per cycle, twice the average at the crests. The load wants constant power, so **something must store the difference**.

Single-Stage PFC LLC Converter Design

2.17 Why they are normally two stages

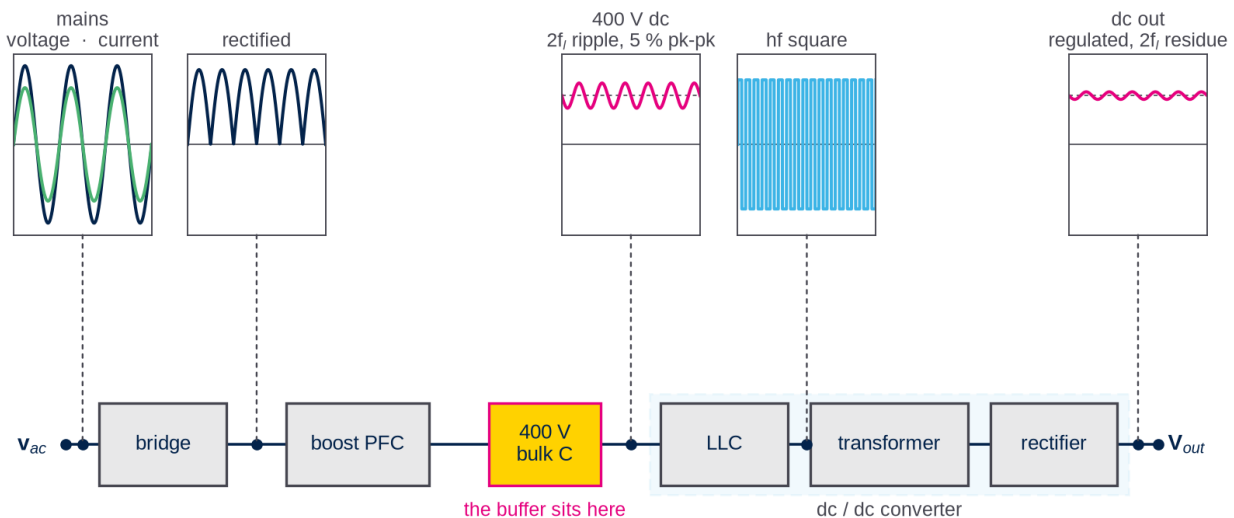


Figure 23: The two-stage arrangement with the waveform at every node. The buffer that absorbs the twice-line-frequency pulsation sits between the stages at 400 V. Ripple on the dc nodes is exaggerated; v_{ac} is the instantaneous mains voltage.

The bus capacitor does two jobs:

- it **buffers** the difference between the pulsating input power and the constant output power;
- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both. The buffering must be done elsewhere, and the tank must work from a rectified sine that goes to zero 100 or 120 times a second.

3 Why single stage, and what it costs

Two steps of the standard LLC sequence (Section 2.13) do not survive here. There is no nominal point, because the input sweeps a half sine; and no single tank covers the gain range, so the bridge itself changes (Sections 4.1 and 5).

3.1 The energy a unity power factor cannot deliver

Unity power factor makes the instantaneous input power follow $\sin^2\theta$, swinging between zero and twice the average at twice the line frequency:

Equation 10

$$p_{in}(t) = V_{ac}I_{ac} [1 - \cos(2\omega_l t)]$$

with V_{ac} and I_{ac} the rms mains voltage and current and $\omega_l = 2\pi f_l$. The load wants constant power, so the difference must be stored and returned a few milliseconds later, whatever the topology.

Panel 2 of Figure 24 draws it: the surplus above the flat load equals the deficit below it, and a capacitor somewhere must absorb the one and return it as the other.

Single-Stage PFC LLC Converter Design

3.2 Moving the buffer from 400 V to the output

In the two-stage converter the buffer is the 400 V bus capacitor. Here there is no bus, so it moves to the output. **The joules do not change; the voltage they sit at does**, and a capacitor returns only the energy between its starting voltage and the lowest the load accepts:

Equation 11

$$E = \frac{1}{2} C (V^2 - V_{min}^2)$$

Hold-up shows the price. Both architectures must store the same energy for the same T_{hold} . On a 400 V bus that may fall to 320 V the window is $400^2 - 320^2$; on an output of a few tens of volts it is $V_{out}^2 - V_{o,min}^2$, smaller by a factor in the hundreds, so the capacitance grows by that factor. Settle this first: if the enclosure cannot house the bank, the output voltage must change. Section 7.14 sizes the bank for this design, where ripple, not hold-up, decides.

The trade. Out: a switch, an inductor, a diode, a 400 V electrolytic and their losses. In: a very large low-voltage capacitor bank, 2f_l ripple on the output, and a tank that must cover a much wider range. Whether it is a good trade depends on how much output ripple the load accepts.

3.3 What the load must tolerate

The output carries the 2f_l ripple directly. It is set by the bank and the load, not by the loop, and it cannot be regulated away: trying puts the distortion on the input current instead (Section 6.30). **The allowed ripple is a property of the load** and must come from the specification. This design allows a few per cent peak-to-peak.

4 Operating principle

4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain. Here it cannot, because **both ports are voltage sources**: the output is a millifarad-class capacitor bank and the input is the rectified mains. The instantaneous gain is forced:

Equation 12

$$M(\theta) = \frac{n V_{o,eff}}{V_{drive}(\theta)} = \frac{2 n V_{o,eff}}{\sqrt{2} V_{ac,eq} \sin \theta}$$

with θ the line phase and V_{drive} the amplitude the bridge applies to the tank: the rectified mains in full bridge, half of it in half bridge. $V_{ac,eq}$ is the mains voltage a half bridge would need for the same drive, so twice the mains in full bridge and the mains itself in half bridge; $V_{o,eff}$ is the output plus the rectifier drop (Equation 19). Both voltages are pinned from outside, so their ratio is fixed moment by moment.

Moving the frequency therefore moves **Q**, that is, power. **Frequency is a power command, not a voltage command.**

4.2 Two divergences that cancel

Near the zero crossing the required gain rises as $1/\sin\theta$ without limit. The converter survives because the load disappears at the same time:

Single-Stage PFC LLC Converter Design

Equation 13

$$Q(\theta) = Q_{pk} \sin^2 \theta$$

with Q_{pk} its value at the line peak, and an unloaded LLC has unlimited gain at f_o . The two infinities cancel, and the operating point walks down towards f_o as the mains approaches zero (Section 7.7). f_o is the **frequency floor of the whole design**; an oscillator clamp below it destroys hardware.

4.3 Frequency modulation is the power factor correction

Figure 24 follows the control law round one line half cycle. The first three steps are forced by the two voltage sources; only the fourth is something the controller does.

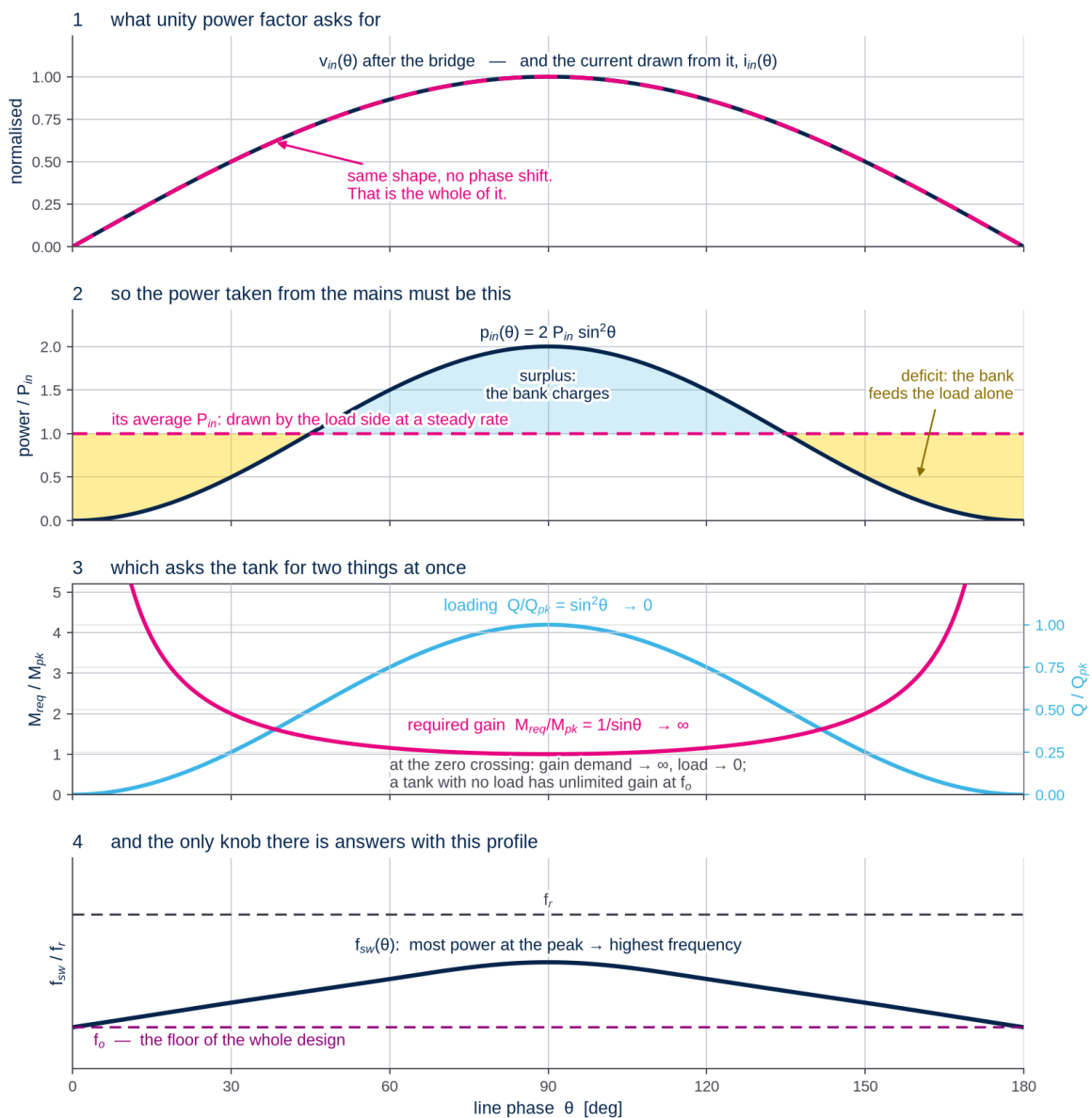


Figure 24: How the power factor is corrected, over one line half cycle. 1 Unity power factor: the current has the shape of the rectified mains. 2 The power drawn is $\sin^2$, while the load takes a steady P_{out} ; the shaded areas are what the bank buffers. 3 The tank is handed a gain and a load, and both run away at the zero crossing. 4 The switching frequency is the only free variable; this profile satisfies 1 to 3. Normalised; with numbers in Section 7.7.

Single-Stage PFC LLC Converter Design

Unity power factor fixes the instantaneous power at $2 P_{in} \sin^2\theta$, which fixes both the gain (Equation 12) and the loading (Equation 13) at every instant. The controller answers with the one variable it has. There is no current loop and no multiplier against a rectified reference: **the frequency profile $f_{sw}(\theta)$ is the power factor correction.**

At the line phase	$\sin\theta$	instantaneous power p/P_{in}	required gain M_{req}/M_{pk}	loading Q/Q_{pk}
$\theta = 90^\circ$, the peak	1.000	2.000	1.000	1.000
$\theta = 60^\circ$	0.866	1.500	1.155	0.750
$\theta = 45^\circ$	0.707	1.000	1.414	0.500
$\theta = 30^\circ$	0.500	0.500	2.000	0.250
$\theta = 10^\circ$	0.174	0.060	5.759	0.030
$\theta \rightarrow 0$	0	0	∞	0

Table 3: The half cycle at six instants, normalised as in Figure 24. Trigonometry only; no part value enters.

In the last two rows the gain demand runs away, but so does the ability to meet it: an unloaded tank has unlimited gain at f_o . The frequency walks down towards f_o and the current goes to zero with the voltage.

The controller has two signals on two time scales:

- **FB says how much.** It carries the error amplifier output through the optocoupler and commands an input *power* (Section 4.6). The loop behind it crosses well below the line frequency, so within one half cycle it is a constant: the *height* of the profile in panel 4.
- **HVSU says where in the cycle.** The start-up pin is also the input-voltage sense, so the controller knows the instantaneous rectified mains without a divider. That is the fast signal, the *shape* of the profile.
- **The internal PFC and THD block combines them** into I_{EA} , the one current that moves the oscillator (Equation 70).

The draft datasheet does not publish the law inside that block. This note therefore computes, from the tank, the profile that *must* come out if the current is to follow the voltage. Measuring $f_{sw}(\theta)$ on the first board is how the block gets characterised.

Zero-crossing dead zone. Very near $\theta = 0$ the profile asks for a frequency below f_{Min} , and the converter stops drawing current until the mains comes back. This is inherent and appears as third-harmonic distortion. The other THD mechanism is the voltage loop (Section 6.30).

4.4 Solving for the profile without a root search

The profile does not need a numerical root search. With $M_{req} = M_{pk}/\sin\theta$, $Q = Q_{pk}\sin^2\theta$ and $x = 1/f_n^2$ the gain equation is a cubic in x :

Equation 14

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{pk}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{pk}^2 u^2$$

Single-Stage PFC LLC Converter Design

One root is negative (the cubic is $+q$ at $x = 0$ and tends to $-\infty$ as $x \rightarrow -\infty$). The other two are the capacitive and the inductive crossing of the same curve, and the inductive one is the **middle** root, because x falls as frequency rises. In the trigonometric form of Cardano it is always the $k = 1$ branch. No positive root means the no-solution case of Section 4.10, not a numerical failure. A grid search would also need an upper bound on f_n , and a bound taken from the ac maximum instead of the morphing corner misses the real operating point.

4.5 Reading the gain chart of a single-stage converter

The chart in Figure 5 is the standard picture: a family of curves, one required-gain line, one crossing. A single-stage converter is drawn the same way and **read differently**. Figure 25 puts the two side by side.

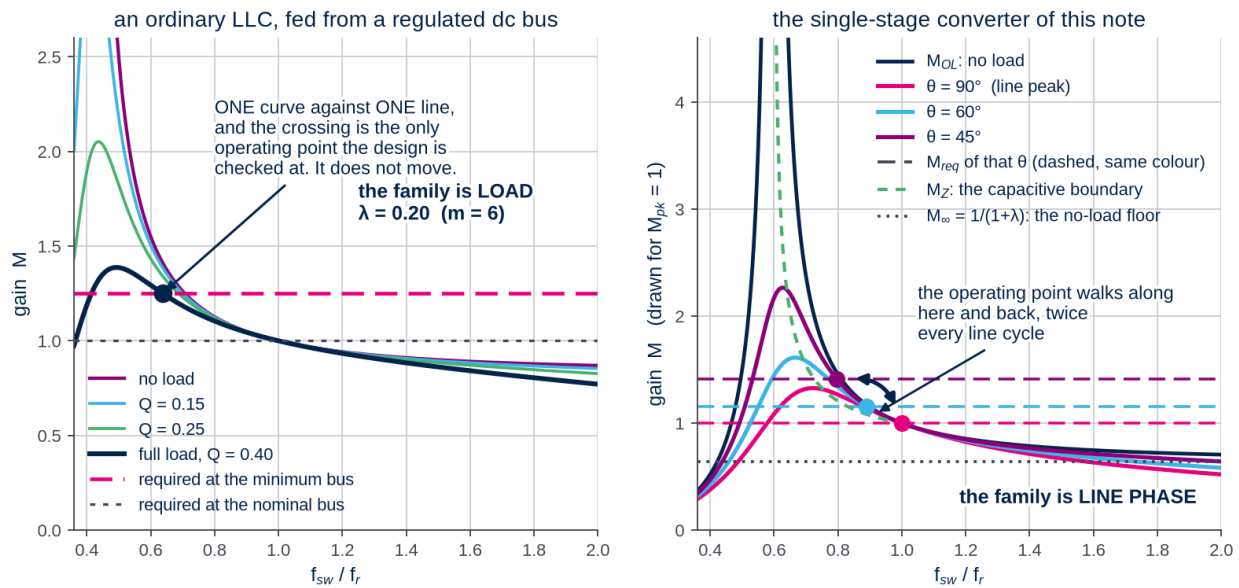


Figure 25: The ordinary chart and the single-stage one: same axes, same equation, read differently. Left, the family is load and the converter sits at one crossing. Right, the family is line phase, each curve has its own required-gain line in its own colour, and the converter travels the family twice per line cycle. The right panel is drawn for the input at which the line peak needs $M = 1$ ($M_{pk} = 1$); a lower input raises every dashed line by the same factor.

Six differences:

- **The family parameter is line phase, not load.** The load is fixed at full and the curves are $\theta = 90^\circ, 60^\circ, 45^\circ, \dots$, because $Q = Q_{pk} \sin^2 \theta$ falls as the mains falls.
- **The operating point moves, 100 or 120 times a second.** There is no single point to check; the design is checked along a path.
- **There is a ladder of required gains,** $M_{req}(\theta) = M_{pk}/\sin \theta$, one line per phase, climbing without limit towards the zero crossing. **Each curve is compared with its own line only.**
- **The curves get taller as the lines get higher.** Halve $\sin \theta$ and the requirement doubles, but Q falls by four and the curve grows far more than twice as tall. The margin grows towards the zero crossing.
- **M_Z , the capacitive boundary, is drawn.** It is where $\arg Z_{in} = 0$ (Section 2.9), a little to the right of each curve's peak, and every crossing must lie to its **right**. It has to be cleared at every phase, so it is a curve of its own.
- **$M_\infty = 1/(1+\lambda)$ is on the chart.** An ordinary LLC never needs a gain that small; a single-stage converter at high input voltage does. Below this line there is **no solution at no load, at any frequency**, and burst mode takes over (Section 4.10).

Single-Stage PFC LLC Converter Design

The mistake this chart invites: checking the tallest curve against the lowest line. There is no such operating point. θ sets the curve *and* the line together, so only the same-coloured pairs mean anything.

At the lowest equivalent input the chart answers “can the tank make the gain and stay inductive?” At the highest it answers “how high does the frequency go, and is there a solution at light load?” Section 7.6 draws it at all six input voltages for that reason.

4.6 The feedback pin is a power command

Combining the burst-mode expression with the multiplier in the block diagram gives the meaning of the feedback voltage V_{FB} :

Equation 15

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_M K_{FF}} R_{CS} P_{in, LLC} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in, LLC}$$

$P_{in, LLC}$ is the power drawn by the bridge: R_{CS} sits in the bridge return, so that is the power the controller senses. V_{os} is the 0.5 V the pin sits at with no power commanded; K_{HV} , K_M and K_{FF} are the input-voltage sense, multiplier and feed-forward gains of the block diagram. The 2.8 V feedback span gives $2.8/0.167 = 16.8 \text{ } \Omega \cdot \text{W}$, the maximum-power rule of the datasheet. **Maximum power, burst entry and overload detection all sit on one scale, R_{CS} .** The optocoupler loop commands an input power; the internal loop distributes it over the line cycle as $\sin^2\theta$.

This is why the current-sense pin tolerates no filter and no series resistor: the power law and the over-current threshold read the same pin.

4.7 Which side of resonance this converter runs on

A single-stage converter does not pick one side. The equivalent input follows a half sine and the required gain follows it, so the operating point can cross f_r and come back within one line cycle. Whether it crosses, and for how long, depends on the mains voltage; at a low input it can stay below f_r all the way. Section 7.7 shows the profile at the six input voltages.

Both sides must be designed for. The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top frequency and takes the secondary out of ZCS, so the rectifier body diode and the SR dead time matter here.

4.8 Why λ must be about 0.5

A classic LLC uses $m = (L_r + L_m)/L_r$ of 5 to 10, that is λ of 0.11 to 0.25. A single-stage converter needs very high gain near the zero crossing, and peak gain falls as λ falls. ST's evaluation board runs $\lambda = 0.500$ and this design comes out close to it (Section 7.4): the value comes from the topology.

The price is circulating current: a small L_m means a large magnetising current with conduction loss at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, inherently.**

Single-Stage PFC LLC Converter Design

4.9 $f_{sw,max}$ does not check the tank — it computes it

Two of the four candidates for λ carry the specified maximum switching frequency in a denominator:

Equation 16

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2}$$

λ_1 is the plain minimum-gain condition; λ_2 and λ_{TD} fold in the frequency ceiling: the tank must reach the lowest required gain *before* it runs out of frequency. With the usual $f_{sw,max} = 1.5 f_r$ the squared ratio is 0.44 and **the value typed as $f_{sw,max}$ roughly doubles the λ asked for**. Set it near f_r and the required λ goes to infinity. The fourth candidate, λ_3 , comes from the specified minimum switching frequency; it is small in this converter and does not bind.

The common $1.5 \times f_r$ is a **choice**, not a limit, and it sits upstream of the tank: a round number typed into it changes L_m .

f_{Max} , the oscillator ceiling, is different. It *is* a limit, fixed by R_T , C_T , $I_{EA,max}$ and the idle time, and it is checked after the fact by k_{ceil} (Section 7.3). One decides the tank; the other reports whether the oscillator can serve it.

4.10 The other bound on λ , and where it has no solution

The ceiling on λ comes from the lowest required gain, at the highest equivalent input and no load. The no-load gain does not fall without limit; raising the frequency only brings it down to an asymptote:

Equation 17

$$M_\infty = \lim_{f \rightarrow \infty} M_{OL} = \frac{1}{1 + \lambda}$$

with M_{OL} the gain curve at no load ($Q = 0$, output open).

If the required minimum gain is below M_∞ , no frequency satisfies it. A spreadsheet solving for that frequency returns an error, and the error looks like a broken formula. It is not: the question has no answer. Whether a tank sits above or below the asymptote is a matter of a per cent, and no full-load check shows it; Section 7.4 reports it for this design.

Losing the solution is not a failure: no load at high input voltage belongs to **burst mode**. But it decides which candidate turns ratios can still regulate by frequency alone at no load.

5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V_{pk} it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

Single-Stage PFC LLC Converter Design

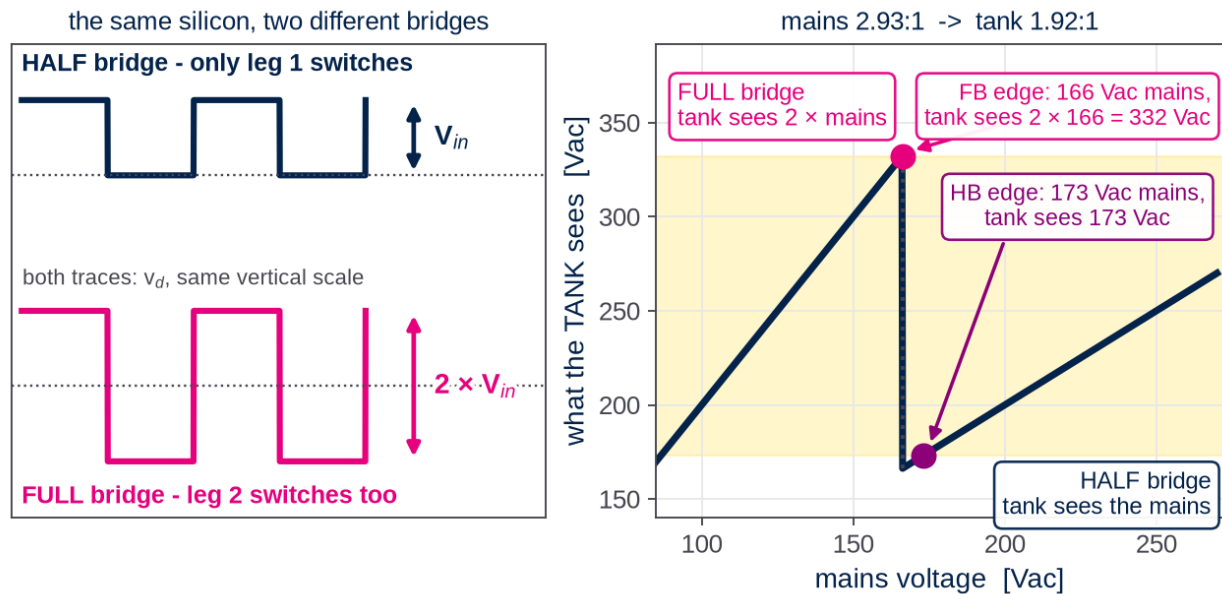


Figure 26: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range. Drawn for a falling mains; Figure 30 shows the band in which the mode depends on the direction.

5.1 Full bridge

Diagonal pairs conduct together, so the drive is a square wave of amplitude V_{in} with fundamental $(4/\pi)V_{in}$. All four devices switch.

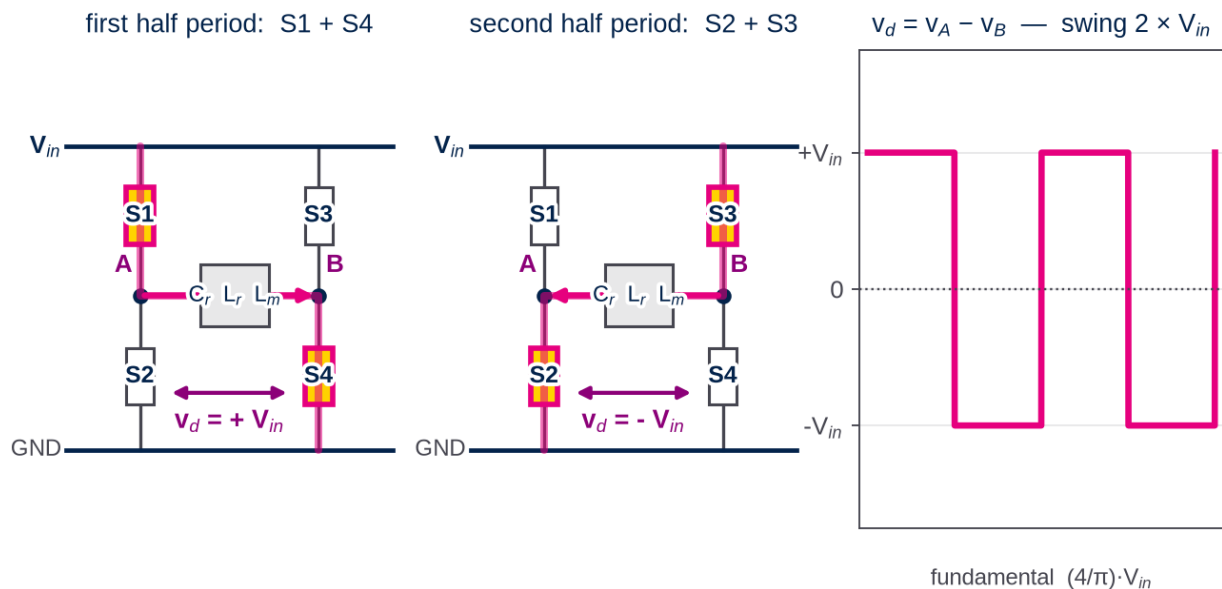


Figure 27: Full bridge: conduction path in each half period, and the resulting tank drive.

5.2 Half bridge

LOUT2 is held high: S3 never turns on and S4 never turns off, so the bridge output swings between 0 and V_{in} . C_r blocks the $V_{in}/2$ of dc, and the tank sees $\pm V_{in}/2$, a fundamental of $(2/\pi)V_{in}$. No extra hardware.

Single-Stage PFC LLC Converter Design

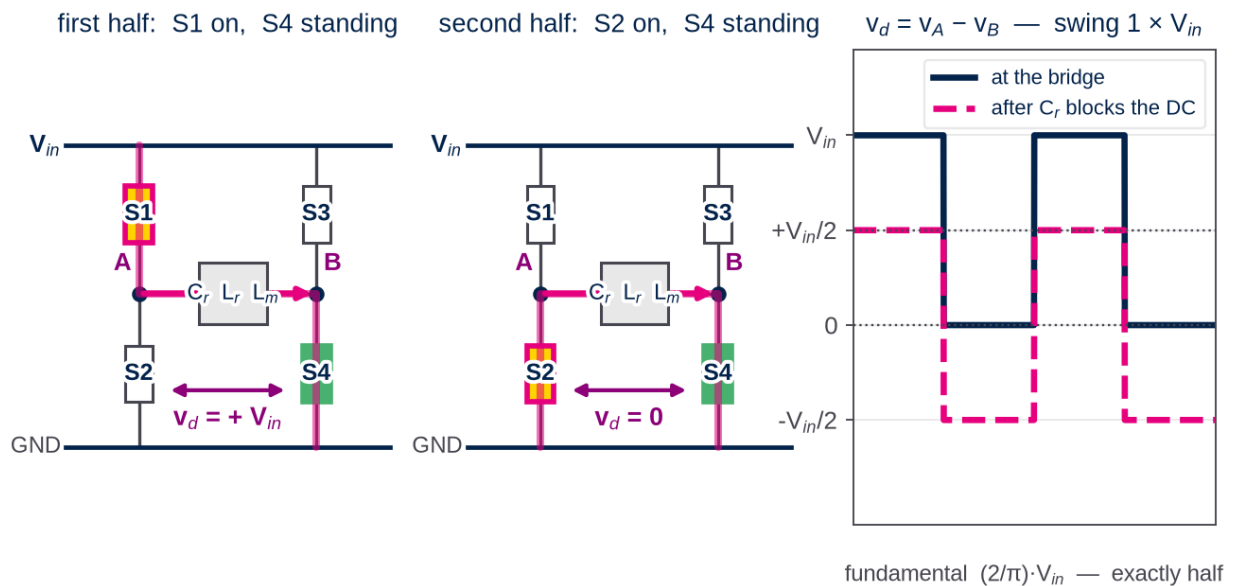


Figure 28: Half bridge: leg 2 stops switching, and C_r removes the dc that the asymmetric drive creates.

The standing device is the hottest one. S4 never switches but conducts the full tank current continuously, and in this design it is the hottest single device (Section 7.15).

5.3 How the controller does it

Only leg 2 changes between the modes: HOUT2 stops and stays low, LOUT2 stops and stays high. The thresholds are **fixed inside the IC**; R_{CFG} only enables morphing and sets the brown-out threshold.

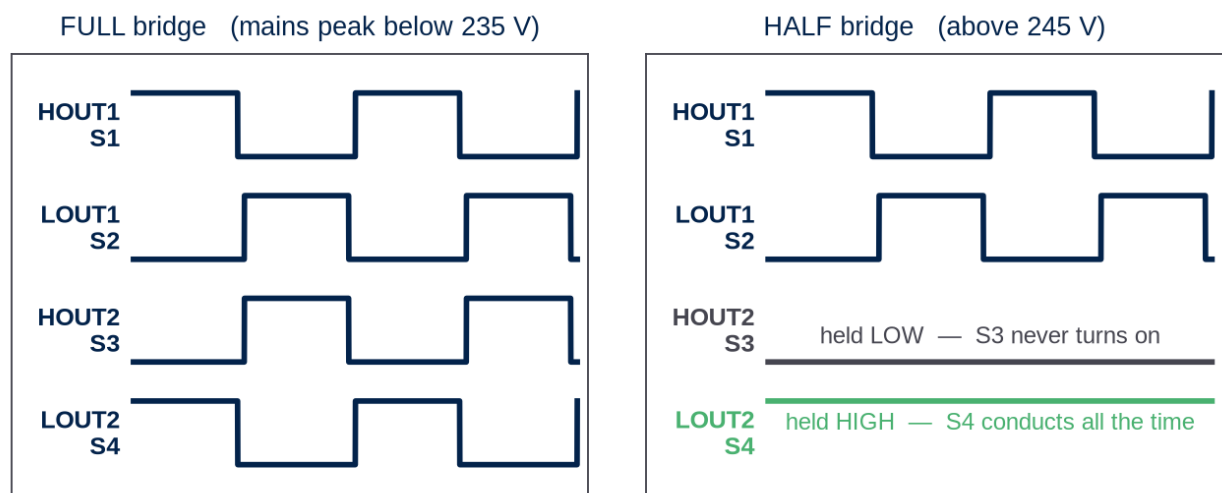


Figure 29: The four bridge drive signals in each mode. Holding LOUT2 high and HOUT2 low is the entire mechanism.

5.4 The hysteresis band, and the range it really implies

The part drops to half bridge at $245 V_{pk}$ going up and returns to full bridge at $235 V_{pk}$ coming down. The hysteresis stops toggling, but **between the two the mode depends on the direction of travel**, not on the voltage.

Single-Stage PFC LLC Converter Design

Taking each threshold where its mode is guaranteed gives 173.2 to 332.3 Vac (1.92:1). Including the band gives **166.2 to 346.5 Vac (2.09:1)**. This design passes at the wider range too; do not use the narrower figure when tightening a specification.

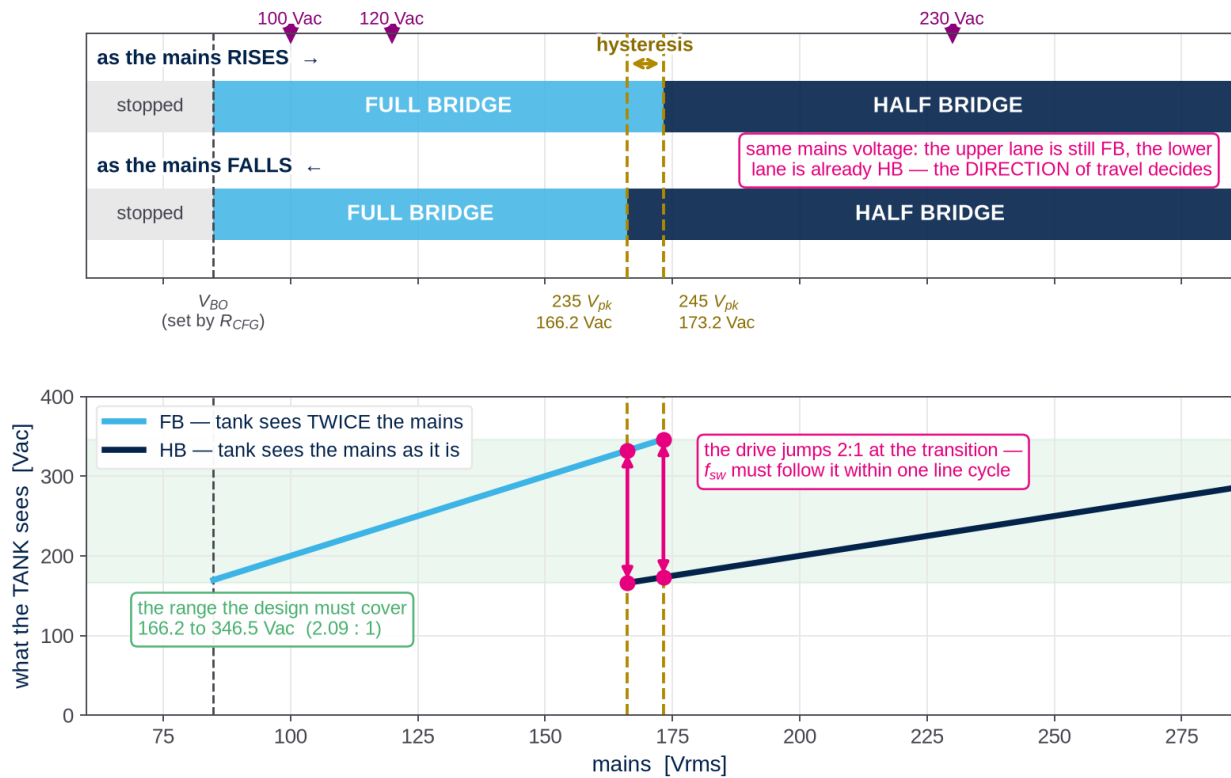


Figure 30: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from. V_{BO} is the brown-out threshold, set by R_{CFG} (Section 7.26).

No nominal mains voltage lies in 166 to 173 Vrms; the band is met in sags, on a programmable ac source and in dip and surge tests. Test it with a **step** across the threshold, not a ramp.

Transition transient. Crossing a threshold changes the drive 2:1 within one line cycle, against a loop that crosses in the tens of hertz, so the output dips (going up) or overshoots (coming down) until the loop recovers. The dip is less severe than hold-up, which is already met, so it is no reason to enlarge C_{out} ; neither has been checked in the time domain.

6 Design procedure

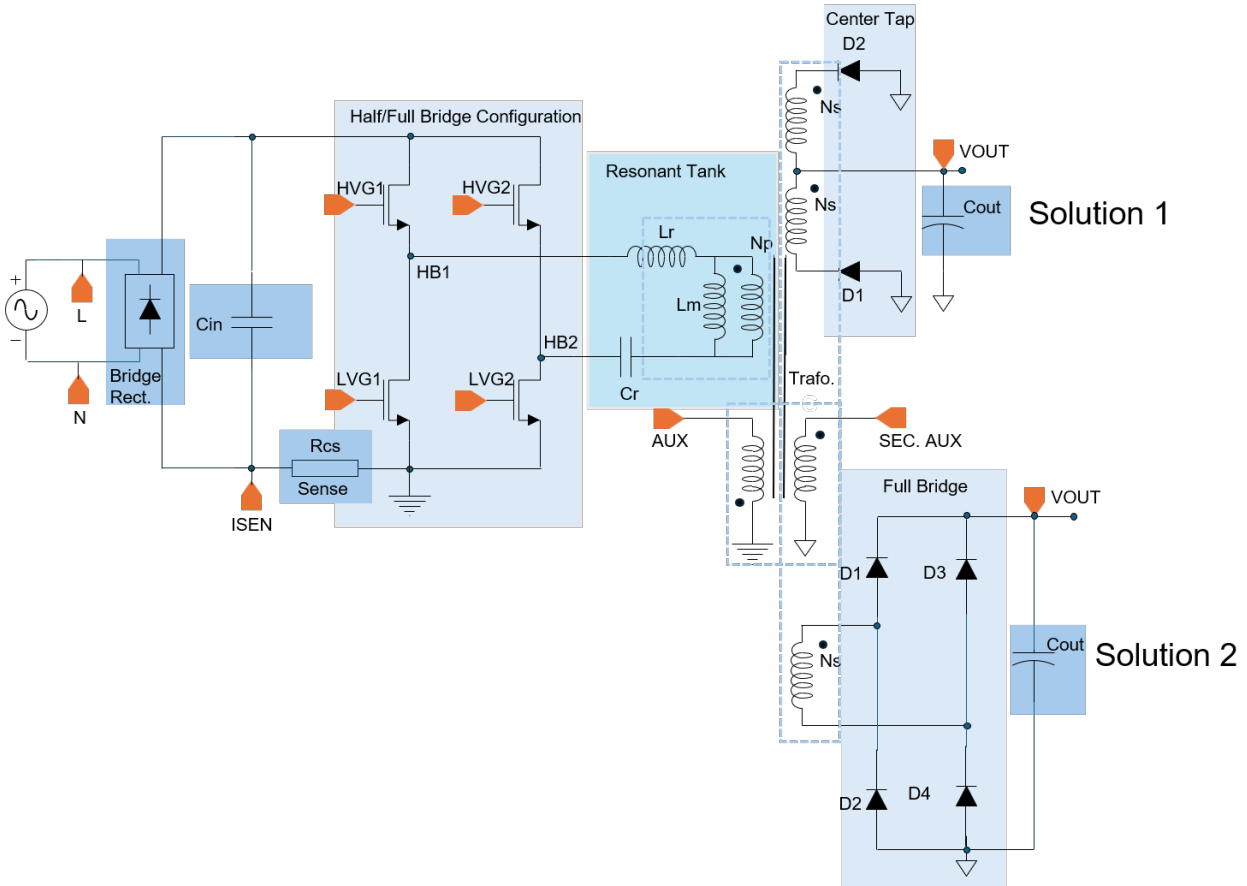


Figure 31: The power stage. The bridge rectifier feeds the leg pair directly: no bulk capacitor, no boost stage. The secondary is drawn as centre tap (Solution 1) and full bridge (Solution 2). Drawing from the ST L6790A design spreadsheet.

Pick the secondary first: it sets N_{rect} , and N_{rect} sets the reflected voltage. A centre tap has one device in the conduction path, a full bridge two, so at a low output voltage the centre tap halves the rectifier conduction loss at the price of twice the reverse voltage per device. This design uses the centre tap. The steps below are in dependency order.

6.1 The specification

The full specification is in Section 7.1. Two qualifiers apply to every step: **the worst line frequency is the lowest**, and **hold-up is specified at the ripple trough**.

6.2 The equivalent input range

With morphing the tank does not see 90 to 264 Vac. It sees an **equivalent** input given by the bridge mode:

Equation 18

$$V_{ac,eq} = 2 V_{ac} \quad (\text{full bridge}) \quad V_{ac,eq} = V_{ac} \quad (\text{half bridge})$$

At the half-bridge edge ($245 V_{pk}$) the tank sees 173.2 Vac; at the full-bridge edge ($235 V_{pk}$) it sees 332.3 Vac. **These edges are the design corners:** the low one for gain and ZVS, the high one for switching frequency. The

Single-Stage PFC LLC Converter Design

mains voltages a supply actually meets map onto the same axis: 90 and 110 Vac in full bridge become 180 and 220 Vac equivalent, 230 and 264 Vac in half bridge stay 230 and 264. All four lie inside the two edges, and this note reports at all six conditions.

6.3 Turns ratio and reflected voltage

The turns ratio appears only multiplied by the effective output voltage, so what matters is the **reflected voltage** V_{refl} :

Equation 19

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f)$$

V_f is the forward drop of one rectifier and N_{rect} the number in the conduction path. n is the model ratio in the gain equation; n_T is the wound ratio with the leakage folded into L_r . At $\lambda \approx 0.5$ they differ by more than 20 % (Section 6.9), so the drawing must carry open- and short-circuit inductance as well as turns.

6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary, evaluated at the line peak, is

Equation 20

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o, eff}^2}{P_{in, LLC}}$$

with $P_{in, LLC}$ the power into the resonant stage. The usual LLC form writes $8/\pi^2$ with the one output power a two-stage converter has. Here the drawn power is $2P \sin^2 \theta$, so R_{ac} is pinned to the line peak, where $p = 2P$: the 8 becomes a 4, and Q means the quality factor at the peak, with $Q(\theta) = Q_{pk} \sin^2 \theta$ elsewhere. Using the two-stage value understates the loading by two.

The gain requirement at the low corner and the ZVS requirement cap the quality factor at Q_{ZVS} . $R_{ac} Q_{ZVS}$ is the impedance the tank is sized from: it fixes C_r through Equations 1 and 2, and C_r with f_r fixes L_r . Three rules matter more than the arithmetic:

- The three calculated values are not one tank. L_r follows from the **selected** C_r , not the calculated one. The calculated L_m is L_r divided by the λ that *no load at the high corner* asks for. A design may knowingly miss that condition (Section 4.10), so do not round towards that L_m .
- **A larger L_m costs ZVS margin.** It lowers λ and the magnetising current that swings the node. C_r above and L_m below their calculated values do the opposite.
- **Stay below the Q_{ZVS} cap.** That gap is most of the ZVS margin. Choose L_m with n so that $n_T = n\sqrt{(1+\lambda_{act})}$ is a ratio that can be wound.

6.5 ZVS verification

The closed-form ZVS estimate in most guides is a fitted approximation at the design Q_{ZVS} , not at the Q the converter runs at. Against a full sweep it is off by tens of per cent, and nothing in it guarantees which way; optimistic is the dangerous direction.

Single-Stage PFC LLC Converter Design

The shortcut is written with a bare λ and means the *design* λ . Read it with λ_{act} and the phase can come out **negative** for a tank well inside the inductive region. The sweep has no such ambiguity.

Sweep θ , input voltage and load with the **selected** tank and take the minimum. Section 7.5 does it for this design; the worst point is not where one would expect.

6.6 Which side of resonance the converter runs on

Below f_r the secondary current is a full resonant half sine followed by the freewheeling interval, and the conduction ratio $d = f_{sw}/f_r$ is less than one; above f_r the half sine is cut short and d clamps at one. The rms figures behind ratings and losses contain d .

The turns ratio decides the side. $M_{req} = 2nV_{o,eff}/(\sqrt{2} V_{ac,eq})$, so a larger n asks for more gain and sits further below resonance. Two candidate transformers on the same tank can end up on opposite sides of f_r at the same mains voltage.

Settle this before the transformer is ordered: below resonance the rectifiers turn off at zero current, above it they do not.

6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses from the line-cycle rms. Two further rules:

- The primary devices and the sense resistor see the **composite** tank current: reflected load current plus magnetising current. Its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for d of each half period. The rms scales with $\sqrt{d}$; omitting d over-estimates it, and the loss by its square. Section 7.8 draws the composite current of this design.

6.8 The transformer

The tank fixed L_r , L_m and a ratio; the transformer has turns, an open-circuit inductance and a leakage. **Two turns ratios and two inductances carry almost the same names**, and that is where most errors are made. This section turns Section 2.12 into numbers a supplier can measure.

Every number on the drawing is read off the primary current, the current in each secondary winding, the secondary winding voltage, or one bench test. Figure 43 marks where on this design's waveforms; Table 4 says what each mark sets.

Mark	Item	Read from — and over what	What it sets
1	Primary peak current	Top of i_p : the half sine of load current riding on the magnetising ramp. Worst switching cycle: line peak, minimum equivalent input, full load.	Primary switch and OCP1 headroom. Not the saturation test.
2	Primary rms current	i_p^2 averaged over the whole switching period, then over the line half cycle. With N_x units each carries all of it: the primaries are in series.	Primary copper: $A_{cu} = I_{rms}/J$ per unit
3	Magnetising peak	The dashed i_{Lm} at the switching instant. Below resonance it depends on neither f_{sw} nor load, only on $V_{o,eff}$, so it is the same on every cycle.	The peak flux; and, scaled to V_{OVP2} , the DC-overlap test current (mark 7)

Single-Stage PFC LLC Converter Design

Mark	Item	Read from — and over what	What it sets
4	Secondary peak current	Top of the pulse in either secondary winding, same worst cycle. With N_x units each carries $1/N_x$ of the rectifier-leg value: the secondaries are in parallel.	Rectifier peak rating; the secondary terminations
5	Secondary rms current, each winding	One pulse per period in each winding, so the rms is taken over the whole period with the idle half counted, then over the line half cycle.	Secondary copper: $A_{cu} = I_{rms}/J$, per winding and per unit
6	Winding volt-seconds	The secondary winding voltage is $V_{o,eff}$ for the resonant half period $T_r/2$ while the rectifier conducts. That area is the flux swing $2N_s A_e B_{pk}$; f_r is the worst case because the interval never gets longer than $T_r/2$.	Core area and N_s : $A_e \geq V_{o,eff}/(4 f_r N_s B_{max})$
7	DC-overlap (saturation) test	Not a waveform. LCR meter across the primary NP1, every other winding open (secondaries NS2 and NS3, auxiliary NAUX); a dc current overlapped on the primary and the primary inductance read against it. The test current is mark 3 scaled to the highest output the controller allows, V_{OVP2}/V_{out} .	Core saturation margin; the number the supplier tests to
—	L_{open} , L_{short}	LCR meter across NP1 at 100 kHz, 1 V. L_{open} : every other winding open. L_{short} : one secondary half shorted, the other half and NAUX open, then the other half: the tank sees the half that conducts. With both halves shorted it reads lower when the two halves do not lie in the same place. Terminal quantities, so a supplier can test them.	$L_m + L_r$ and L_r : the tank itself

Table 4: What each mark in the figure sets. Currents are per unit: for one transformer that is the whole part; split into N_x units (Section 6.21), the primaries are in series and the secondaries in parallel.

6.9 Two ratios, two inductances

With all leakage referred to the primary the tank sees an ideal transformer of ratio n , a series L_r and a shunt L_m . That n is **not** the wound ratio. With k the coupling coefficient,

Equation 21

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$

so $n_T = n\sqrt{1+\lambda}$: more than 20 % apart at $\lambda \approx 0.5$, far more than any gain margin.

The inductances split the same way. The tank model uses L_r and L_m ; the physical transformer has a magnetising inductance L_μ with leakage on both sides:

Equation 22

$$L_\mu = \sqrt{L_m(L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2}$$

with L_{L1} the primary leakage and L_{L2} the secondary leakage, each on its own side of the ratio.

Single-Stage PFC LLC Converter Design

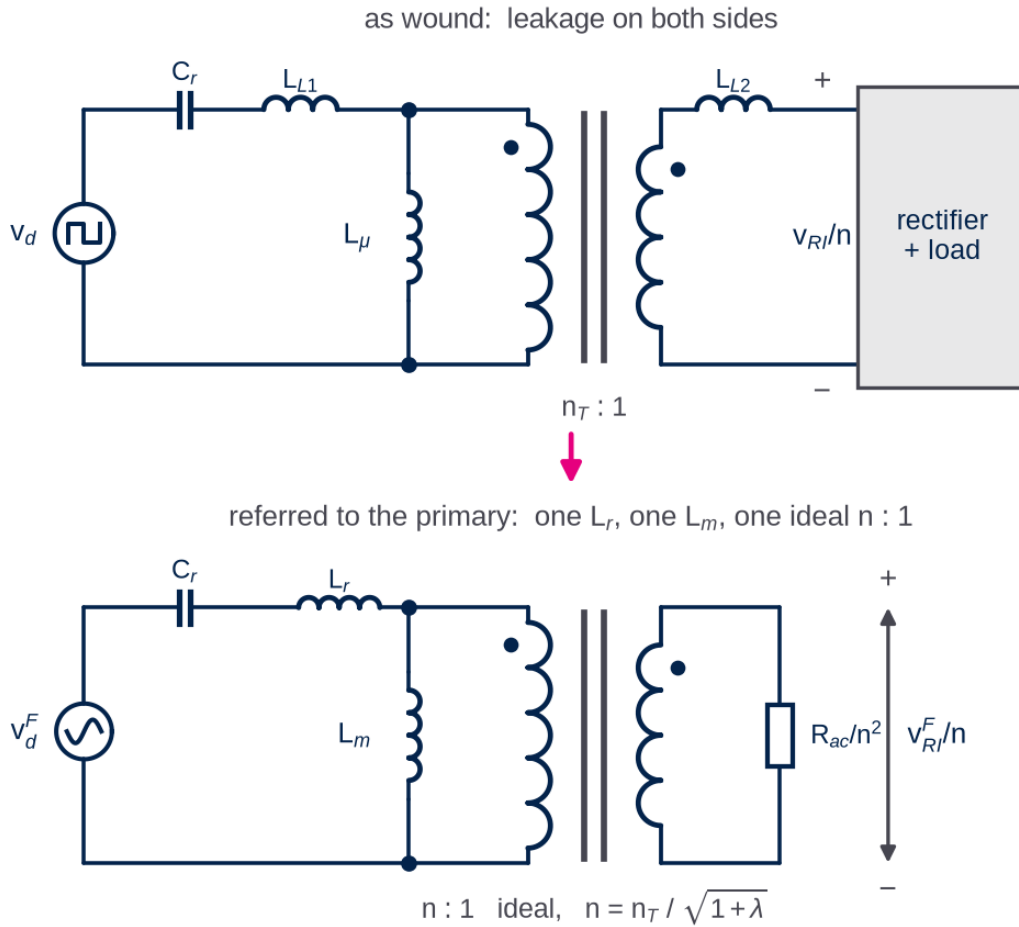


Figure 32: The same transformer drawn both ways. Above, as wound: leakage on both sides, the physical L_μ in shunt, ratio $n_T : 1$, driven by v_d . Below, referred to the primary and driven by the fundamental of v_d : one L_r , one L_m , ratio $n : 1$. Neither the shunt elements nor the ratios are the same numbers.

The as-wound form (L_{L1} , L_μ , L_{L2}) assumes the leakage splits evenly. The real split does not matter: the tank sees only the two quantities a meter can read.

6.10 What the transformer specification must say

Only two inductances are measurable at the terminals, and they are the two the tank needs:

Equation 23

$$L_{open} = L_\mu + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_\mu n_T^2 L_{L2}}{L_\mu + n_T^2 L_{L2}} = L_r$$

Both are read across the primary. L_{open} : every other winding open. L_{short} : the secondary shorted, the auxiliary open. On a centre-tapped part short **one half at a time**: in service only one half conducts, so each half is what the tank sees. Both halves shorted read lower when the two halves do not lie in the same place; record that value, do not hold it to L_r . **Specify those two and the turns, never L_μ** : no terminal pair exposes it. State the turns as

Single-Stage PFC LLC Converter Design

numbers, because with one or two secondary turns the lead-out loop dominates that winding's inductance, and the root of the open-circuit inductances read at the primary and at one secondary, $\sqrt{L_1/L_2}$, does not give the ratio.

6.11 Peak flux is set by the secondary, not the primary

Mark **6** of Figure 43: the secondary winding voltage is a rectangle of height $V_{o,eff}$ and width $T_r/2$, and the flux swings $\pm B_{pk}$ across it. Above resonance the secondary conducts for a shorter switching half period, so f_r is the worst case everywhere on the line cycle:

Equation 24

$$B_{pk} = \frac{V_{o,eff}}{4f_r N_s A_e} \implies A_e \geq \frac{V_{o,eff}}{4f_r N_s B_{max}}$$

N_p does not appear. Only secondary turns reduce the flux. For a low-voltage, high-current output, where the secondary wants to be one turn, halving N_s doubles the core area.

From an inductance the same flux is $B_{pk} = L_{\mu, \mu, pk} i_{\mu, pk} / (N_p A_e)$ with the **physical** L_{μ} . Using the tank L_m understates it by $\sqrt{(1+\lambda)}$.

6.12 Choosing the core: two areas, and the one that usually decides

A core must satisfy two independent conditions: enough cross-section A_e for the flux at the chosen turns, and enough window A_N for the copper. Their product is the **area product**, and either can bind.

Check the flux against A_{min} , not A_e . The narrowest section saturates first: the real peak is $B_{pk} A_e / A_{min}$.

In a single-stage PFC LLC **a third condition normally decides**: the leakage must come out at L_r . As a fraction of what a meter reads,

Equation 25

$$\frac{L_{short}}{L_{open}} = \frac{L_r}{L_m + L_r} = \frac{\lambda}{1 + \lambda}$$

At $\lambda \approx 0.5$ that is **about a third**. A conventional transformer leaks a few per cent; this one must leak ten times more, on purpose. Leakage that large is a winding **geometry**, and geometry costs window. A core that passes on A_e and copper can still fail here.

6.13 The winding arrangement is the leakage

Leakage is set by how the windings are arranged and how far apart they sit:

- **Interleaved** (primary, secondary, primary): one or two per cent. What a forward or flyback transformer wants; the opposite of what is needed here.
- **Concentric, not interleaved**: a few per cent. Still short.
- **Side by side**, primary on one half of the width and secondary on the other: tens of per cent, adjustable by the gap between them. **This is what a single-stage tank asks for**, and why the window must be generous.

Single-Stage PFC LLC Converter Design

- **Split**, the primary in two parts, one on either side of the secondary: each gap carries only its own part's ampere-turns, so the leakage is a fraction of the side-by-side value and the gaps still trim it. The ratio of the parts sets where the target falls in the range the gaps can reach.
- **A magnetic shunt** in the window: leakage without winding width, at the cost of a part and a harder tolerance.

The leakage of a sectioned winding can be estimated before anything is wound. Along the bobbin the ampere-turns rise across part A, stay flat across the first gap, fall through the secondary and come back to zero across part B; the field crosses the window depth h_w , and its energy gives

Equation 26

$$L_{short} \approx \frac{\mu_0 l_N}{h_w} \left[N_A^2 \left(\frac{w_A}{3} + g_1 \right) + (N_A^2 - N_A N_B + N_B^2) \frac{w_S}{3} + N_B^2 \left(g_2 + \frac{w_B}{3} \right) \right]$$

with $N_A + N_B = N_p$ the turns of the two primary parts, w_A, w_S, w_B the axial widths of part A, the secondary and part B, g_1 and g_2 the gaps and l_N the mean turn. $N_B = 0$ is the side-by-side winding. It is an upper estimate: it takes the whole turn as inside the core, and the part outside, with no outer leg to return the field, leaks less. A field solution of the actual layout narrows it; the first samples decide.

The usual transformer advice is to interleave. Here L_r is a design value, and a supplier who “improves” the coupling breaks the converter. Say so on the drawing, next to L_{short} .

6.14 The gap, and why the drawing must not name it

L_m is far below what an ungapped core gives, so the core is gapped:

Equation 27

$$A_L = \frac{L_{open}/N_x}{N_p^2} \quad g \approx \frac{\mu_0 A_e}{A_L}$$

with N_x the number of units in the assembly (1 for a single transformer; Section 6.21), g the total centre-leg gap and μ_0 the permeability of free space. The gap expression ignores fringing, which makes the gap actually needed larger. **Specify A_L , or better L_{open} , and leave the gap to the supplier**: that is what they grind to and what a meter can check.

6.15 The window: wire, current density and what actually fits

Marks **2** and **5** of Figure 43. Copper is sized on the **line-cycle** rms: the rms of each switching period, averaged in I^2 over the line half cycle. The secondary value is per winding and per unit, with the idle half inside the average.

Each winding needs a copper area set by its rms current and the current density the design can cool:

Equation 28

$$A_{cu} = \frac{I_{rms}}{J} \quad \sum_w N_w A_{cu,w} \leq k_u A_N$$

with N_w the turns of winding w and $A_{cu,w}$ its copper area. J is an assumption: 4 to 5 A/mm² in free air, less enclosed. k_u , the window utilisation, is **0.3 or less** with Litz wire once insulation, bobbin wall, margins and tape are

Single-Stage PFC LLC Converter Design

counted. Sizing on bare copper overstates what fits by three times.

Split into N_x units (Section 6.21), every unit carries the whole primary current, because the primaries are in series; only the secondaries share. That is why splitting helps a low-voltage, high-current output.

6.16 Skin depth, and why the wire is not a wire

At the switching frequency current crowds into a surface layer of depth

Equation 29

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

with ρ the resistivity of copper at the winding temperature and f the frequency it carries, here f_r . Copper deeper than δ below the surface carries little current, so a conductor thicker than 2δ gains little. The field from the other turns also drives circulating current in each conductor (the proximity effect), and in a side-by-side winding that term can dominate.

So a winding is not one wire. There are two ways out:

- **Litz wire:** a bundle of many fine wires, twisted so that each takes a turn at every position, and each insulated by its own enamel. One fine wire is a **strand**; d_s is its diameter and a_s its copper cross-section. The rule is $d_s \leq 2\delta$ **per strand**.
- **Copper foil** on a low-voltage, high-current secondary: thickness t_f near δ , width does the carrying. It terminates well into a centre tap, but **only in a concentric winding**, where the leakage field runs along the foil. Between the sections of a side-by-side or split winding the field crosses the window, straight through the face of the foil, and drives eddy currents across its whole width. There the secondary is Litz too.

Either way the ac resistance heats the winding; the dc resistance is a floor, not an answer.

Litz. Pick a standard strand at or below 2δ ; the strand count is the copper needed over what one strand gives:

Equation 30

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$

with A_{cu} from Equation 28. The bundle is wider than its copper: gaps between round strands, enamel and serving are collected into a fill factor k_{litz} of about 0.5 to 0.6, and the outside diameter is

Equation 31

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$

Foil. Choose the thickness from the skin depth and what the supplier stocks; the width follows. If it exceeds the bobbin width, use several narrower strips in parallel, stacked radially:

Single-Stage PFC LLC Converter Design

Equation 32

$$w_f = \frac{A_{cu}}{t_f}, \quad n_f = \left\lceil \frac{w_f}{w_{f,max}} \right\rceil$$

6.17 Safety insulation: what the winding has to carry

The transformer is the isolation barrier, and a product sold in several markets needs one construction that passes all of them. Take every requirement at its strictest market: the insulation grade of a Class II set, **reinforced** between everything mains-referenced and the secondary (it covers Class I too); the highest altitude any market assumes, because clearances grow with it; and the working voltage at the top of the input range.

- **Clearance** follows the mains transient of overvoltage category II. Reinforced insulation is taken one withstand step higher, and above 2000 m the result is multiplied by the altitude factor.
- **Creepage** follows the rms working voltage, the pollution degree and the material group of the surface; reinforced is twice basic. Without a tracking index for the bobbin, take the lowest group.
- **Solid insulation** — tape, bobbin wall, sleeving — needs either a minimum thickness or several layers that each pass the reinforced electric-strength test.
- **Between the sections, either the distance or the wire carries the reinforced insulation.** By distance, the gap between a primary and a secondary section is the creepage along the tube and the clearance across it, so it can be no narrower than the reinforced creepage. It is also the leakage, so the insulation then puts a floor under L_{short} : check that floor with Equation 26 before committing to it. By the wire, triple-insulated wire (made as Litz too) carries the reinforced insulation itself, and the gaps are free to be whatever the leakage asks.
- **Treat the core as part of the side that sits closest to it.** With plain wire the primary fills the window almost to the outer legs, so the core is primary, and the secondary needs reinforced distances to it: through the tube wall, over the flange, to the outer legs and at the leads. With a triple-insulated primary the core is on the secondary's side, and what is left to check is where the insulation ends: the stripped wire ends at the primary pins, against the core and the secondary leads.
- **An auxiliary winding that feeds V_{CC} and senses the output belongs over the secondary**, where it follows the output. It is mains-referenced, so there it must carry reinforced insulation itself; triple-insulated wire does that without tape.

Where the windings are stacked instead, margin tape at both flanges buys the creepage and costs winding width. Triple-insulated wire needs no margin but is thicker for the same copper and costs more.

6.18 Loss, and the temperature the drawing has to survive

- **Core loss** comes off the material curve at the operating flux and frequency, not off the headline figure. The flux is fixed below resonance, so the worst loss is near f_r , the top of the sweep below resonance; above it the flux falls as $1/f_{sw}$.
- **Copper loss** is $I^2 R_{ac}$ for both windings, and in a side-by-side arrangement the proximity term is not small.
- **Temperature rise** is what limits the design, and it depends on surface area and airflow. The computed loss is an input to a thermal measurement, not an answer.

Single-Stage PFC LLC Converter Design

6.19 The saturation test is not the peak winding current

What the line on the specification says. A transformer specification usually carries a line of the form “DC overlap: inductance more than 90 % of initial, test current I_{sat} , normal temperature”. The name is the measurement. An LCR meter reads the primary inductance with a small ac signal (100 kHz, 1 V is a common setting, the same one the leakage test uses) while a bias source overlaps a dc current on the same winding; every other winding is open. The dc current is stepped up and the inductance is read at each step. The same test is called dc bias or dc superposition.

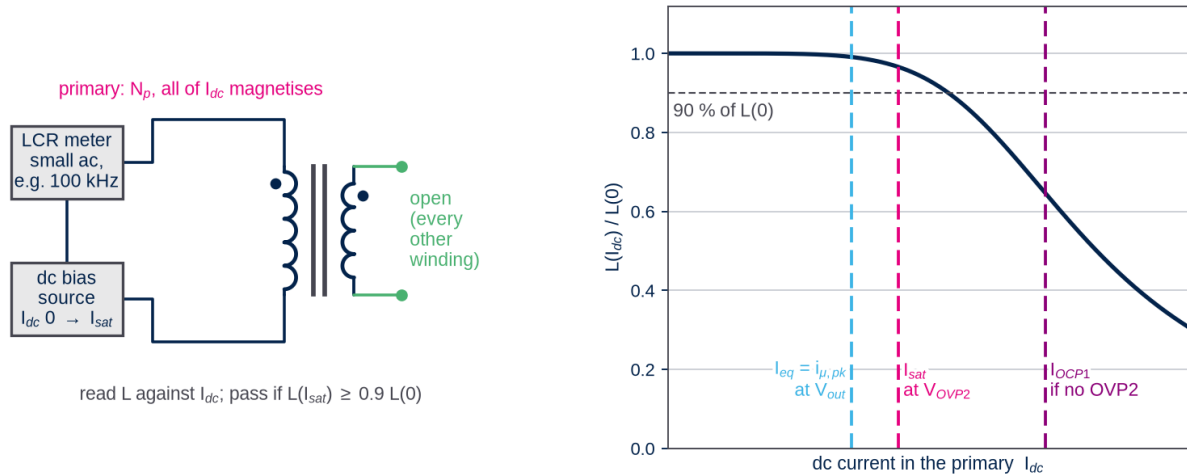


Figure 33: The DC-overlap test. Left: the bench. Right: inductance against the dc current, an illustrative shape, with this design's three currents marked: the operating magnetising peak, the test current the specification asks for, and the current that would have to be used if the controller had no voltage ceiling.

What it reveals. With the other windings open the dc current alone sets the flux, so the inductance stays flat while the core is linear and falls as the core approaches saturation; the 90 % point marks the knee. It is the only test at the terminals that reads the flux margin. Turns and A_L do not: the same A_L comes from a different gap, a different A_{min} or a different material, and L_{open} confirms the inductance, not the margin. A part with the right L_{open} and the wrong core is caught here and nowhere else.

Which current to write on the line. Mark 3 and the bench panel of Figure 43. The current to ask for is the peak of the magnetising trace, scaled to the over-voltage ceiling, not the peak winding current at mark 1.

In the test nothing cancels the primary ampere-turns, so the same current makes far more flux than in operation. Ask for the current that reproduces the operating flux in that test:

Equation 33

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_{\mu}}$$

The denominator is L_{μ} , not L_{open} : the primary leakage links no core. Dividing by L_{open} understates the current by a factor $\sqrt{(1+\lambda)}$. A free check: the answer **must** equal $i_{\mu, pk}$.

Testing at the peak tank current would ask for a flux the converter never produces. No arbitrary margin is wanted on I_{eq} either: the flux ceiling is already defined by the highest output the controller allows before it shuts down.

Single-Stage PFC LLC Converter Design

Why the ceiling is OVP2. Write the same peak flux twice: from the volt-seconds the clamped secondary applies (Equation 24), and from the current that magnetises the core (Equation 33). A_e cancels:

Equation 34

$$B_{pk} = \frac{V_{o,eff}}{4f_r N_s A_e} = \frac{L_\mu i_{\mu,pk}}{N_p A_e} \implies i_{\mu,pk} = \frac{n_T V_{o,eff}}{4f_r L_\mu}$$

with $n_T = N_p/N_s$. Nothing that the load or the input sets is left in it. Below resonance the secondary is clamped for $T_r/2$ whatever f_{sw} is; above resonance for less, so the flux is lower. **The magnetising peak, and the flux with it, is proportional to the output voltage and to nothing else in operation.** The worst case is therefore the highest output voltage at which the bridge is still switching.

The L6790A has two levels on the ZCD pin. At OVP1 (2.3 V on the pin) it keeps switching at reduced power, so the output can sit between OVP1 and OVP2 with the transformer still driven. At OVP2 (2.5 V) it stops switching for 100 ms and restarts with a soft start. While the core is driven, the output therefore stays at or below V_{OVP2} , and Equation 34 gives the largest magnetising peak the core ever carries.

On the bench every other winding is open, so the dc current is the only ampere-turns on the core and makes $B = L_\mu I_{dc}/(N_p A_e)$, the same relation as in operation. The test current that reproduces the flux at OVP2 is the magnetising peak scaled by the voltage ratio, and the ratio of the two voltages is the whole margin. No arbitrary factor goes on top of it:

Equation 35

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out}}$$

With $V_f = 0$ this is the exact flux ratio. With a rectifier drop the flux ratio is $(V_{OVP2} + N_{rect} V_f)/(V_{out} + N_{rect} V_f)$, a little smaller, so Equation 35 errs on the safe side.

Overload, start-up, burst mode and the switching frequency do not raise the flux (Section 2.12); an output over-voltage does, and temperature lowers the ceiling. B_s of a MnZn power ferrite falls by about a fifth between 25 and 100 °C, so a test made at normal temperature is read against the material's hot B_s , and the design flux at OVP2 has to sit under it with margin. The 90 % criterion is that margin.

If the controller has no OVP2-like ceiling. Then something else has to bound the output voltage, and the specification follows whichever bound exists.

- **A protection that stops the drive**, on either side of the barrier: a secondary crowbar or latch, an SR-controller OVP, a primary OVP that latches. Use its threshold in place of V_{OVP2} .
- **A protection that only reduces power**, like OVP1, is not a ceiling: the output can rest at it while the core is driven. Do not use it.
- **No voltage ceiling at all.** The output is then bounded only by what the tank can deliver at the oscillator floor into the lightest load, and with the floor near f_o that bound is not useful. What remains is the current limit: with nothing clamping the secondary the primary current is the magnetising current, so the core must not saturate at the largest current the controller ever lets through the primary, $I_{sat} = I_{OCP1}$. Safe, and expensive; Section 7.13 gives the ratio for this design.
- **The rule of thumb**, 1.2 to 1.5 times $i_{\mu,pk}$ or the winding peak at normal temperature, is what a specification carries when nobody asked the question. It names no event, so it is neither safe nor economical. Replace it with one of the three above.

Single-Stage PFC LLC Converter Design

6.20 The specification, and what it is not allowed to leave out

Check the open-circuit inductance tolerance against the frequency floor. f_o goes as $1/\sqrt{L_{open}}$, so a low part raises f_o towards the oscillator floor f_{Min} ; unless the floor margin covers the fall, a part that meets $\pm 10\%$ can hard switch. Section 6.26 gives the check.

Core, bobbin, wire and winding order can be left to the supplier as long as L_{open} , L_{short} and the insulation of Section 6.17 are met, because all of those are measurable at the terminals. The design example fixes them anyway: its L_r is the leakage of one particular winding.

6.21 If one transformer is not practical

At low voltage and high current the secondary is one heavy turn and the core comes out large. The transformer can then be built as N_x identical units with **primaries in series and secondaries in parallel**; series primaries carry the same current, so the secondaries share without balancing resistors.

- Open-circuit and leakage inductance divide by N_x , and so do the secondary currents. **The primary current does not.**
- The DC-overlap test current does not change.
- The wound ratio becomes $N_x N_p / N_s$, so **achievable ratios are quantised in steps of N_x / N_s** . Respect that when the tank is chosen.

6.22 The input capacitor

C_{in} is a film capacitor that absorbs switching ripple only. It has a floor, the ST rule of thumb that the evaluation board also follows:

Equation 36

$$C_{in} = 3 \frac{\text{nF}}{\text{W}} \times P_{in}$$

That is a **lower** bound; round up to the next standard value and rate it for the full input voltage. Do not go far above it: a large value holds charge across the zero crossing and distorts the line current, so it is a THD term.

6.23 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

Equation 37

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

and the hold-up condition is

Equation 38

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

Single-Stage PFC LLC Converter Design

Hold-up starts at the worst line phase. If the mains fails at the ripple trough the bank is already half a ripple down, hence the $-\frac{1}{2}\Delta v_{pp}$ term; it shortens the ride-out. Δv_{pp} is the ripple the bank produces, not the allowance, so this is checked once the bank is chosen.

Both conditions scale as $1/V_{out}^2$, so which wins depends only on the specification. With $k = V_{o,min}/V_{out}$, ripple dominates when

Equation 39

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

using the *allowed* Δv , since it is asked before the bank exists. Section 7.14 evaluates it.

- Include the **$2f_l$ component** in the ripple current; it is comparable to the switching part and adds in quadrature.
- With a centre-tapped secondary, judge capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the frequency of each current component: the **switching-frequency** ESR for the switching part, the **120 Hz** figure from $\tan\delta$ for the $2f_l$ part. Here the two parts are of similar size.

The controller start-up window is finite and this bank is a far heavier start-up load than a conventional design. Cold start into the full bank should be measured early.

6.24 Semiconductor requirements

State requirements rather than pick parts. Four rules:

- Rate the primary switches on the **composite tank peak**; the load component alone under-rates them.
- Divide by the number of devices in parallel. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current.
- Compute loss from $R_{DS(on)}$ at $T_{j,max}$. The datasheet maximum is 25 °C process spread; temperature is a separate multiplier, about two for a superjunction device.
- A curve against T_a may be read as T_j **only if it is a pulse test**.

The standing device in half-bridge morphing (Section 5.2) never switches and carries the whole tank current; Section 7.15 shows what that cost here.

6.25 Driving the bridge: external gate drivers

The controller's HOUTx and LOUTx are logic outputs. Each leg needs a half-bridge driver, which brings four checks of its own.

The high side lives on a bootstrap. Its capacitor C_{BOOT} is recharged from V_{CC} while the low side of the same leg is on, for T_{charge} . A driver that does this through an integrated switch of resistance R_{BS} loses

Single-Stage PFC LLC Converter Design

Equation 40

$$V_{drop} = \frac{Q_g R_{BS}}{T_{charge}}$$

which grows with frequency; when it is a large part of V_{CC} , an external fast diode from V_{CC} to BOOT replaces it. The high-side supply then sits a diode drop V_F and the per-cycle ripple below the rail; the ripple is the gate charge plus what the floating section draws, I_{QBO} , over the longest high-side pulse $T_{on,max}$:

Equation 41

$$\Delta V_{boot} = \frac{Q_g + I_{QBO} T_{on,max}}{C_{BOOT}}$$
$$V_{BO} = V_{CC} - V_F - \Delta V_{boot}$$

The driver sets a floor on V_{CC} . Both its own supply and the floating one have a recommended minimum; below them the driver may still switch, but nothing in its datasheet is guaranteed. The start-up sizing therefore uses

Equation 42

$$V_{CC,floor} = \max(V_{CC,off}, V_{CC,drv,min}, V_{BO,min} + V_F + \Delta V_{boot})$$

The gate power is shared. Each driver moves the charge of two switches per period. The power $Q_g V_{CC} f$ splits between the driver output R_{drv} , the gate resistor R_G and the MOSFET's internal $R_{g,int}$, so the driver keeps

Equation 43

$$s_{drv} = \frac{1}{2} \left(\frac{R_{so}}{R_{so} + R_G + R_{g,int}} + \frac{R_{si}}{R_{si} + R_G + R_{g,int}} \right)$$
$$P_{drv} = 2 Q_g V_{CC} f_{sw} s_{drv} + (I_{QCC} + I_{QBO}) V_{CC}$$

with R_{so} and R_{si} the source and sink resistances of the output, taken as V_{CC} over the short-circuit currents when the datasheet gives nothing better, and I_{QCC} , I_{QBO} the quiescent currents of the low and floating sections.

Timing. The dead time reaches the gates shortened by the driver's delay mismatch MT , so $(t_D - MT)$ must still cover the swing T_T . And the midpoint slope S_{mid} during the swing, the commutation current over the output capacitance at the top of the swing, must stay under the driver's allowed OUT slew rate $S_{OUT,max}$:

Single-Stage PFC LLC Converter Design

6.26 Controller network

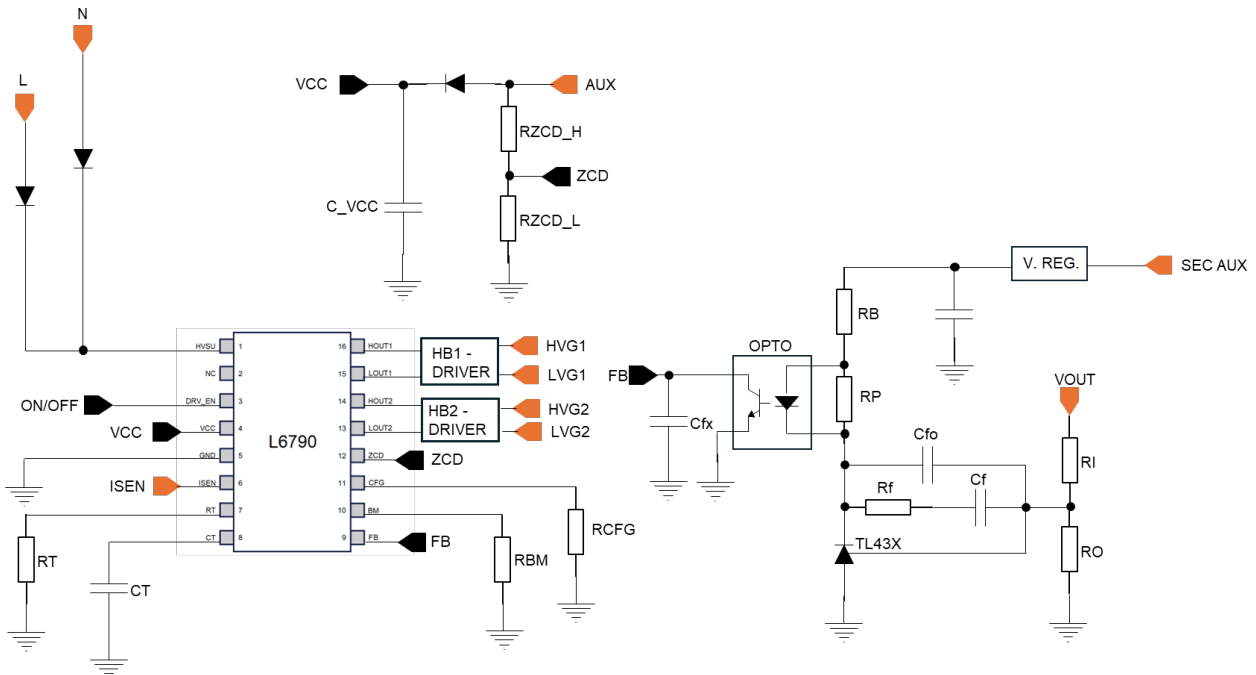


Figure 34: The controller and its passive network. HVSU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider; R_T and C_T set the oscillator limits; R_{CFG} and R_{BM} are read at power-up. The drawing feeds V_{CC} from the auxiliary winding through a diode alone; the worked design puts a Zener-referenced regulator in between (Section 7.28). Drawing from the ST L6790A design spreadsheet.

Decide first what the auxiliary winding is for. Wired straight to V_{CC} , its voltage at OVP2 must stay under the V_{CC} rating, which caps its ratio. With a regulator in between the cap moves to the regulator output, and the ratio is set by two other things (Section 6.27).

Section 7.22 sizes the parts in order. R_T deserves the most attention: it sets the oscillator floor, and that floor must stay above f_o .

That floor is usually the thinnest margin. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so it must be confirmed by measuring $f_{sw}(\theta)$ on hardware.

The transformer tolerance must fit inside the same margin. A **low** L_{open} raises f_o ; the fall that can be tolerated is

Equation 44

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2}$$

with $k_{floor} = f_{Min}/f_o$ the floor margin. A floor margin of only a few per cent allows a fall of only a few per cent, and then the usual $\pm 10\%$ does **not** fit: at the bottom of the tolerance f_o rises above f_{Min} and the bridge can hard switch near the zero crossing. Either tighten the low-side tolerance or lower R_T to lift f_{Min} ; a higher floor widens the zero-crossing dead zone (Section 4.3). **Check before the transformer is ordered.**

Single-Stage PFC LLC Converter Design

6.27 Supplying V_{CC} from the auxiliary winding

The winding charges a capacitor through a diode. An NPN emitter follower, its base held by a Zener, passes that voltage on to the pin. Between them sits the bypass diode the datasheet asks for; it keeps the start-up current out of the regulator. The pin then sits at $V_{CC,reg}$, the Zener voltage V_{DZ} less the base-emitter drop V_{BE} and the diode drop V_D :

Equation 45

$$V_{CC,reg} = V_{DZ} - V_{BE} - V_D$$

and the Zener tolerance puts it in a band. The bottom of the band is kept above $V_{CC,HVSUon}$, the level below which the start-up unit turns its charge current on; the draft datasheet does not say whether it still does after its start-up timeout, and above that level the question never arises. The top must stay under the pin's operating limit. The load I_{VCC} is the controller's own I_{CC} , the quiescent current of the gate drivers $I_{drv,q}$, and the gate charge Q_g of each of the N_{sw} switch positions, once per period:

Equation 46

$$I_{VCC} = I_{CC} + I_{drv,q} + N_{sw} Q_g f_{sw}$$

Q_g is the charge at the voltage the driver really applies, the rail itself, not the 10 V of the datasheet headline. Past the Miller plateau the drain is already down and the gate sees a fixed capacitance, so the gate-charge curve is a straight line there and two points on it give the charge at any drive voltage:

Equation 47

$$Q_g(V) = Q_{g,10} + C_g (V - 10 \text{ V})$$

The winding ratio has a floor and a cost. At the end of hold-up, output $V_{o,min}$, the winding must still lift the base above the highest Zener voltage $V_{DZ,max}$ with enough current left for the base, I_{VCC}/β_{min} , and the Zener bias $I_{Z,min}$; that caps the feed resistor R_{BZ} ,

Equation 48

$$R_{BZ} \leq \frac{n_{aux} V_{o,min} - V_D - V_{DZ,max}}{I_{VCC}/\beta_{min} + I_{Z,min}}$$

and at OVP1 the pass transistor burns the difference between the capacitor and the pin, times I_{VCC} . The lowest whole-turn ratio that meets the floor therefore also minimises the cost.

Start-up is the other sizing case. The start-up unit fills C_{VCC} to V_{CCon} ; switching begins, and until the output is high enough for the winding to hold the pin, C_{VCC} feeds the controller and the drivers: alone down to $V_{CC,HVSUon}$, then helped by the start-up charge current I_{HVSU} down to $V_{CC,floor}$. The floor is not the controller's V_{CCoff} but the highest of it and the lowest supply the gate drivers are specified at (Equation 42). That has to last t_{hand} , until the output reaches $V_{out,UV}$, the level at which the winding holds the pin at the floor. The drive current falls as the rail does, $I_{VCC}(V) = I_{VCC,SU} - B(V_{CCon} - V)$ with $B = N_{sw} C_g f_{sw}$ (Equation 47), and an oscillator that starts above its run range spends an extra charge ΔQ_{OSC} at the top. With the output bank charged at rated current and no load, integrating $C_{VCC} dV/dt = -I_{VCC}(V)$ over the two stretches gives

Single-Stage PFC LLC Converter Design

Equation 49

$$t_{hand} = \frac{C_{out} V_{out, UV}}{I_{out}}$$

$$C_{VCC} \geq \frac{B(t_{hand} + \Delta Q_{OSC}/I_{VCC, SU})}{\ln \frac{I_{VCC}(V_{CCon})}{I_{VCC}(V_{CC, HVSUon})} + \ln \frac{I_{VCC}(V_{CC, HVSUon}) - I_{HVSU}}{I_{VCC}(V_{CC, floor}) - I_{HVSU}}}$$

A larger C_{VCC} is paid for in the delay before the first pulse, $C_{VCC} V_{CCon}/I_{HVSU}$.

6.28 Voltage loop and compensation

The loop is the chain of Figure 35: a divider measures the output, a TL431 compares it with its reference and drives an optocoupler LED, the optocoupler transistor pulls the FB pin, the FB voltage commands power (Section 4.6), and that power goes into the output capacitor.

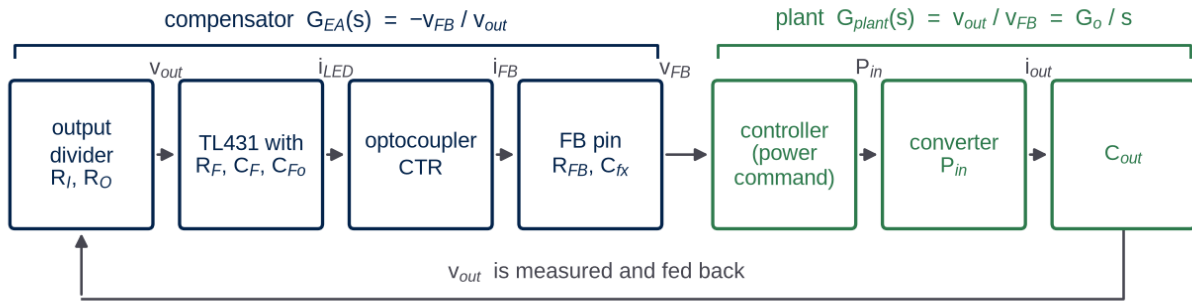


Figure 35: The voltage loop as blocks. Everything left of the FB pin is the compensator $G_{EA}(s)$; everything right of it is the plant $G_{plant}(s)$. The loop gain is $T(s) = G_{plant}(s) G_{EA}(s)$.

The plant. The FB voltage sets the power, the power over the output voltage is the current into C_{out} , and a current into a capacitor integrates. In small signal, with v_{out} and v_{FB} the changes of V_{out} and V_{FB} :

Equation 50

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$

V_{FB} is the feedback voltage above its 0.5 V offset at rated power (Equation 15). An integrator falls at 20 dB per decade with -90° everywhere and would cross 0 dB at $f_{cto} = G_o/2\pi$. There is no second pole and no right-half-plane zero, so the loop is easy to stabilise. What makes it hard to place is the output ripple (Section 6.30).

The integrator assumes a load that draws constant power as the output moves, which a downstream converter does. A resistive load R_L draws less as the output falls, and in small signal the integrator becomes a pole at $1/(\pi R_L C_{out})$. Above that pole the plant is the same; below it the loop gain is lower, and at the crossover the phase is up to 90° less negative, so the integrator is the worst case for phase margin.

Single-Stage PFC LLC Converter Design

6.29 What the loop must achieve, and what goes wrong when it does not

Three numbers are read from the Bode plot of $T(s)$: the **crossover** f_c where $|T| = 1$, the **phase margin** $180^\circ + \arg T$ at f_c , and the **gain margin**, how far $|T|$ is below 0 dB at f_{180} where $\arg T = -180^\circ$. Table 5 gives the aims.

Quantity	Aim	If it is too low	If it is too high
Crossover f_c	15 to 20 Hz in this converter (Section 6.30). The general rules are below $f_{sw}/5$ and below the ripple frequency the loop must not follow.	The loop is slow. The output dips further and recovers later after a load step or a morphing transition.	The loop follows the $2f_l$ output ripple. The power command, and so the input current, is modulated at $2f_l$; third harmonic on the input current, and chatter across the burst threshold.
Phase margin Φ_M	45° is the floor; 50 to 60° is the target. At 76° the step response has no overshoot; at 45° it rings ($Q \approx 1.2$) [onsemi TND381].	Ringing after every disturbance. At 0° the loop oscillates at f_c .	The response is over-damped and slow. Not a fault, but no longer free: the crossover has to drop to get it.
Gain margin GM	6 dB is the floor; 10 dB is comfortable.	Part spread moves $ T $ up. The optocoupler CTR alone spreads by 2:1 over bins, current and temperature; a loop with a small gain margin oscillates at f_{180} on a warm board with a high-CTR part.	—
G_{EA} at $2f_l$	At or below the value that keeps the third harmonic inside its budget (Equation 53).	Nothing wrong with the loop; the crossover is lower than it needs to be.	Third harmonic above budget, burst chatter.

Table 5: What the loop must achieve.

6.30 Why the crossover must be low: the $2f_l$ ripple

The output ripple at $2f_l$ (Section 3.1) has the peak-to-peak value

Equation 51

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_l C_{out}}$$

The compensator passes a fraction $G_{EA}(2f_l)$ of it to the FB pin, which is a power command, so the input current is modulated at $2f_l$. A sine modulated at twice its own frequency gains a third harmonic; a modulation of the command by m puts $m/2$ into it, and the ripple amplitude is half the peak-to-peak, so

Equation 52

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

Turned around, the third-harmonic budget fixes the largest compensator gain at $2f_l$:

Single-Stage PFC LLC Converter Design

Equation 53

$$G_{EA}(2f_i) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$

A small gain at $2f_i$ means a crossover well below it; with 5 % third harmonic allowed, 15 to 20 Hz. A faster loop would give less output ripple and a quicker load-step response, and more input distortion. The distortion limit wins. onsemi TND381 calls the failure “tail chasing”.

6.31 The compensator: TL431, optocoupler and the FB pin

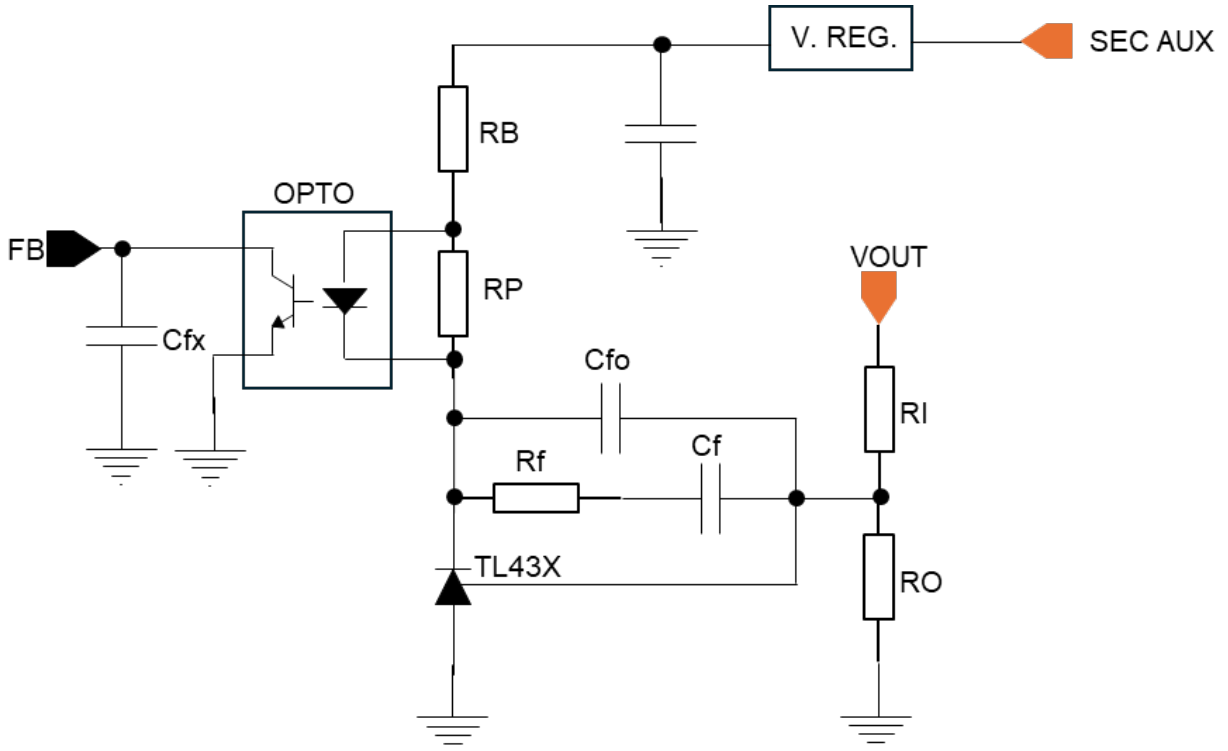


Figure 36: The compensator on the secondary side. R_I and R_O set the regulated output; the TL431 is the error amplifier; R_F , C_F and C_{F0} (R_f , C_f , C_{fo} on the drawing) shape its gain; R_B feeds the optocoupler LED from the regulated V_Z rail; R_P keeps the TL431 above its minimum current; C_{fx} sits on the FB pin. Drawing from the ST L6790A design spreadsheet.

The divider R_I , R_O brings the output down to the TL431 reference $V_R = 2.495$ V:

Equation 54

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$

When the output rises the TL431 sinks more cathode current through the LED, which R_B feeds from a regulated rail V_Z ; the optocoupler transistor pulls FB down against the internal pull-up R_{FB} and the power command falls. R_P across the LED carries the minimum cathode current the TL431 needs, so it has an upper limit:

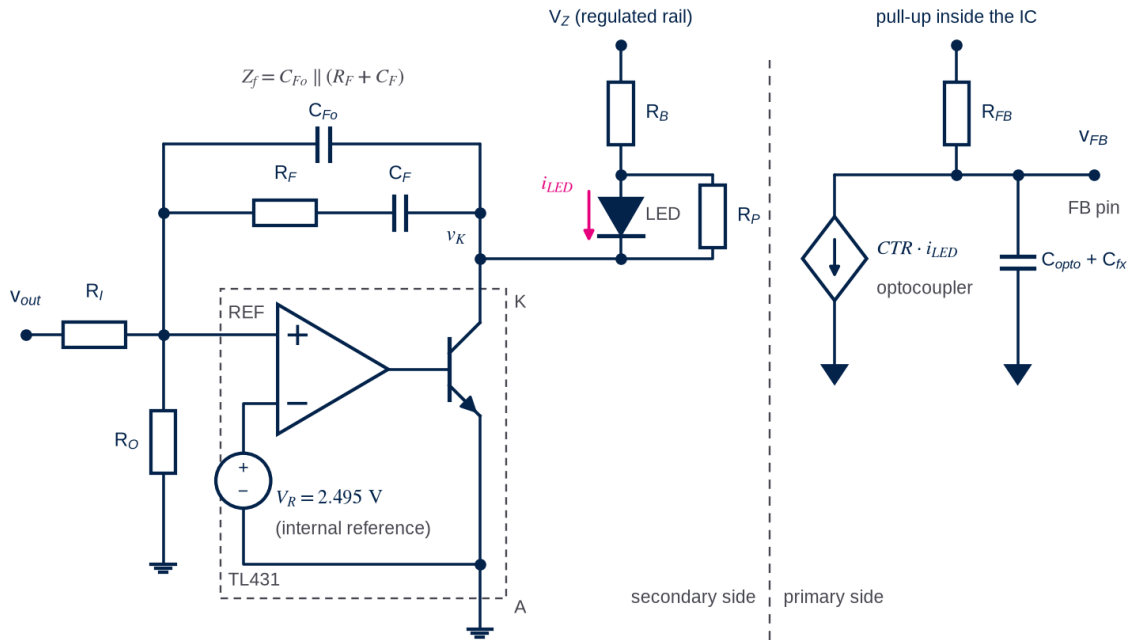
Single-Stage PFC LLC Converter Design

Equation 55

$$R_P \leq \frac{V_{Fo}}{I_{min}}$$

with V_{Fo} the LED forward voltage and I_{min} the least cathode current the TL431 needs. Because the LED is fed from V_Z , the output ripple reaches it only through the TL431: there is no second path (TND381's "fast lane").

The same network as an op-amp. Figure 37 redraws it. Inside the TL431 an amplifier compares REF with the internal V_R and drives an NPN whose collector is the cathode; the transistor inverts, so from REF to cathode the part is a high-gain inverting amplifier that holds REF at V_R . R_I is its input resistor and C_{Fo} in parallel with $R_F + C_F$ its feedback impedance Z_f , so the cathode moves by $-Z_f/R_I$ times the output change; R_O sets only the dc point. The cathode current is the LED current, and the optocoupler transistor is a current-controlled current source sinking $CTR i_{LED}$ from the FB node into R_{FB} with the pole of $C_{opto} + C_{fx}$. The three blocks multiply to a negative gain, which is the negative feedback. G_{EA} is written without that sign: $G_{EA} = -v_{FB}/v_{out}$.



$$\frac{v_K}{v_{out}} = -\frac{Z_f}{R_I}, \quad i_{LED} = -\frac{v_K}{R_B}, \quad v_{FB} = -CTR i_{LED} \left(R_{FB} \parallel \frac{1}{s(C_{opto} + C_{fx})} \right)$$

R_O carries no signal: the REF input is held at V_R

$$G_{EA}(s) = -\frac{v_{FB}}{v_{out}} = \frac{Z_f}{R_I} \cdot \frac{CTR R_{FB}}{R_B} \cdot \frac{1}{1 + s R_{FB}(C_{opto} + C_{fx})}$$

Figure 37: The TL431 compensator as an op-amp circuit: R_I and Z_f make an inverting amplifier; R_B , R_P and the LED turn the cathode voltage into i_{LED} ; the optocoupler is a current source $CTR i_{LED}$ into R_{FB} and $C_{opto} + C_{fx}$. The two sides return to different grounds across the isolation barrier.

Its transfer function from output to FB pin is a Type II network with one more pole:

Single-Stage PFC LLC Converter Design

Equation 56

$$G_{EA}(s) = \frac{EA_o}{s} \cdot \frac{1 + s/\omega_z}{(1 + s/\omega_p)(1 + s/\omega_{px})}$$

Equation 57

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \quad [\text{rad/s}], \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

$\omega = 2\pi f$ throughout, and C_{ser} is C_F and C_{Fo} in series. $C_F + C_{Fo}$ make the pole at the origin (zero static error); $R_F C_F$ the zero f_z that returns the phase; R_F with C_{Fo} the pole f_p that brings the gain down again by $2f_i$; $C_{opto} + C_{fx}$ with R_{FB} a third pole f_{px} at about a kilohertz that removes switching noise. CTR multiplies the whole gain, which is why its spread matters.

The bias window. The LED must drive the FB pin at its steady-state current with the lowest CTR (upper bound), and must not exceed the pin's maximum current with the highest CTR when the TL431 is fully on (lower bound). $I_{FB,steady}$ and $I_{FB,max}$ are those two FB-pin currents:

Equation 58

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,steady}/CTR_s}$$

$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,max}/CTR_m}$$

EA_o uses the CTR at the steady-state LED current, CTR_s ; the lower bound of R_B uses the CTR at maximum LED current, CTR_m . Take both from the bin that will be fitted, and re-check the loop at the top of the bin: the crossover rises almost in proportion to CTR, because the loop falls at close to 20 dB per decade there, and the phase margin falls.

6.32 Placing the zero and the pole: the K-factor method

The Venable K-factor method sets EA_o , the zero and the pole from two targets: the phase margin Φ_M and the gain allowed at $2f_i$. A spread factor K_v comes from the phase margin, with α_v a weighting constant (1.2 in the ST tool):

Equation 59

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$

The zero goes K_v below the crossover and the pole K_v above it. Above the pole the gain is $EA_o K_v^2/\omega$, so the limit at $2f_i$ fixes EA_o :

Single-Stage PFC LLC Converter Design

Equation 60

$$EA_o = \frac{2\pi(2f_l) G_{EA}(2f_l)}{K_v^2}$$

The centre point follows from the plant gain, with $\Gamma_v = 0.744 V_{eq,max}/V_{eq,min}$ an input-voltage margin factor of the ST tool (Appendix 10.3). The tool takes $V_{eq,min}$ as the lowest equivalent input and $V_{eq,max}$ as the mains maximum, not the highest equivalent input:

Equation 61

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v EA_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$

The parts follow from Equation 57. The high-frequency pole is placed at a chosen f_{pHF} , and C_{fx} is what is left after the optocoupler capacitance:

Equation 62

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_I R_B EA_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$

$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$

Round each value to a standard part, then compute the zero, pole and gain the standard parts give and check the loop on those.

6.33 Checking the loop: crossover, phase margin and gain margin

The loop gain (Equations 50 and 56) is two integrators, one zero and two poles:

Equation 63

$$|T(\omega)| = \frac{G_o EA_o}{\omega^2} \cdot \frac{\sqrt{1 + (\omega/\omega_z)^2}}{\sqrt{1 + (\omega/\omega_p)^2} \sqrt{1 + (\omega/\omega_{px})^2}}$$

$$\arg T(\omega) = -180^\circ + \arctan \frac{\omega}{\omega_z} - \arctan \frac{\omega}{\omega_p} - \arctan \frac{\omega}{\omega_{px}}$$

The -180° is the two integrators, so the **phase margin is the compensator phase at the crossover**. With $A(\omega)$ the fraction in Equation 63, the crossover condition $\omega_c^2 = G_o EA_o A(\omega_c)$ is solved by repeating

Equation 64

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

a few times; each pass cuts the error by more than half, because the square root halves the change in A, so no solver is needed. Then

Single-Stage PFC LLC Converter Design

Equation 65

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

Figure 38 shows where the three numbers are read, in units of the loop's own crossover, so it is the shape of every loop of this kind. Section 7.21 puts this design's numbers on the same plot.

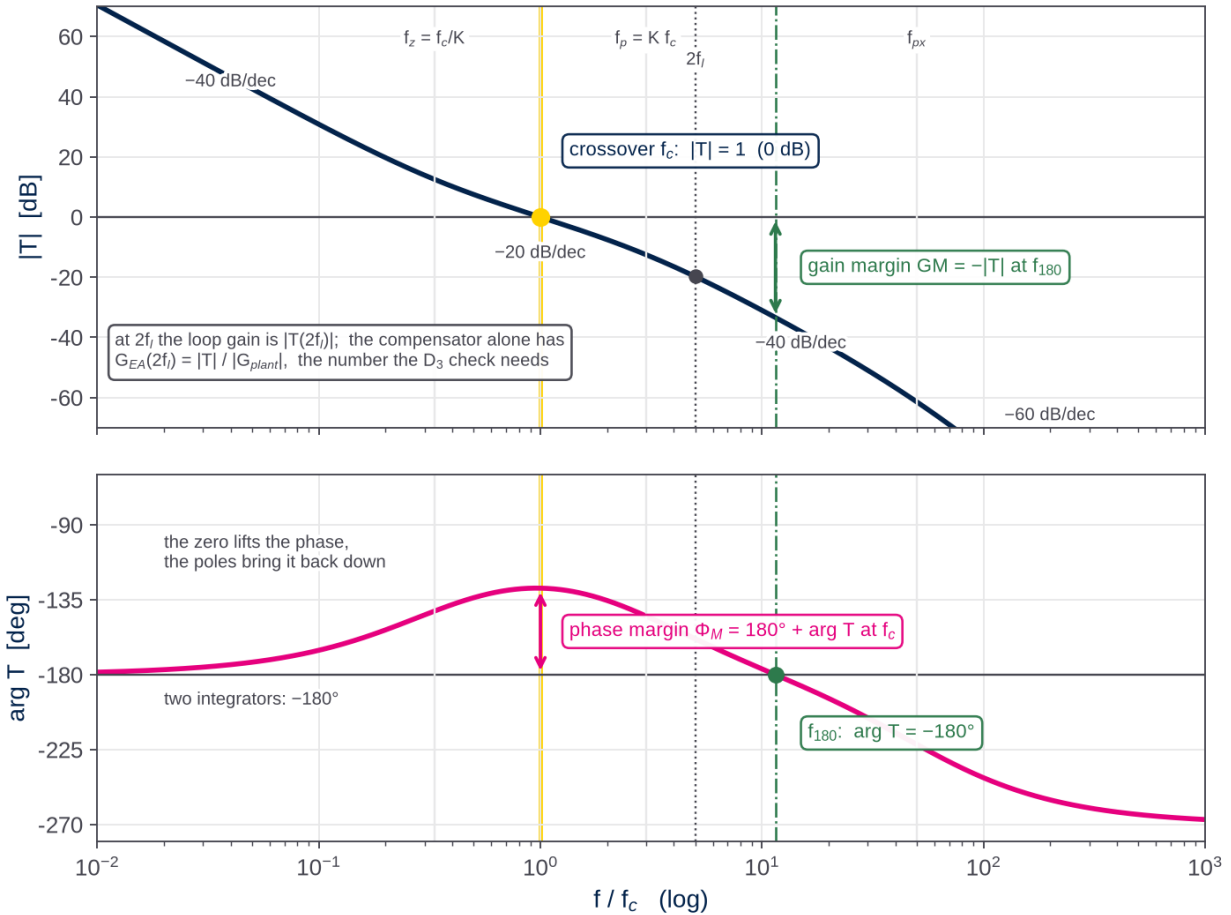


Figure 38: A two-integrator loop with a Type II compensator, in units of its crossover. Phase margin is the distance from -180° at f_c ; gain margin is $|T|$ below 0 dB at f_{180} . The dot at $2f_l$ is the gain the third-harmonic check reads.

6.34 Gain margin, and why it needs checking

Above f_{px} one zero faces two poles, so the phase keeps falling and -180° is crossed at a finite frequency, which also needs no search:

Equation 66

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})} , \quad GM = -20 \log_{10} |T(f_{180})|$$

The root is real whenever the zero sits well below both poles, which the K-factor placement gives; if its argument were negative the phase would sit below -180° at every frequency, and the phase margin would be negative at any crossover. Judge against 6 dB as a floor and 10 dB as comfortable. With a crossover in the tens of hertz the margin is usually large; that is a reason to compute it, not to assume it.

Single-Stage PFC LLC Converter Design

The same expression at $2f_l$ gives the third harmonic through Equation 52, which closes the check.

6.35 Feedback ripple against the burst threshold

A crossover this low leaves $2f_l$ ripple on the FB pin, which moves the commanded power every half cycle. If it crosses the burst-entry threshold the converter chatters. At the burst point the ripple is

Equation 67

$$\Delta V_{FB} = \frac{P_{in,BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

with $P_{in,BM}$ the input power at the burst entry point. Lowering R_{BM} moves the burst entry below that ripple, at 0.01 V per k Ω , and the ripple must clear it on both sides:

Equation 68

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

Raising the crossover would shrink the ripple but worsen the distortion, so R_{BM} is the practical answer.

A loop this slow cannot recover from a large load step by itself; the anti-saturation circuit does, and it needs the FB pin to reach its full current range. That is why R_B must sit inside its window (Equation 58).

7 Design example

One design from specification to component values; every computed value shows its arithmetic. **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor, no boost stage.

7.1 The specification, sorted by what kind of number it is

Each number is **given** by the load and the mains, **chosen** by the designer, read from a part's **datasheet**, or **assumed** in place of a measurement not yet made.

Item	Symbol	Value	Kind	Note
Mains input	V_{ac}, f_l	90 to 264 Vac, 47 to 63 Hz	given	the worst line frequency is the lowest, so every f_l term uses 47 Hz
Output	V_{out}, I_{out}	25 V, 26.3 A (657.5 W)	given	$V_{o,eff} = V_{out} + N_{rect} V_f = 25.0 \text{ V}$ with $V_f = 0$
Output ripple, $2f_l$ pk-pk	Δv	$\leq 5\%$ of V_{out} (1.25 V)	given	a property of the load, not of the converter
Hold-up	$T_{hold}, V_{o,min}$	12 ms down to 19 V, at 100 Vac	given	starts at the ripple trough, the worst line phase
Secondary rectifier	N_{rect}	1 (centre tap, SR)	chosen	halves the rectifier conduction loss; doubles the device reverse voltage
Target series resonance	f_r	150 kHz	chosen	the starting point; the leakage of the transformer sets the final f_l , 75.0 kHz (Section 7.4)

Single-Stage PFC LLC Converter Design

Item	Symbol	Value	Kind	Note
Specified maximum f_{sw}	$f_{sw,max}$	225 kHz (1.5 f_r)	chosen	an input to λ , Section 4.9
Bridge dead time	t_D	220 ns	chosen	what the ZVS sweep has to beat
V_{CC} source	—	auxiliary winding, 2 T, through a Zener regulator	chosen	no external rail; the regulator, not the turns ratio, keeps the pin under its rating
LLC stage efficiency	η_{HB}	98 %	assumed	a 95 % outcome moves R_{ac} and R_{CS} by about 3 %; nothing downstream is sensitive
Oscillator idle time	T_{idle}	249 ns	assumed	the draft datasheet also implies 350 and 700 ns; at 700 ns they still do, but the floor margin falls to 1.032 — the first thing to measure
$R_{DS(on)}$ temperature factor	k_T	1.9 / 1.5	datasheet	primary / secondary at $T_j = 125^\circ\text{C}$, both from the vector curves of their datasheets
Burst entry point	r_{BM}	11 % of P_{in} (75 W)	assumed	sets R_{BM} , then checked against the feedback ripple
Primary switches	—	STO60N045DM9, one per position, four in all	chosen	600 V superjunction with a fast-recovery body diode (Section 7.17)
Primary gate and output charge	$Q_{g,10}, C_{o(tr)}$	90 nC (0–10 V), 897 pF (0–400 V)	datasheet	Q_g sets the V_{CC} load, $C_{o(tr)}$ the swing the dead time must cover
Gate drivers	—	2 × L6498LD (SO-14), one per leg, external bootstrap diodes	chosen	the controller has no driver of its own (Section 7.18)
SR MOSFETs	—	STL160N10F8, 2 per leg, four in all	chosen	100 V STRipFET F8 (Section 7.19)
Synchronous rectifier	—	TEA2095TE	chosen	connection and limits only (Section 7.20)
Regulator parts	$V_{BE}, V_D, \beta_{min}, I_{Z,min}$	0.7 V, 0.7 V, 50, 1 mA	assumed	junction drops assumed; gain and bias are requirements put on the parts
Safety conditions	—	US, EU, Japan, Korea, China; household AV; 5000 m	given	sets the transformer insulation (Section 7.12)
Transformer construction	—	section winding on a catalogue coil former with a partition added: NP1 in triple-insulated Litz in one section, the Litz secondary in the other, tape over every layer	chosen	the insulation in the wire; the window shape and the partition set L_{short} (Section 7.10)

Single-Stage PFC LLC Converter Design

Item	Symbol	Value	Kind	Note
Winding figures	J, k_{litz} , k_w	4.5 A/mm ² , 0.55, 1.10; 0.2 mm of triple insulation	assumed	the copper and the diameters of the transformer; k_w is the winding pitch over the bundle, from the first sample; the insulation thickness comes from the wire vendor

Table 6: The specification. Everything else in this chapter is computed from these.

Harmonic class. IEC 61000-3-2 Class D reaches only 600 W. This design draws 685 W at the input, so the absolute Class A limits apply.

7.2 What the design came out as

Block	Values
Tank	C_r 180 nF, L_r 25 μ H, L_m 45 μ H, λ 0.556, f_r 75.0 kHz, f_o 44.8 kHz
Transformer	$N_p:N_s$ 15:2, n 6.013, n_T 7.50, L_{open} 70.0 μ H, L_{short} 25.0 μ H, $A_e \geq 208$ mm ² at B_{pk} 0.20 T
Frequency	f_{sw} at the line peak, full load: 62.2 kHz at the low corner, 118.5 kHz at the high corner
Output	470 μ F $\times$ 160 = 75.2 mF, ripple 1.18 V (4.74 %), hold-up 13.42 ms
Controller	R_T 21.5 k Ω , C_T 470 pF, R_{CS} 24.0 m Ω , R_{CFG} 30 k Ω , R_{BM} 30 k Ω
Loop	f_c 17.57 Hz, Φ_M 49.6°, third harmonic 4.23 %
Switches	STO60N045DM9 $\times$ 4; 2 $\times$ L6498LD, R_G 4.7 Ω , C_{BOOT} 470 nF with an external diode; TEA2095TE, 2 STL160N10F8 per leg
V_{CC}	N_{aux} 2 T, Zener 15 V, R_{BZ} 560 Ω , C_{VCC} 1000 μ F, $V_{CC,reg}$ 13.60 V
Insulation	reinforced, primary to secondary, carried by triple-insulated NP1 and NAUX; clearance 4.5 mm and creepage 6.4 mm at the pins, 3000 V ac for 60 s

Table 7: Principal values. The rest of the chapter shows where each one comes from.

7.3 What a verification margin is

Every check is what the design has divided by what it needs:

Equation 69

$$k = \frac{X_{act}}{X_{req}} \implies \text{pass when } k > 1$$

$k = 1.05$ is five per cent of room; the smallest k is the thinnest part of the design.

Check	$k = \text{has} / \text{needs}$	Substituted	k
ZVS at the worst point of the sweep	$T_{ZC,min} / t_D$	636 ns / 220 ns	2.891
Oscillator floor clears the lower resonance	f_{Min} / f_o	48.29 kHz / 44.84 kHz	1.077

Single-Stage PFC LLC Converter Design

Check	k = has / needs	Substituted	k
Oscillator ceiling clears the highest operating frequency	$f_{\text{Max}} / f_{\text{sw}} \text{ (FB edge, line peak)}$	285.59 kHz / 118.49 kHz	2.410
Over-current threshold clears the tank peak	$I_{\text{OCP1}} / I_{\text{Lr,pk}}$	22.92 A / 18.12 A	1.265
Hold-up achieved against hold-up required	$t_{\text{hold}} / T_{\text{hold}}$	13.42 ms / 12 ms	1.119
Secondary loss per rectifier leg against its budget	$P_{\text{budget}} / P_{\text{SR,leg}}$	3.00 W / 1.95 W	1.539
Loss of the standing primary device against its budget	$P_{\text{budget}} / P_{\text{mos,dc}}$	3.00 W / 9.27 W	0.324
V_{CC} stays above the start-up threshold	$V_{\text{CC,reg,min}} / V_{\text{CC,HVSUon}}$	12.85 V / 12 V	1.071
The Zener feed still regulates at the end of hold-up	$R_{\text{BZ,max}} / R_{\text{BZ}}$	656 Ω / 560 Ω	1.172
C_{VCC} bridges the start-up hand-over	$C_{\text{VCC}} / C_{\text{VCC,req}}$	1000 μF / 884 μF	1.131
The high-side driver supply stays in its range	$V_{\text{BO}} / V_{\text{BO,min}}$	11.54 V / 9.3 V	1.241
The dead time at the gates covers the swing	$(t_{\text{D}} - \text{MT}) / T_{\text{T}}$	190 ns / 170 ns	1.118
The midpoint slope stays under the driver limit	$S_{\text{OUT,max}} / S_{\text{mid}}$	50 V/ns / 40.9 V/ns	1.222
Gate driver dissipation against its rating	$P_{\text{drv,max}} / P_{\text{drv}}$	1 W / 0.537 W	1.864
The SR gate drive reaches the $R_{\text{DS(on)}}$ test voltage	$V_{\text{G,SR}} / 10 \text{ V}$	10.4 V / 10 V	1.040

Table 8: The verification margins.

$k_{\text{Ploss}} = 0.324$ is a result, not a failed calculation: the standing primary device spends 9.27 W against a 3 W budget; no single 600 V device meets that budget in the standing position, so the design goes ahead on a heatsink (Section 7.17) and the rest is settled by measurement (Section 7.32). The thinnest margin that passes is the SR gate drive, $k = 1.040$. It is thin by construction: the SR controller clamps its gate drive at 10.4 V and the SR $R_{\text{DS(on)}}$ is quoted at 10 V, so the clamp only has to clear the test voltage. $k_{\text{floor}} = 1.077$ rests on unknowns: the transformer tolerance and the idle time both move it, and neither is measured yet.

7.4 Following the numbers through

The design in the order it was computed; each step shows its equation and the numbers in it. Table 9 collects the results.

Step 1 — the power the tank passes. The bridge, the EMI filter and the LLC stage each get a loss budget, the last as η_{HB} :

$$P_{\text{in}} = P_{\text{out}} + P_{\text{LLC}} + P_{\text{EMI}} + P_{\text{BR}} = 657.5 + 13.42 + 8.34 + 5.52 = \mathbf{684.8 \text{ W}}$$

Single-Stage PFC LLC Converter Design

$$P_{in,LLC} = P_{in} - P_{BR} - P_{EMI} = 684.8 - 5.52 - 8.34 = \mathbf{670.9 \text{ W}}$$

Step 2 — the two corners of the equivalent input. They are the morphing thresholds, not the mains limits:

Equation 18, repeated

$$V_{ac,eq} = 2 V_{ac} \text{ (full bridge)} \quad V_{ac,eq} = V_{ac} \text{ (half bridge)}$$

$$V_{ac,eq,low} = \frac{245 V_{pk}}{\sqrt{2}} = \mathbf{173.2 V_{ac}} \quad V_{ac,eq,high} = \frac{2 \times 235 V_{pk}}{\sqrt{2}} = \mathbf{332.3 V_{ac}}$$

Step 3 — the turns ratio. The wound ratio 15:2 is the input, because it is what can be built (the plain calculation gives $n = 6.046$); the model ratio follows from it and the realised λ (step 8):

Equation 21, repeated

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$

$$n = \frac{n_T}{\sqrt{1 + \lambda_{act}}} = \frac{7.500}{1.2472} = \mathbf{6.013}$$

Equation 19, repeated

$$V_{refl} = n V_{o,eff} = n(V_{out} + N_{rect} V_f)$$

$$V_{refl} = n V_{o,eff} = 6.013 \times 25.0 = \mathbf{150.3 \text{ V}}$$

Step 4 — the gain the tank is asked for, at the line peak ($\theta = 90^\circ$) of each corner: M_{low} at the low corner, M_{high} at the high one:

Equation 12, repeated

$$M(\theta) = \frac{n V_{o,eff}}{V_{drive}(\theta)} = \frac{2 n V_{o,eff}}{\sqrt{2} V_{ac,eq} \sin \theta}$$

$$M_{low} = \frac{2 \times 6.013 \times 25.0}{\sqrt{2} \times 173.24} = \mathbf{1.2272} \quad M_{high} = \frac{2 \times 6.013 \times 25.0}{\sqrt{2} \times 332.34} = \mathbf{0.6397}$$

Step 5 — the load the tank sees. The 4, not the textbook 8, because $P_{in,LLC}$ is the line-peak power (Section 6.4):

Single-Stage PFC LLC Converter Design

Equation 20, repeated

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o,eff}^2}{P_{in,LLC}}$$

$$R_{ac} = \frac{4}{\pi^2} \frac{6.013^2 \times 25.0^2}{670.9} = \mathbf{13.65 \, \Omega}$$

Step 6 — the Q cap and the design impedance. At the low corner, the Q at which the tank still reaches M_{low} and the Q at which ZVS still completes inside t_D ; the smaller is kept (Equation 1). The design impedance $Z_{0,design}$ follows from it:

$$Q_{ZVS} = \mathbf{1.7172} \quad Z_{0,design} = R_{ac} Q_{ZVS} = 13.65 \times 1.7172 = \mathbf{23.44 \, \Omega}$$

Step 7 — C_r and L_r . C_r is taken up on purpose for Q margin. The L_r that pairs with the selected C_r at the target f_r is computed below; the L_r fitted is the leakage of the transformer (Section 7.10), and f_r follows it:

Equation 2, repeated

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

$$C_r = \frac{1}{2\pi f_r Z_{0,design}} = \frac{1}{2\pi \times 150 \text{ kHz} \times 23.44 \, \Omega} = 45.26 \text{ nF} \rightarrow \mathbf{180 \text{ nF}}$$

$$L_r = \frac{1}{(2\pi f_r)^2 C_r} = \frac{1}{(2\pi \times 150 \text{ kHz})^2 \times 180 \text{ nF}} = 6.25 \, \mu\text{H} \rightarrow \mathbf{25 \, \mu\text{H}}$$

Step 8 — λ and L_m . Four candidates for λ , the largest binding:

Equation 16, repeated

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2}$$

$$\lambda_1 = 0.563, \quad \lambda_2 = 1.014, \quad \lambda_3 = 0.125, \quad \lambda_{TD} = \mathbf{1.247} \implies L_m = \frac{L_r}{\lambda_{TD}} = \frac{25}{1.247} = 20.05 \, \mu\text{H}$$

That 20.1 μH is the no-load condition at the high corner, which this design knowingly does not meet (Section 4.10), so it is not rounded to. L_m is chosen with n so that n_T is windable:

$$L_m = \mathbf{45 \, \mu\text{H}} \quad \lambda_{act} = \frac{L_r}{L_m} = \frac{25}{45} = \mathbf{0.556}$$

Single-Stage PFC LLC Converter Design

Step 9 — what the selected parts make. Every later check is measured against these:

Equation 3, repeated

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

$$f_r = \frac{1}{2\pi\sqrt{25 \mu\text{H} \times 180 \text{ nF}}} = \mathbf{75.0 \text{ kHz}} \quad f_o = \frac{1}{2\pi\sqrt{70 \mu\text{H} \times 180 \text{ nF}}} = \mathbf{44.8 \text{ kHz}}$$

$$Z_0 = \sqrt{\frac{25 \mu\text{H}}{180 \text{ nF}}} = 11.79 \Omega \quad Q_{pk} = \frac{Z_0}{R_{ac}} = \frac{11.79}{13.65} = \mathbf{0.863}$$

$$n_T = n\sqrt{1 + \lambda_{act}} = 6.013 \times 1.2472 = \mathbf{7.500} = 15 : 2$$

Step	Quantity	Eq.	Result
1	P _{in} , P _{in,LLC}	—	684.8 W, 670.9 W
2	V _{ac,eq} , low and high corner	18	173.2, 332.3 Vac
3	n, V _{refl}	21, 19	6.013, 150.3 V
4	M _{low} , M _{high}	12	1.2272, 0.6397
5	R _{ac}	20	13.65 Ω
6	Q _{ZVS} , Z _{0,design}	1	1.7172, 23.44 Ω
7	C _r , L _r	2	45.3 → 180 nF, 6.25 → 25 μH
8	λ _{req} , L _m , λ _{act}	16	1.247, 45 μH, 0.556
9	f _r , f _o , Q _{pk} , n _T	2, 3, 1	75.0 kHz, 44.8 kHz, 0.863, 7.500

Table 9: The design, step by step: the results of the nine steps. In a → b, a is the calculated value and b the part selected.

Steps 7 and 8 are the judgement calls. C_r is 180 nF against a calculated 45.3: the larger capacitor lowers Q_{pk}, which is where the ZVS margin comes from. L_m is 45 μH against 20.1: it lets the wound 15:2 give the n the tank needs, and gives up the no-load solution (next note). L_r is 25 μH against 6.25: it is the leakage of the filled transformer window (Section 7.10), so f_r falls from the 150 kHz target to 75.0 kHz, and L_m keeps λ with it.

No-load solution: none. The no-load gain of this tank bottoms out at M_∞ = 0.6429, and the high corner asks for 0.6397, below it. No frequency reaches that gain without load, and the margin is under one per cent, so no full-load check would show it. Burst mode owns that region (Section 4.10).

Single-Stage PFC LLC Converter Design

7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 589 ns on this tank against a swept 636 ns (7 % pessimistic), and it also reads low, by tens of per cent, on the other tanks computed for this project. Being low there is no guarantee for another tank. Read with $\lambda_{\text{act}} = 0.556$ instead of the design $\lambda = 1.247$ it returns -941 ns, a capacitive answer for an inductive tank. This design is judged by the sweep.

The capacitive edge of the full-load curve ($\arg Z_{\text{in}} = 0$) is at 56.6 kHz; the gain peak is lower, at 54.1 kHz, with 7.8° of capacitive phase still there (Section 2.9). Neither is used for the verdict.

Load	173 Vac HB edge	90 Vac FB	110 Vac FB	230 Vac HB	264 Vac HB	166 Vac FB edge
100 %	636	814	1254	1293	1339	1258
75 %	1317	1379	1507	1500	1420	1173
50 %	2024	2024	1893	1834	1585	1054
25 %	2869	2824	2470	2360	1923	854

Table 10: $T_{\text{ZC,min}}$ in nanoseconds, the smallest over the line half cycle, at each load and input voltage; FB is full bridge (the tank sees twice the mains), HB half bridge; $t_{\text{D}} = 220$ ns.

The worst point in Table 10 is full load at the HB edge, $T_{\text{ZC}} = 636$ ns against $t_{\text{D}} = 220$ ns, a margin of 2.89 times — the same corner the tank was designed at.

7.6 The gain chart of this design

Figure 39 is the chart of Section 4.5 on this tank ($\lambda_{\text{act}} = 0.556$, $Q_{\text{pk}} = 0.863$), one panel per input voltage: the two morphing edges and the four mains voltages a supply meets. Each panel names the mains voltage and the bridge; in full bridge the tank sees twice the mains, so each panel also says what the tank sees. The curves are the same in every panel; only the required-gain lines move. The marked crossings are the operating points the sweep is built from.

Single-Stage PFC LLC Converter Design

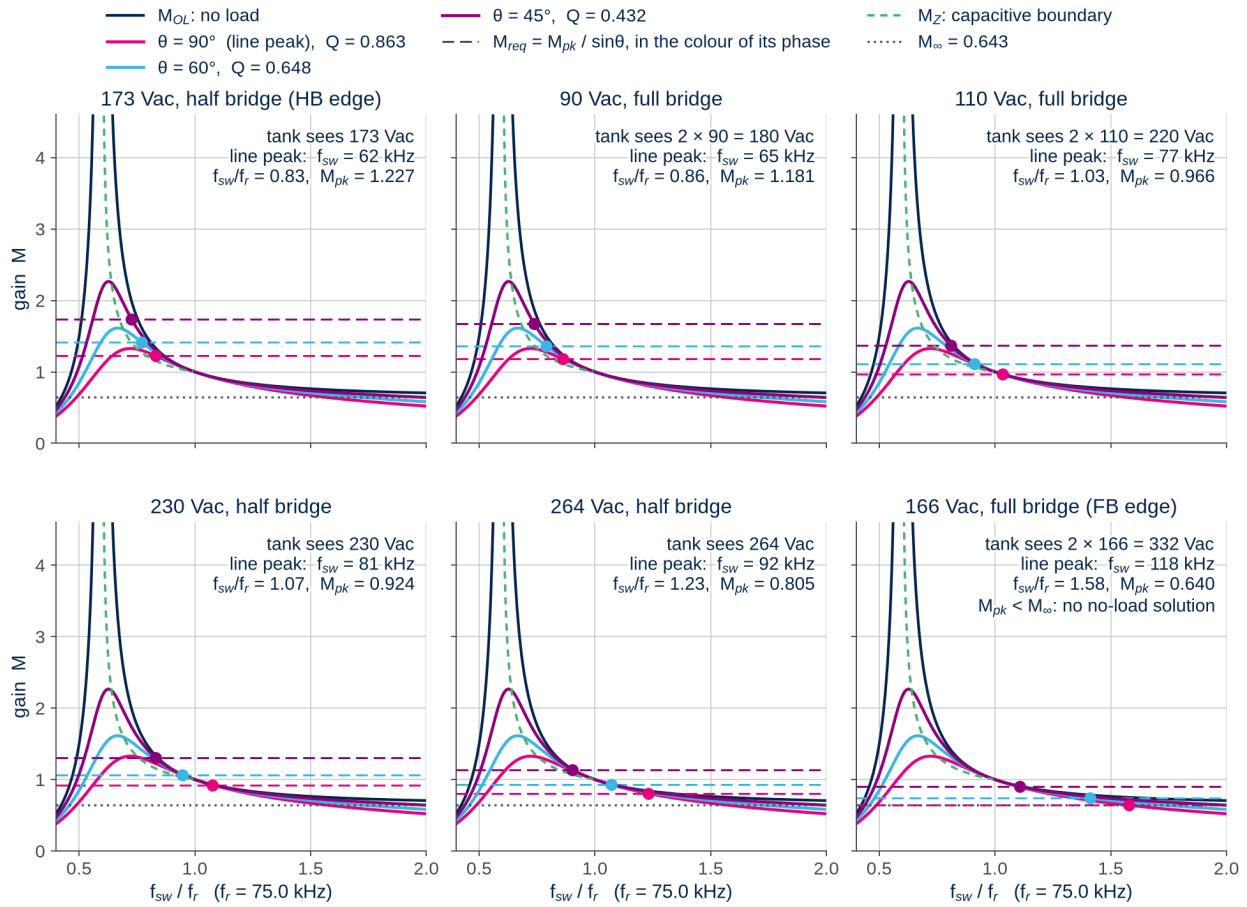


Figure 39: The gain chart of this design at the six input voltages, low to high as the tank sees them. Top left, the HB edge (173 Vac, half bridge): highest required gain, crossings below f_r . Bottom right, the FB edge (166 Vac, full bridge, so the tank sees 332 Vac): lowest required gain, crossings above f_r . Each phase is compared with the dashed line of its own colour only.

Input voltage	θ	$Q(\theta)$	$M_{req}(\theta)$	f_{sw}/f_r	f_{sw}
173 Vac, half bridge (HB edge)	90°	0.863	1.227	0.829	62.2 kHz
	60°	0.648	1.417	0.769	57.7 kHz
	45°	0.432	1.735	0.727	54.5 kHz
90 Vac, full bridge	90°	0.863	1.181	0.861	64.6 kHz
	60°	0.648	1.364	0.790	59.3 kHz
	45°	0.432	1.670	0.738	55.4 kHz
110 Vac, full bridge	90°	0.863	0.966	1.032	77.4 kHz
	60°	0.648	1.116	0.912	68.5 kHz
	45°	0.432	1.367	0.810	60.8 kHz
230 Vac, half bridge	90°	0.863	0.924	1.075	80.6 kHz
	60°	0.648	1.067	0.946	70.9 kHz

Single-Stage PFC LLC Converter Design

Input voltage	θ	$Q(\theta)$	$M_{\text{req}}(\theta)$	f_{sw}/f_r	f_{sw}
264 Vac, half bridge	45°	0.432	1.307	0.829	62.2 kHz
	90°	0.863	0.805	1.230	92.3 kHz
	60°	0.648	0.930	1.072	80.4 kHz
166 Vac, full bridge (FB edge)	45°	0.432	1.139	0.903	67.7 kHz
	90°	0.863	0.640	1.579	118.5 kHz
	60°	0.648	0.739	1.409	105.7 kHz
	45°	0.432	0.905	1.107	83.0 kHz

Table 11: The marked crossings of Figure 39, at every input voltage. $Q = Q_{\text{pk}} \sin^2\theta$ and $M_{\text{req}} = M_{\text{pk}}/\sin\theta$; M_{pk} scales as $1/V_{\text{ac,eq}}$, from 1.227 at the HB edge to 0.640 at the FB edge.

Four readings matter:

- **The HB edge sets the tank.** 173 Vac in half bridge is the lowest voltage the tank ever sees, so its line-peak requirement $M_{\text{pk}} = 1.227$ is the largest of the six. It is met at $f_{\text{sw}}/f_r = 0.829$, well to the right of M_z : inductive.
- **The FB edge sets the frequency.** At 166 Vac in full bridge the tank sees 332 Vac, the most it ever sees; the line-peak crossing is 118 kHz, and that, not the mains maximum, is what the oscillator ceiling must clear (Section 7.23).
- **The four mains voltages sit between the edges.** 90 Vac in full bridge puts $2 \times 90 = 180$ Vac on the tank, only 4 % above the HB edge, so a 90 Vac system runs close to the gain worst case. 110 Vac in full bridge (220 Vac on the tank) and 230 Vac in half bridge (230 Vac on the tank) are within 5 % of each other: morphing makes the two mains systems nearly alike to the tank. 264 Vac in half bridge is still 21 % below what the tank sees at the FB edge.
- **The FB edge also judges λ .** The line-peak requirement there is 0.640 and the no-load floor $M_{\infty} = 0.643$; below it there is no no-load solution and burst mode takes over (Section 4.10).

The six panels are six places on one axis, not six designs. The curves do not depend on input voltage: $M(f_n, Q)$ is fixed by λ_{act} and Q_{pk} . **The input voltage enters only through M_{req} .**

7.7 Which side of resonance this design runs on

With $n_T = 7.50$, at four of the six input voltages the tank spends part of every line cycle above f_r , where the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time need checking.

Single-Stage PFC LLC Converter Design

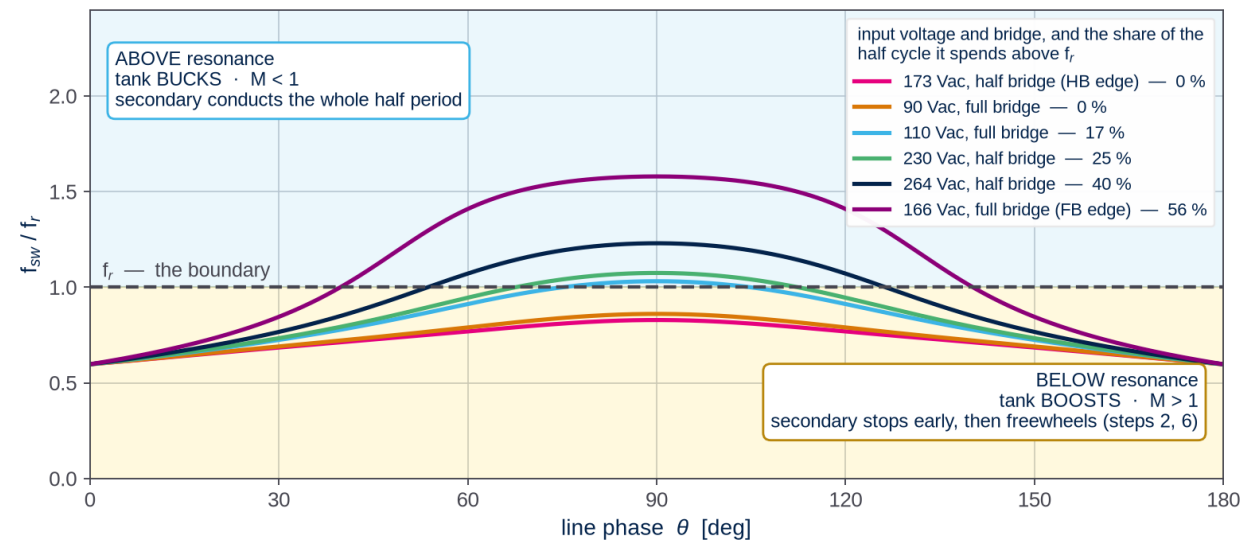


Figure 40: Which side of f_r the converter is on over a line half cycle, at the six input voltages. Every curve converges on f_o at the zero crossing. The percentage in each legend entry is the fraction of the half cycle spent above f_r , without zero-current turn-off.

Input voltage and bridge	the tank sees (Vac rms)	peak f_{sw}	side of f_r
173 Vac, half bridge (HB edge)	173 Vac	62.2 kHz	below
90 Vac, full bridge	180 Vac	64.6 kHz	below
110 Vac, full bridge	220 Vac	77.4 kHz	above
230 Vac, half bridge	230 Vac	80.6 kHz	above
264 Vac, half bridge	264 Vac	92.3 kHz	above
166 Vac, full bridge (FB edge)	332 Vac	118.5 kHz	above

Table 12: Peak f_{sw} over the half cycle against $f_r = 75.0$ kHz. 4 of the 6 input voltages cross into above-resonance operation for part of the cycle.

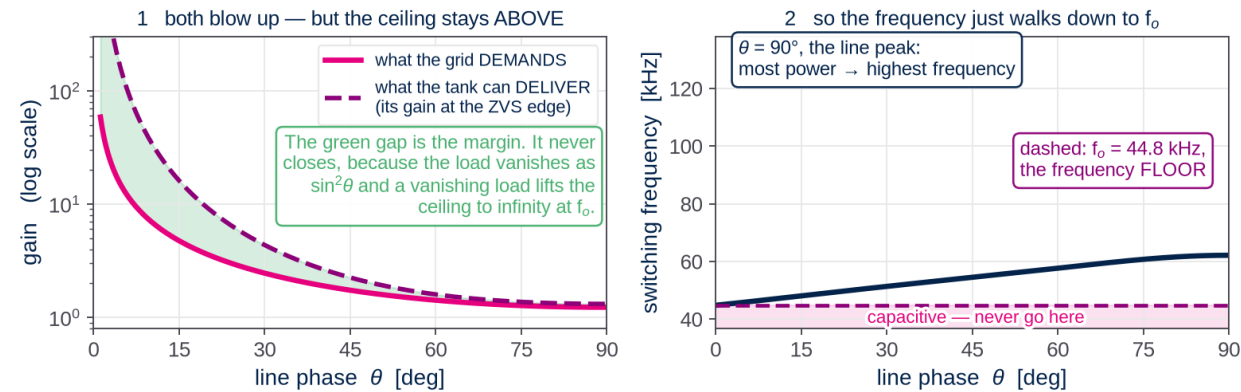


Figure 41: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on $f_o = 44.8$ kHz (Section 4.2).

Single-Stage PFC LLC Converter Design

7.8 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed. The peaks are read off one cycle at the HB edge, line peak ($\theta = 90^\circ$ in Table 11). The composite peak is the peak of the sum, because the two components peak at different instants:

$$I_{Lr,pk} = \max_t |i_{Lm}(t) + i_{trafo}(t)| = 18.12 \text{ A} < 11.13 + 16.58 = 27.71 \text{ A}$$

The rms values are the rms of one switching period, averaged in I^2 over the line half cycle by the five-point Simpson rule:

$$I_{lc} = \sqrt{\frac{2}{\pi} \int_0^{\pi/2} I_{rms}^2(\theta) d\theta} \simeq \sqrt{\frac{1}{12} \sum_k w_k I_{rms}^2(\theta_k)}, \quad w = \{1, 4, 2, 4, 1\}$$

That is why the line-cycle primary rms, 10.41 A, is lower than the worst cycle's 13.50 A.

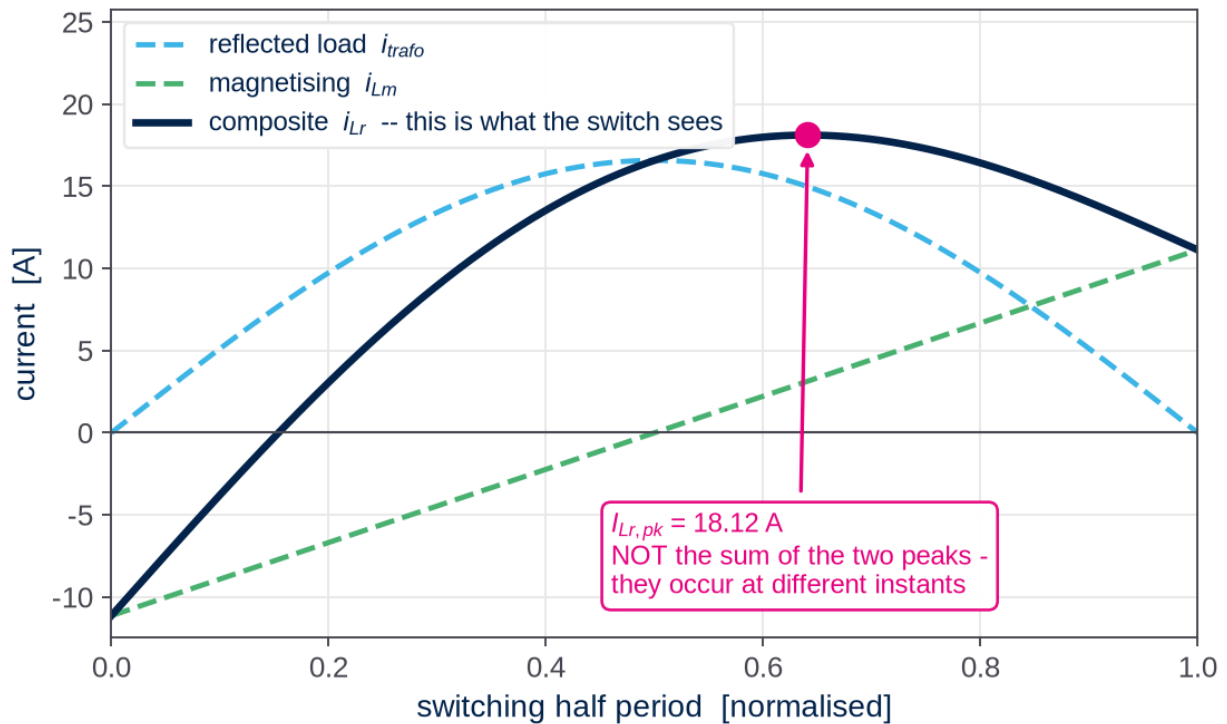


Figure 42: The composite tank current at the HB edge, full load, and why its peak is not the sum of the two component peaks: they occur at different instants. Drawn as the calculation models it: the reflected load current a half sine over the whole switching half period, the magnetising current a linear ramp.

Quantity	Value	Where it is used
f_{sw}	62.2 kHz	the switching cycle the peaks below are read in
$I_{sec,pk}$	99.7 A	secondary rectifier peak
$I_{trafo,pk}$	16.58 A	reflected load component
$I_{Lm,pk}$	11.13 A	magnetising component

Single-Stage PFC LLC Converter Design

Quantity	Value	Where it is used
composite tank peak	18.12 A	primary device rating, OCP margin
$I_{pri,rms}$ (worst cycle)	13.50 A	primary rms rating
I_{pri} (line-cycle rms)	10.41 A	primary conduction loss
I_{rect} per leg (line-cycle rms)	28.50 A	secondary conduction loss
I_{Cout} (line-cycle rms)	30.54 A	output bank ripple current

Table 13: Currents at the HB edge (173 Vac, half bridge), full load. Peaks and worst-cycle rms are taken in the switching cycle at the line peak, $\theta = 90^\circ$; line-cycle rms values are over the whole line half cycle.

7.9 What the transformer must provide

The tank asks for $L_r = 25 \mu\text{H}$, $L_m = 45 \mu\text{H}$ and $n_T = 7.50$. The transformer is one part: a 15-turn primary and two 2-turn secondaries, NS2 and NS3, joined at the centre tap. NS2 is N_{s1} of Figure 1 and NS3 is N_{s2} , so NS3 conducts while S1 and S4 are on. Several units in series (Section 6.21) are not needed here.

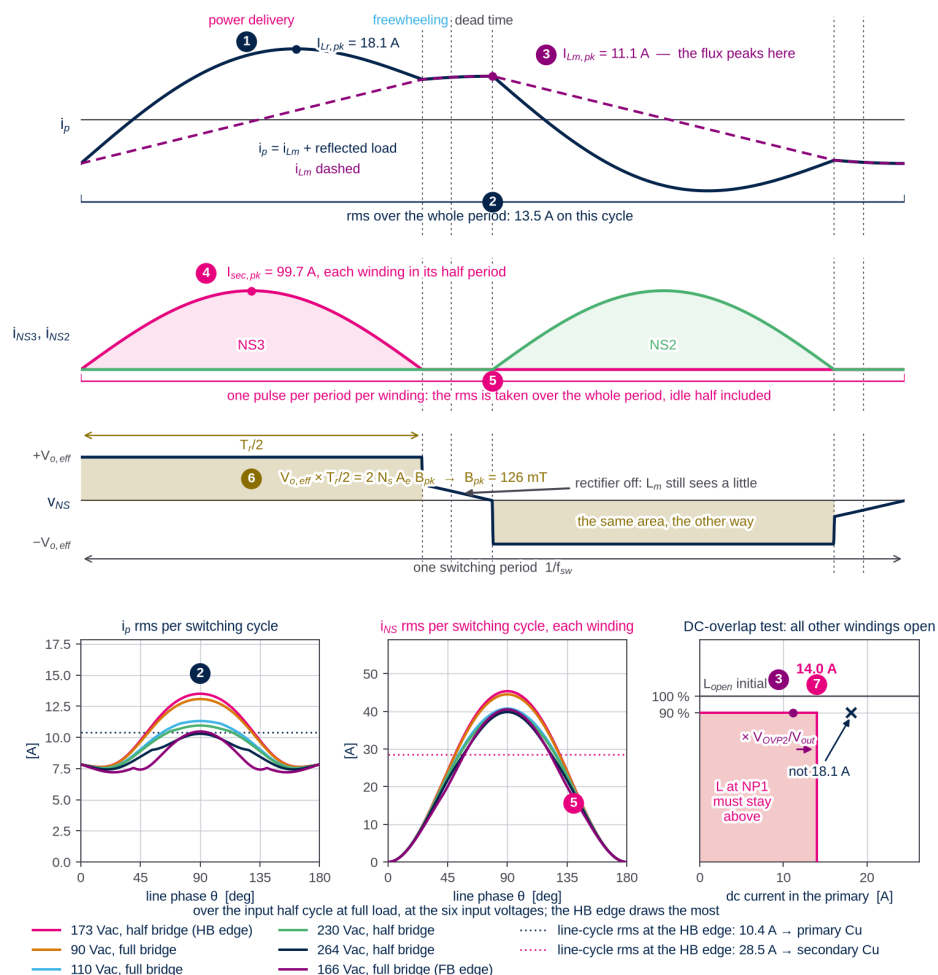


Figure 43: Where each number is read. Top: primary current, the two secondary winding currents and the secondary winding voltage over one switching period at the worst cycle: line peak at the HB edge (173 Vac, half bridge), full load. Lower left and centre: the rms of the two winding currents over the input half cycle at the six input voltages; the HB edge draws the most in both windings, which is why the table is read there. Lower right: the DC-overlap bench test. The circled marks are the rows of Table 4.

Single-Stage PFC LLC Converter Design

Quantity	Value	Note
Turns	N_p 15 T; NS2 2 T, NS3 2 T; NAUX 2 T	$n_T = 15 / 2 = 7.5$; the auxiliary feeds V_{CC} and ZCD
Open-circuit inductance	70.0 μ H	$L_r + L_m$; across NP1, every other winding open
Short-circuit inductance	25.0 μ H	this is L_r , no separate resonant inductor; across NP1, NS2 shorted alone, then NS3 alone; the other half and NAUX open
A_L	311 nH	open-circuit inductance / N_p^2 ; the core is gapped to it (Section 7.10)

Table 14: What the tank asks of the transformer.

7.10 Choosing the core

Three requirements come before any core is picked: the flux sets the core area, the copper sets the window, and the leakage sets the winding width.

Core area. The flux at f_r , with B_{pk} held to 0.20 T:

Equation 24, repeated

$$B_{pk} = \frac{V_{o, eff}}{4 f_r N_s A_e} \implies A_e \geq \frac{V_{o, eff}}{4 f_r N_s B_{max}}$$

$$A_e \geq \frac{25.0 \text{ V}}{4 \times 75.03 \text{ kHz} \times 2 \times 0.20 \text{ T}} = \mathbf{208 \text{ mm}^2}$$

Copper. Each winding carries its line-cycle rms current (Table 13) at $J = 4.5 \text{ A/mm}^2$. Summed over the windings, that is the copper the window must hold:

Equation 28, repeated

$$A_{cu} = \frac{I_{rms}}{J} \quad \sum_w N_w A_{cu, w} \leq k_u A_N$$

$$\text{NP1 : } A_{cu} = \frac{10.41 \text{ A}}{4.5 \text{ A/mm}^2} = 2.31 \text{ mm}^2, \quad \text{NS2 : } A_{cu} = \frac{28.50 \text{ A}}{4.5 \text{ A/mm}^2} = 6.33 \text{ mm}^2$$

$$\sum_w N_w A_{cu, w} = 15 \times 2.31 + 4 \times 6.33 = 60.0 \text{ mm}^2 \leq 0.30 A_N \Rightarrow A_N \geq \mathbf{200 \text{ mm}^2}$$

The conductor for each follows from the skin depth.

Skin depth at f_r , with copper at 100 °C ($\rho = 2.26 \times 10^{-8} \Omega\text{m}$):

Equation 29, repeated

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

Single-Stage PFC LLC Converter Design

$$\delta = \sqrt{\frac{2.26 \times 10^{-8}}{\pi \cdot 75.0 \times 10^3 \cdot 4\pi \times 10^{-7}}} = 0.276 \text{ mm}$$

A conductor thicker than $2\delta = 0.55 \text{ mm}$ gains little, so **both windings are Litz**. The standard strand safely inside that is $d_s = \text{ø}0.10 \text{ mm}$, 5.5 times smaller than 2δ . For the primary, its copper area and the strand count are Equation 30:

Equation 30, repeated

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$

$$a_s = \frac{\pi \cdot (0.10)^2}{4} = 0.00785 \text{ mm}^2, \quad n_s = \left\lceil \frac{2.31}{0.00785} \right\rceil = 295$$

As one served bundle, with $k_{litz} = 0.55$, that would be:

Equation 31, repeated

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$

$$d_{litz} = \sqrt{\frac{4 \cdot 2.31}{\pi \cdot 0.55}} = 2.31 \text{ mm}$$

The secondary is Litz too. Between the two sections of the winding chosen below the field crosses the secondary (Section 6.16), and foil would carry eddy currents across its width. One turn of NS2 or NS3 needs 6.33 mm^2 :

$$n_s = \left\lceil \frac{6.33}{0.00785} \right\rceil = 807, \quad d_{litz} = \sqrt{\frac{4 \cdot 6.33}{\pi \cdot 0.55}} = 3.83 \text{ mm}$$

Each winding is one conductor a turn: no bundles in parallel, so nothing can share current unevenly and every end is one conductor on one pin. The sizes are rounded: NS2 and NS3 **ø4.0 mm Litz**, above the 3.83 mm the copper asks; NP1 **ø2.9 mm over its triple insulation**, above the 2.51 mm the copper asks (2.31 mm plus 0.2 mm of insulation), so that its layers fill the same radial room as the secondary's (below).

The arrangement. L_{short} must come out at $25.0 \mu\text{H}$. First try plain wire, side by side, with the gap between the sections carrying the reinforced insulation: the gap is then at least the 6.4 mm creepage (Section 7.12). Take the window chosen below (depth $h_w = 12.00 \text{ mm}$, mean turn $l_N = 100.5 \text{ mm}$), the primary as one plain Litz bundle in three layers ($w_A = 11.57 \text{ mm}$) and the secondary group ($w_S = 9.74 \text{ mm}$). Equation 26 with $N_B = 0$ gives

$$L_{short} \approx \frac{4\pi \times 10^{-7} \times 100.5 \text{ mm}}{12.00 \text{ mm}} \times 15^2 \left(\frac{11.57}{3} + 6.4 + \frac{9.74}{3} \right) \text{ mm} = \mathbf{32 \mu\text{H}}$$

That is 1.3 times the target. So **the insulation goes into the wire**: NP1 is triple-insulated Litz and NAUX is triple-insulated wire. The winding is then the **section winding** of the usual LLC reference transformer: a

Single-Stage PFC LLC Converter Design

two-section coil former, NP1 in one section, the secondary group in the other, the **partition** between them, and a wrap of tape over every layer.

The window this winding needs. When each section fills the window depth, its width is its copper area over the depth, so Equation 26 goes as I_N times the copper area over h_w^2 : the leakage of a filled section winding is set by the SHAPE of the window: a deep, short window leaks little, a shallow, long one a lot. Here the window is that of a catalogue coil former, filled with nothing left empty, and the tank's L_r is set to what it gives.

The build. A real winding is not packed tight: turns in a layer, and layers on each other, sit a little apart. The build is laid out at a pitch of $k_w = 1.10$ times the conductor from layer to layer, with a 0.05 mm wrap of tape over every layer; k_w is an assumption that the first sample's measured build replaces. NP1 is 3 layers of 5 turns; NS2 is one layer of its 2 turns side by side, NS3 the next layer, and NAUX (ø0.6 mm TIW) the last. The two sections come out nearly as deep as each other (h_A for NP1, h_S for the secondary section with NAUX), and the radial room h_r from the tube to the arc of the outer legs is 10.40 mm, 9.95 mm at the tolerance limits (the arc at its least, the tube at its most). Along the layer the pitch k_{ax} spreads the two sections over the winding width less the 3.0 mm partition, so nothing is left empty:

$$h_A = 3 \times (2.90 \times 1.10 + 0.05) = 9.72 \text{ mm}$$

$$h_S = 2 \times (4.00 \times 1.10 + 0.05) + (0.6 \times 1.10 + 0.05) = 9.61 \text{ mm}$$

$$J_p = \frac{10.41 \text{ A}}{3.15 \text{ mm}^2} = \mathbf{3.31 \text{ A/mm}^2}, \quad J_s = \frac{28.50 \text{ A}}{6.91 \text{ mm}^2} = \mathbf{4.12 \text{ A/mm}^2}$$

$$k_{ax} = \frac{30.4 - 3.0 \text{ mm}}{5 \times 2.90 + 2 \times 4.00 \text{ mm}} = \mathbf{1.218}$$

The build is 9.72 mm deep: 0.68 mm clear of the leg arc, 0.23 mm at the tolerance limits, and about level with the rim of the flanges (not dimensioned). The primary bundle is 401 strands against the 295 the copper asks, the secondary 880 against 807; both run under the 4.5 A/mm² of this note.

The leakage. Equation 26 with the widths of Table 15 and the 3.00 mm partition, the whole turn taken inside the core, gives the upper estimate

$$L_{short} \approx \frac{4\pi \times 10^{-7} \times 100.5 \text{ mm}}{12.00 \text{ mm}} \times 15^2 \left(\frac{17.66}{3} + 3.00 + \frac{9.74}{3} \right) \text{ mm} = \mathbf{28.7 \mu H}$$

Two things bring it down. The back plate of a PQ core is not a rectangle but a bow tie, narrow at the centre leg (Figure 44): at the middle of the winding only 47 % of a turn has an outer leg beside it, 11 % lies under the plate with no outer leg, and 42 % is outside the core. And the winding is deep: a turn of the outer secondary layer is 32 % longer than one of the inner. A field solution of the drawn layout takes both in: the section is solved through the outer legs (the ferrite, the gap, the plates) and where the turn leaves the core (the centre leg alone), and every part of it is weighted with the length of turn at its radius that lies in each; the part under the plate alone is counted half in each.

Single-Stage PFC LLC Converter Design

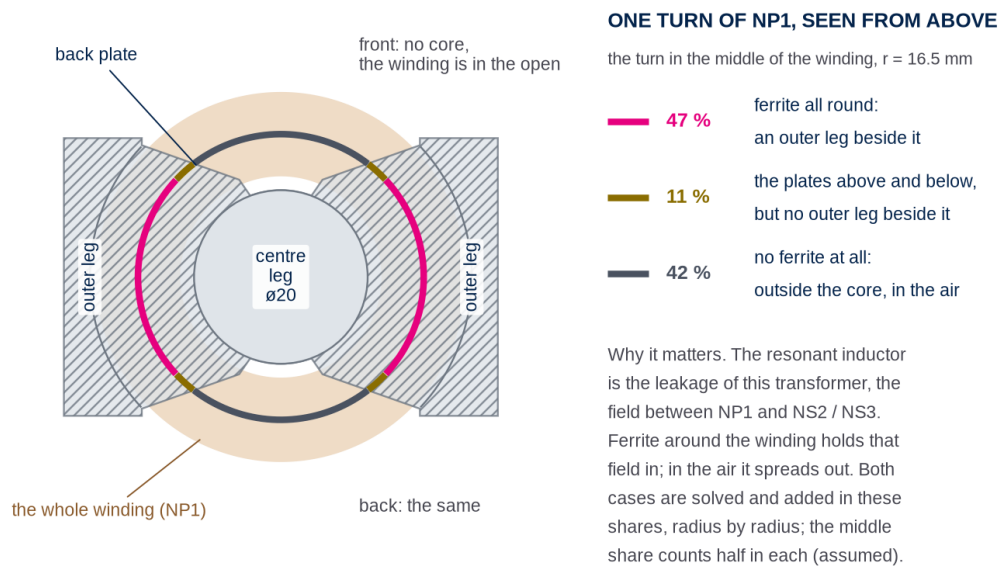


Figure 44: The core seen from above, to scale: the outer legs, the bow-tie back plate and the centre leg, the whole of NP1 as a pale ring under the plate, and the turn in the middle of it coloured by what surrounds it. The leakage is solved for the ferrite-surrounded and the open case and added in these shares, radius by radius.

What the tank sees. In service only one half of the centre tap conducts at a time. With NS2 shorted alone the field solution gives **24.50 μH** , with NS3 alone **25.14 μH** : NS3 lies further out, where the turns are longer. Their mean, 24.82 μH , is within 1 % of the tank's L_r of 25.0 μH ; counting the part under the plate fully out, or fully in, moves the halves to 23.35 and 26.40 μH at the extremes. The higher half is 0.6 % over L_r , inside the $\pm 10 \%$ the specification holds each half to, and its resonant frequency is 1.3 % below the other's: the half with less leakage carries a little more of the load. With both halves shorted, the specification's other reading, the figure is lower, 23.09 μH (21.89–24.29), because the two layers then share the secondary ampere-turns. It is recorded, not held to L_r .

The partition is the knob after the first samples, in one direction. The 3.0 mm wall can be moulded thinner, which lowers the leakage, but not usefully thicker; 2.5 mm lowers the both-shortened figure to 22.50 μH , and the pitch along the layer takes up the width. Wound tight ($k_w = 1$) the same winding reads 23.20 μH : the looseness moves it by 0.5 %. All of these are estimates; the first samples decide.

The ac loss. The dc resistance is a floor: 2.27 W at the line-cycle rms currents, every turn at its own radius. Every strand also sits in the leakage field, up to 22.3 mT rms in the primary section and 17.7 in the secondary, and a strand of diameter d_s in a field of peak B loses $\pi\omega^2\sigma d_s^4 B^2/128$ per metre. Summed over the strands of the drawn layout, the line-cycle rms current taken as a sine, that is 1.5 to 2.2 W in NP1 and 0.8 to 1.1 W in NS2 and NS3 between 62 kHz (the line peak at the half-bridge edge) and f_r : a first-order estimate, **1.0 to 1.5 times the dc loss**. It is the price of using the leakage field as the resonant inductor, and the strand sets it: for the same copper it goes as d_s^2 . The thermal measurement of the first sample decides the strand.

What the core must offer. A_e of at least 208 mm²; a window of at least $60.0/0.30 = 200 \text{ mm}^2$ for the copper of NP1, NS2 and NS3 (NAUX carries only the V_{CC} load, 144 mA on average, and is left out) at $k_u = 0.30$; and, for the leakage, a window deep and short enough that the filled section winding comes out at L_r .

The core. TDK **B65981A**, a PQ 50/50 in TDK N97 ferrite: $A_e = 332 \text{ mm}^2$, 1.6 times what the flux needs, so B_{pk} is 125 mT. Its window is 12.0 mm deep, from the round centre leg of $\phi 20 \text{ mm}$ to the arc that forms the inside of each outer leg ($\phi 44 \text{ mm}$), and 36.1 mm high. Its coil former is the catalogue part **B65982E**, one section and 12 pins; the only change asked of the bobbin maker is a 3.0 mm partition. The winding takes the 30.4 mm between the flanges,

Single-Stage PFC LLC Converter Design

the partition included. A_N is 340 mm²; the copper as built is 74.9 mm². The catalogue gives the core loss at one point only, 3.50 W at most at 100 kHz, 100 mT and 100 °C; the loss at the operating flux comes from the TDK N97 material curve. Table 15 is the arithmetic.

The gap is one piece, and the winding is close to it. The data sheet gives no relation between A_L and the gap for this core; gapped cores are made to order, ground to A_L . Without fringing, $\mu_0 A_e / A_L$ gives 1.34 mm for $A_L = 311$ nH; the field solution, fringing included, **about 1.78 mm**. The gap is at the middle of the centre leg, 15.2 mm from the primary-section flange, under the primary section, 2.5 mm short of the partition, with the first layer of NP1 1.60 mm away across the tube. Its fringing field reaches into the inner layers; the field solution counts it in the strand loss above.

Step	Primary NP1	Secondary NS2
Line-cycle rms current	10.41 A	28.50 A
Copper asked at $J = 4.5$ A/mm ²	$10.41 / 4.5 = \mathbf{2.31\text{ mm}^2}$, 295 strands	$28.50 / 4.5 = \mathbf{6.33\text{ mm}^2}$, 807 strands
Conductor as built	one triple-insulated Litz bundle a turn, $401 \times \varnothing 0.10$ mm: $\varnothing 2.70 + 0.2 = \mathbf{\varnothing 2.90\text{ mm}}$	one Litz bundle a turn, $880 \times \varnothing 0.10$ mm: $\varnothing 4.00\text{ mm}$
Current density as built	$10.41 / 3.15 = \mathbf{3.31\text{ A/mm}^2}$	$28.50 / 6.91 = \mathbf{4.12\text{ A/mm}^2}$
Turns and layers	15 T in 3 layers of 5, 5, 5 turns	2 T side by side in one layer; NS3 the next layer (from the tube: NS2, NS3)
Pitch	$k_w = 1.10$ layer to layer (assumed), $k_{ax} = 1.218$ along the layer, tape 0.05 mm a layer	the same; NAUX $0.6 \times 1.10 + 0.05$ mm on top
Width on the former	$5 \times 2.90 \times 1.218 = \mathbf{17.66\text{ mm}}$	$2 \times 4.00 \times 1.218 = \mathbf{9.74\text{ mm}}$
Radial build, tape included	$3 \times (2.90 \times 1.10 + 0.05) = \mathbf{9.72\text{ mm}}$	$2 \times (4.00 \times 1.10 + 0.05) = 8.90\text{ mm}$, + NAUX = 9.61 mm
Radial room, tube to leg arc	10.40 mm nominal, 9.95 mm at the tolerance limits; 0.23 mm left at the limits	
Former, B65982E	$17.66 + \text{partition } 3.00 + 9.74 = \mathbf{30.40\text{ mm}}$, the width between its flanges; nothing left free	
Copper in the window	asked: $15 \times 2.31 + 4 \times 6.33 = 60.0\text{ mm}^2$, which at $k_u = 0.30$ needs 200 mm ² of window; as built 74.9 mm ² ; $A_N = 340\text{ mm}^2$	

Table 15: The winding, from current to copper. NS3 is the same as NS2, one layer further out.

7.11 The winding and the pins

Figure 45 shows the section and the winding; Table 16 names each item.

Single-Stage PFC LLC Converter Design

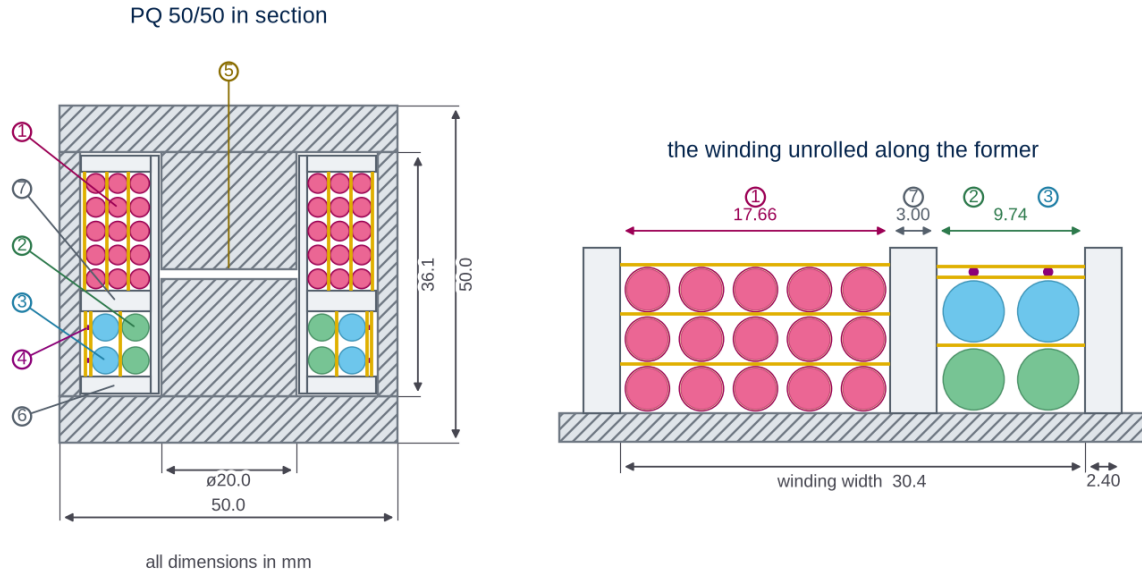


Figure 45: The core in section (left, to scale) and the winding unrolled along the coil former (right, to scale). NP1 in one section, NS2 and NS3 in the other, a 3.00 mm partition between them; the yellow lines are the layer tape. The circled numbers are the rows of Table 16.

Mark	Item	Turns	Conductor, placement
1	NP1 primary, pins 1 – 3	15 T	one triple-insulated Litz bundle a turn, $401 \times \varnothing 0.10$ mm ($\varnothing 2.90$ mm), 3 layers of 5, 5, 5 turns in the primary section; tape over each layer
2	NS2 secondary, pins 7 – 9	2 T	one Litz bundle a turn, $880 \times \varnothing 0.10$ mm ($\varnothing 4.00$ mm), both turns side by side in the first secondary layer; tape over it
3	NS3 secondary, pins 10 – 12	2 T	the same, the next layer out, wound in the same sense as NS2
4	NAUX auxiliary, pins 4 – 6	2 T	triple-insulated wire, $\varnothing 0.6$ mm assumed, one layer over the secondaries; feeds V_{CC} and ZCD
5	Centre-leg gap	—	one gap, about 1.78 mm from the field solution, ground to $A_L = 311$ nH
6	Coil former	—	TDK B65982E: tube $\varnothing 23.2$ mm, flanges 2.40 mm at the tube; no insulation role
7	Partition	—	3.00 mm, added to the catalogue former; the wall between the sections, sets L_{short} with the winding, no insulation role

Table 16: Items of the figure.

Single-Stage PFC LLC Converter Design

Pins. The former B65982E has 12 pins in two rows of 6, rows 45.72 mm apart; in a row they are two groups of three, 7.62 mm apart within a group and 12.7 mm between the groups. The data sheet numbers pins 1, 6, 7 and 12 in its view **from below**, the pin side; its drawing does not say so, but that is the view that shows the pin ends, and the view from above shows none. From the component side of the board, where the footprint is drawn, left and right are exchanged: Figure 46 draws both. NP1 and the auxiliary NAUX take the pin-1 row, NS2 and NS3 the other. Every winding is one conductor, one pin an end. **The centre tap is made on the board**, joining NS2's finish (pin 9) and NS3's start (pin 10), so each half can be measured on its own. A secondary pin carries the whole current of its winding, 28.5 A rms line-cycle: the bobbin maker confirms that rating, or the secondary ends are brought straight out to the board. The NAUX and secondary leads pass the primary section on their way to the pins; NP1 carries the insulation there.

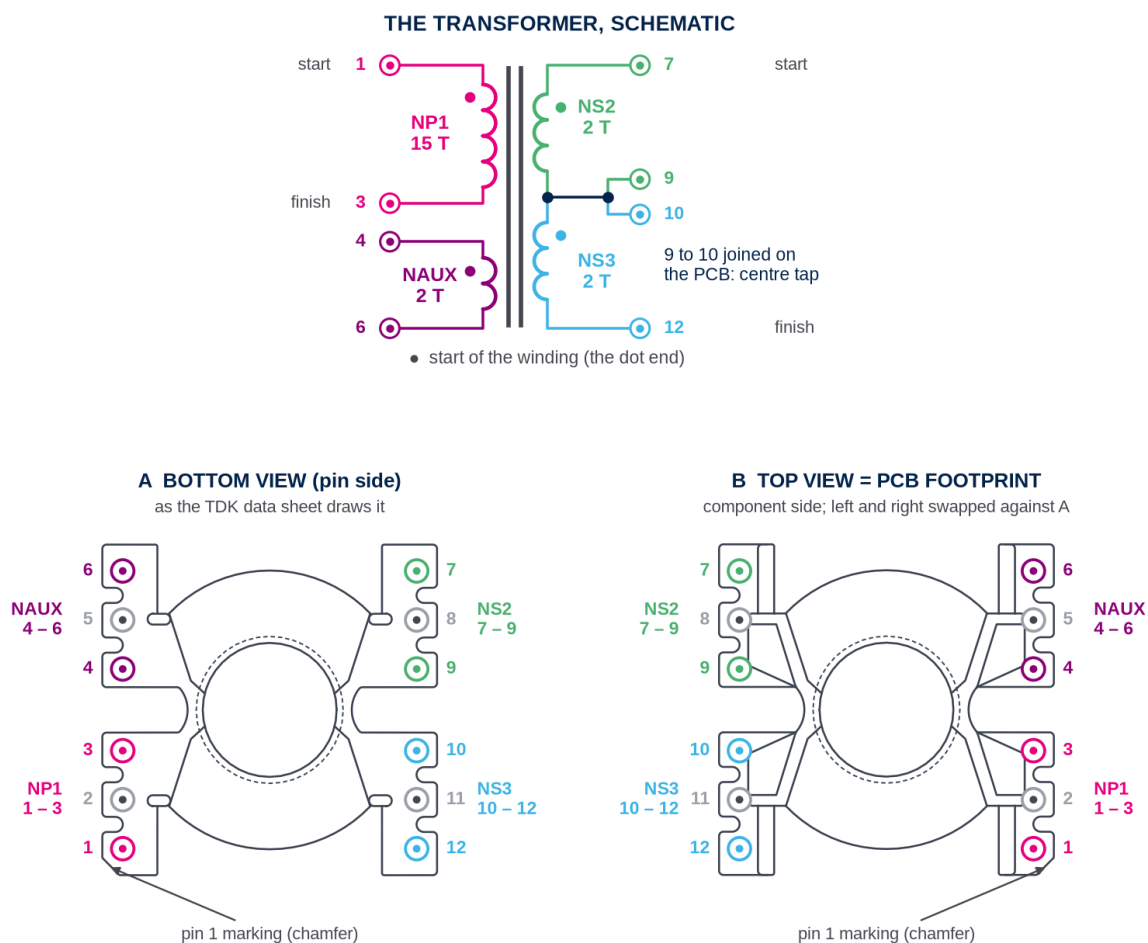


Figure 46: The schematic symbol with its pin numbers, and the same pins on the B65982E coil former (12 pins, rows 45.72 mm apart), its outline traced from the data sheet drawing: (A) from below, the pin side, as the data sheet draws it, and (B) from above, the PCB footprint on the component side. A ring in a winding's colour marks its pins. The dot end is the start of each winding. Pin numbers as the data sheet gives them in its view from below, the pin side: pin 1 at the chamfered corner, 1 to 6 in one row and 7 to 12 in the other, 6 facing 7 and 1 facing 12. The data sheet prints 1, 6, 7 and 12; the pins between are counted. Seen from the component side of the board, left and right are exchanged.

Single-Stage PFC LLC Converter Design

Winding	Pins (start – finish)	Turns	Conductor	Row
NP1 primary	1 – 3	15 T	triple-insulated Litz, 401 × Ø0.10 mm	pin-1 row 1-6
NAUX auxiliary (V_{CC} , ZCD)	4 – 6	2 T	triple-insulated wire over the secondary	pin-1 row 1-6
NS2 secondary	7 – 9	2 T	Litz, 880 × Ø0.10 mm	other row 7-12
NS3 secondary	10 – 12	2 T	the same, the next layer	other row 7-12
Centre tap	9 and 10	—	NS2 finish and NS3 start, joined on the board	other row 7-12
Free	2, 5, 8, 11	—	not connected	both rows

Table 17: Winding-to-pin assignment.

Polarity. The datasheet asks for ZCD positive while the low-side switch of leg 1 (LOUT1) is on. The primary dot is then negative, so the auxiliary start, pin 4 (the dot), goes to ground and its finish, pin 6, to $R_{ZCD,H}$ and to the V_{CC} rectifier. NS2 and NS3 are wound in the same sense; the tap joins the finish of NS2 to the start of NS3.

7.12 Safety insulation of this transformer

The set is a TV sold in the US, the EU, Japan, Korea and China, so each requirement is taken where it is strictest: **reinforced** insulation, because a Class II set needs it and it covers Class I; **5000 m**, because GB 4943.1-2022 assumes that altitude unless the manufacturer specifies otherwise [GB]; and the working voltage at **264 Vac**, the top of the design range. Primary means NP1 and NAUX; secondary means NS2, NS3 and the core, which sits on the secondary side because the primary carries its own insulation (Section 6.17).

The working voltage between the primary winding and the secondary is estimated from the design: the mains neutral and the secondary at earth, the bridge taking the primary ground to the line in one half-cycle, the leg voltages and the C_r swing of the sweep at every point of the line cycle. At 264 Vac it comes to 187 V rms and 373 V peak, below the mains itself, so the mains sets the creepage row. The certifying body measures it; the row taken covers 320 V.

Item	Requirement	Basis
Grade	reinforced, primary to secondary	Class II set [TUV]
Conditions	pollution degree 2, overvoltage category II, material group IIIb, 5000 m	IIIb because the bobbin's tracking index is not known; altitude [GB]
Clearance	$3.0 \times 1.48 = 4.44 \rightarrow \mathbf{4.5\ mm}$	mains transient 2500 V; reinforced at the next step, 4000 V, which reads 3.0 mm [ATIS, TUV]; the temporary-overvoltage route gives only 2.54 mm [PI]; $\times 1.48$ for 5000 m, rounded up [PI]
Creepage	$2 \times 3.2 = \mathbf{6.4\ mm}$	basic at the 320 V row [PI]; reinforced is twice basic [TI]; the same rows a CB report applies [TUV]
Solid insulation	thickness $\geq 0.4\ mm$, or ≥ 2 tape layers	each layer must pass the reinforced electric-strength test [TUV]
Electric strength	3000 V ac for 60 s, or 4242 V dc	type test, NP1 + NAUX against NS2 + NS3 and the core; the value a CB report applies to a Class II 100–240 V product at 5000 m [TUV]; one minute after humidity conditioning [PI]

Single-Stage PFC LLC Converter Design

Item	Requirement	Basis
NP1 and NAUX wire	triple-insulated wire approved as reinforced; Litz for NP1	approvals exist to EN IEC 62368-1 and to IEC 62368-1:2018 Annex J at 1000 V rms and 500 kHz, insulation Class B (130 C), the Litz version on the same approvals [TIW]

Table 18: The insulation requirements and where each number comes from. The sources are listed under References.

Table 19 takes every path between the two sides and names what carries it. The wire carries all of them inside the winding; the partition carries none.

Path	Carried by	Required	Status
NP1 to NS2, NS3 and the core	the triple insulation of the NP1 Litz, one wire from pin to pin	reinforced: triple-insulated wire, rated 1000 V rms against a working voltage of 187 V rms	ok
NAUX to NS2, NS3 and the core	the triple insulation of NAUX, from pin to pin	reinforced: triple-insulated wire, rated 1000 V rms against a working voltage of 187 V rms	ok
the partition between the sections (3.00 mm)	nothing: it sets the leakage only; the wire carries the insulation	none	ok
NS2, NS3 to the core	the same side of the barrier: the core is secondary here	functional	ok
primary pin row to secondary pin row, across the coil former	44.5 mm between the rows (45.72 apart, less one pin)	creepage 6.4 mm, clearance 4.5 mm	ok
primary pin to the core, through the air	6.01 mm: rows 45.72 mm apart, core 32.5 mm deep at most, pin 1.2 mm	clearance 4.5 mm	ok
primary pins and the stripped TIW ends to the core yoke and to secondary leads	the coil former and the lead dress, not this drawing	creepage 6.4 mm, clearance 4.5 mm, or sleeving qualified as reinforced	open
the TIW approval against the frequencies it sees	approved to 500 kHz	every switching frequency, start-up included (477 kHz)	ok

Table 19: What carries the insulation on each path, on the PQ 50/50.

NP1 and NAUX stay in their triple insulation from pin to pin, crossings included; only the ends at the pins are stripped, and that is where the drawing stops. What it cannot settle goes to the certifying body and the vendors:

- The primary pins and the stripped wire ends against the core yoke and the secondary leads: set by the coil former and the lead dress. Reinforced distances, or sleeving qualified as reinforced insulation.
- The triple-insulated Litz itself: one bundle of $401 \times \varnothing 0.10$ mm, $\varnothing 2.9$ mm over its insulation, is the size this design asks for; the 0.2 mm of insulation is assumed, and the size and its approval come from the wire vendor.
- The electric-strength value under the edition the certificate will cite, and the production test voltage and time.
- Clearance for the high-frequency part of the working voltage: the sources quoted do not give that table. A CB report applied the ordinary values at 133 kHz and 588 V peak [UL].
- The TIW approval quoted stops at 500 kHz; the highest switching frequency, the start-up frequency, is 477 kHz, under it. Its Class B (130 C) rating also caps the winding hot spot; the thermal measurement has to show it.

Single-Stage PFC LLC Converter Design

- L_{short} is a field-solution estimate, 24.50 and 25.14 μH for the two halves with the 3.00 mm partition at the assumed winding pitch; the first samples decide.

7.13 Checking the flux, and the specification

Peak flux density on the chosen core, at f_r (Equation 24):

$$B_{pk} = \frac{25.0 \text{ V}}{4 \times 75.03 \text{ kHz} \times 2 \times 332.0 \text{ mm}^2} = \mathbf{126 \text{ mT}}$$

Magnetising peak, over L_μ (it equals $i_{\mu,pk}$):

Equation 33, repeated

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu}$$

$$I_{eq} = \frac{0.1255 \text{ T} \times 15 \times 332.0 \text{ mm}^2}{56.12 \text{ } \mu\text{H}} = \mathbf{11.13 \text{ A}} = i_{\mu,pk}$$

The same current from the output voltage alone (Equation 34), with no core data in it:

$$i_{\mu,pk} = \frac{7.50 \times 25.0 \text{ V}}{4 \times 75.03 \text{ kHz} \times 56.12 \text{ } \mu\text{H}} = \mathbf{11.13 \text{ A}}$$

Saturation test current, the magnetising peak at the OVP2 output voltage:

Equation 35, repeated

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out}}$$

$$I_{sat} = 11.13 \text{ A} \times \frac{31.45 \text{ V}}{25.0 \text{ V}} = 11.13 \times 1.2580 = \mathbf{14.0 \text{ A}}$$

Single-Stage PFC LLC Converter Design

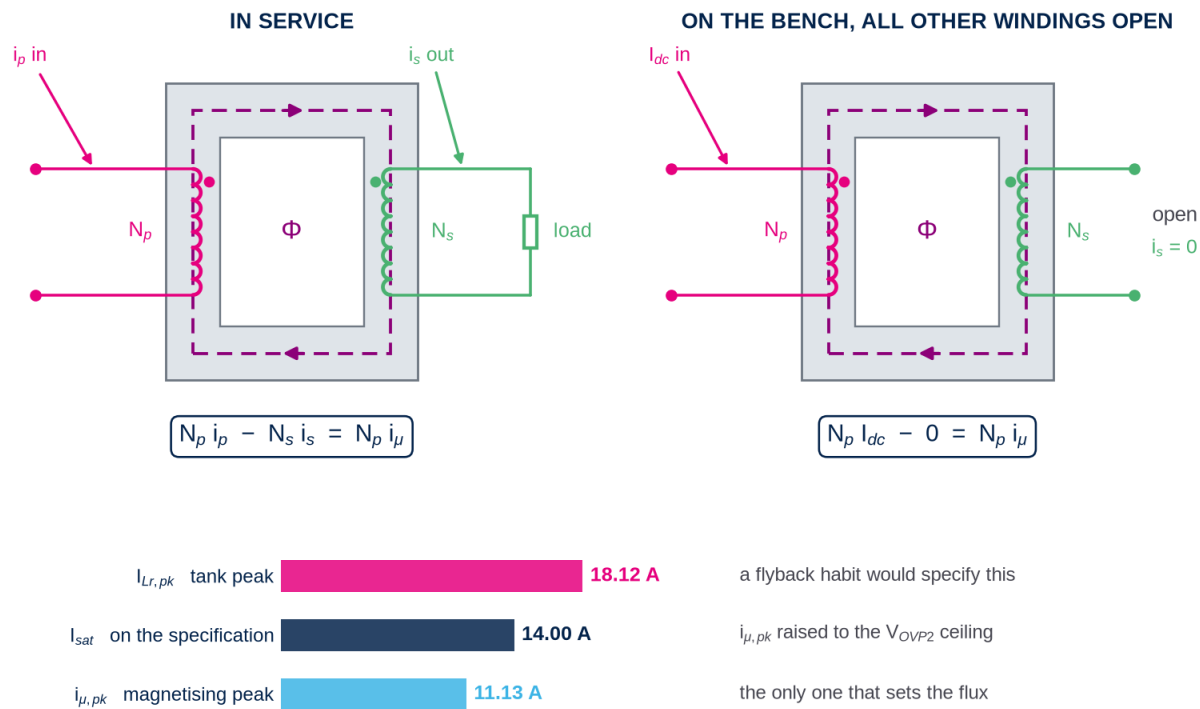


Figure 47: In service the secondary cancels most of the primary ampere-turns; with every other winding open, the whole test current magnetises the core. The design flux is therefore reached at 11.13 A on the bench, not at the 18.12 A tank peak.

Why OVP2, in this design's numbers. OVP1 is set at 28.93 V, 1.157 times the output, and OVP2 at 31.45 V, 1.258 times. At OVP1 the controller keeps switching, so the flux can reach 145 mT with the core still driven; at OVP2 it stops, and 158 mT is the largest flux this core is ever asked to carry. That is the number to hold against the hot B_s of TDK N97 from the material data. The line on the specification reads: DC overlap, inductance at least 90 % of initial at 14 A (14.0 A rounded up), normal temperature (Table 20).

How to run it on this part. LCR meter across the primary (NP1, the pins of Table 17) with both secondaries, the centre tap and the auxiliary winding open. Small signal 100 kHz, 1 V, the same as the leakage test; bias current stepped from zero to past 14 A, inductance recorded at each step. Pass if L at 14 A is at least 0.9 of L at zero, where L at zero is $L_{open} \approx 70.0 \mu H$. Keep each step short: the dc heats the winding through its resistance, and a warm core saturates earlier. A part whose curve knees below 14 A has too little gap or the wrong core, even when L_{open} is on target; the margin against a hot core is in B_{pk} , not in the test.

If there were no OVP2. The current limit would be the only ceiling: OCP1 trips at 22.92 A, 1.64 times the OVP2-based 14.0 A. At the same L_μ that is 258 mT on this core, which has 1.6 times the area the flux target needs. On a core sized to the 0.20 T target it would be 412 mT; holding it to the 252 mT that OVP2 allows would take 1.64 times the core area or N_s . The rule of thumb of 1.3 times $i_{\mu,pk}$ would give 14.5 A, near the OVP2 figure by coincidence and with no event behind it.

Reduced to what a supplier can measure, this is the specification sheet. L_μ is deliberately absent (Section 6.10).

Item	Value	Condition
Core	PQ 50/50, TDK N97, A_L 311 nH	TDK B65981A; coil former B65982E with a 3.0 mm partition added; one centre-leg gap, ground to A_L

Single-Stage PFC LLC Converter Design

Item	Value	Condition
Turns	N_p 15 T; NS2 2 T and NS3 2 T; NAUX 2 T	section winding: NP1 in one section, one triple-insulated Litz bundle a turn, 3 layers of 5 turns; NS2 and NS3 in the other, one Litz bundle a turn, NS2's 2 turns side by side in the first layer and NS3's in the second, wound in the same sense; a 3.00 mm partition between the sections; tape over every layer; one conductor on a pin; the centre tap (pins 9 and 10) joined on the board; NAUX in triple-insulated wire over the secondary
Open-circuit inductance	70.0 μ H, no more than 13.8 % low (Equation 44)	measured across NP1; every other winding (NS2, NS3, NAUX) open; LCR meter 100 kHz, 1 V
Leakage inductance, each half	25.0 μ H $\pm$ 10 %	measured across NP1; NS2 shorted alone, NS3 and NAUX open; then NS3 shorted alone, NS2 and NAUX open; LCR meter 100 kHz, 1 V; estimated at 24.50 and 25.14 μ H (Section 7.10). Both halves must pass: the tank sees the one that conducts
Leakage inductance, both halves shorted	record	as above with NS2 and NS3 both shorted; estimated at 23.09 μ H, lower than each half alone
DC overlap	≥ 90 % of initial inductance	measured across NP1, every other winding open; dc 14 A overlapped on the 100 kHz, 1 V signal; normal temperature (mark 7)
Primary current	10.4 A rms / 18.1 A pk	line-cycle rms (mark 2) and composite peak (mark 1), both at the HB edge (173 Vac, half bridge), full load
Secondary current, each winding	28.5 A rms / 100 A pk	line-cycle rms (mark 5) and peak (mark 4), both at the HB edge, full load
Core area	$A_e \geq 208$ mm ²	holds B_{pk} at or below 0.20 T at f_r (mark 6)
Switching frequency	62 to 118 kHz	at the line peak, full load, from the HB edge to the FB edge; near the zero crossings it falls towards $f_o = 44.8$ kHz
Insulation	reinforced, NP1 + NAUX to NS2 + NS3 and the core: by the triple-insulated wire; at the pins clearance ≥ 4.5 mm, creepage ≥ 6.4 mm	Table 19; the partition carries no insulation
Electric strength	3000 V ac for 60 s (4242 V dc)	NP1 + NAUX against NS2 + NS3 and the core; type test, no breakdown

Table 20: Transformer specification for the worked design.

7.14 The output bank, as sized

Both conditions of Section 6.23; the larger wins. **Ripple**, with Δv as a fraction:

Single-Stage PFC LLC Converter Design

Equation 37, repeated

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

$$C_{out} \geq \frac{657.5 \text{ W}}{2\pi \times 47 \text{ Hz} \times 0.05 \times 625 \text{ V}^2} = \mathbf{71.25 \text{ mF}}$$

Hold-up, starting half a ripple below V_{out} :

Equation 38, repeated

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

$$C_{out} \geq \frac{2 \times 657.5 \text{ W} \times 0.012 \text{ s}}{24.408^2 - 19^2} = \mathbf{67.22 \text{ mF}} \quad (\text{started at } V_{out} : 59.77 \text{ mF})$$

Which wins, at $k = V_{o,min}/V_{out} = 0.760$:

Equation 39, repeated

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

$$0.373 > 0.354 \implies \text{ripple decides}$$

The left side is only 5.3 % larger; allow more than 5.2 % ripple and hold-up would decide instead. The bank fitted is the next assembly above 71.2 mF: $\mathbf{470 \mu F \times 160 = 75.2 \text{ mF}}$.

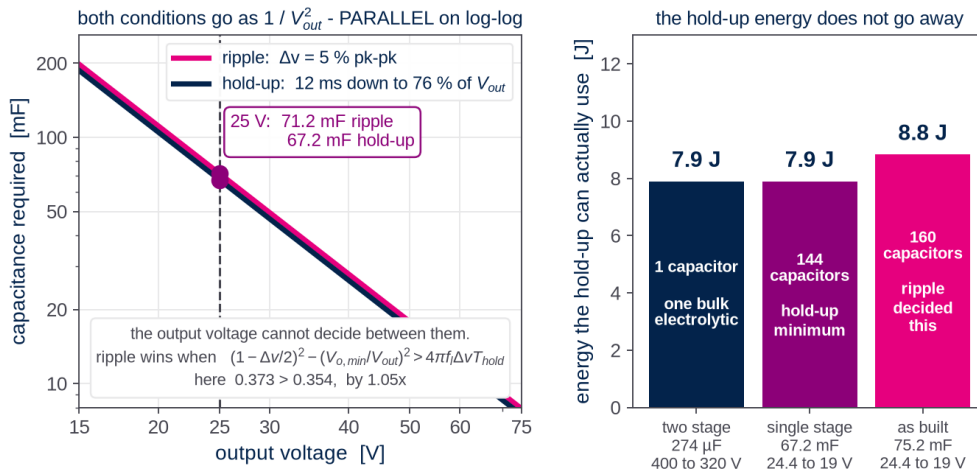


Figure 48: Left: both conditions fall as $1/V_{out}^2$; at 25 V ripple asks for 71.2 mF and hold-up for 67.2 mF. Right: the same 7.9 J of hold-up energy is a 274 μF part on a 400 V bus and 67.2 mF here, 245 times more. That ratio is the price of the architecture.

Single-Stage PFC LLC Converter Design

Quantity	Value	Against
Achieved ripple	1.18 V (4.74 %)	5 % allowed
Achieved hold-up	13.42 ms	12 ms required at 100 Vac, from the ripple trough down to 19 V — 15.10 ms if started at V_{out} instead of at the trough
Ripple current, switching component	24.2 A rms	—
Ripple current, $2f_l$ component	18.6 A rms	orthogonal to the above
Ripple current, total	30.54 A rms	0.19 A in each of 160 parts
Bank ESR	0.1875 m Ω	30 m Ω each, 160 in parallel
Ripple and noise	18.2 mV	with the 10 μ F ceramic bypass

Table 21: What the selected bank then delivers, and what it has to survive.

7.15 Where the power goes

The specification allowed 27.3 W of loss, split three ways before any component existed. The device losses computed afterwards need not agree with that split, and here they do not.

Item	Value	Note
Budget — input bridge	5.52 W	synchronous bridge assumed
Budget — EMI filter	8.34 W	—
Budget — LLC stage	13.42 W	$\eta_{HB} = 98\%$ of $P_{in,LLC}$
Primary conduction, standing device	9.27 W	the half-bridge leg that never switches — the hottest single device
Primary conduction, the two switching positions	9.27 W	S1 and S2 in half bridge, 4.64 W each; conduction only, switching loss not in it
Secondary rectifiers	3.90 W	0.974 W per device, 2 in parallel per leg
Current-sense resistors	2.60 W	line-cycle average; 4.38 W at the worst switching cycle
Output capacitor ESR	0.170 W	all of the ripple current in the switching-frequency ESR; the $2f_l$ part in the higher 120 Hz ESR is not in it ($\tan\delta$ of the fitted part not known)
V_{CC} pass transistor	1.44 W	at the nominal output; the gate drive itself, up to $I_{VCC}V_{CC,reg} = 1.96$ W at f_{Max} , is spent in the drivers and gate resistors and is not in the total
Itemised total	26.65 W	devices only: the transformer windings add 4.6 to 5.6 W (dc and strand loss, Section 7.10); the core loss is not computed here

Table 22: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.

The standing primary device, the low-side switch of the idle leg in half-bridge morphing, carries the whole line-cycle rms continuously, with $R_{DS(on)}$ **hot** at $k_T = 1.9$ times the 25 °C 45 m Ω (Section 6.24):

Single-Stage PFC LLC Converter Design

$$P_{mos,dc} = I_{pri}^2 k_T R_{DS(on),25} = (10.413)^2 \cdot 1.9 \cdot 45 \times 10^{-3} = 9.27 \text{ W}$$

Each secondary rectifier carries the leg current over the 2 devices in parallel, with $k_T = 1.5$ on $3.2 \text{ m}\Omega$ (STL160N10F8, Section 7.19):

$$P_{SR} = \left(\frac{28.50}{2} \right)^2 \cdot 1.5 \cdot 3.2 \times 10^{-3} = 0.974 \text{ W per device}$$

The efficiency assumption is optimistic. The itemised device losses are 26.7 W against the 13.4 W that $\eta_{HB} = 98 \%$ allows, before the transformer. Nothing electrical depends on it; the thermal design does, and that is settled by measurement.

7.16 What the semiconductors have to be

The requirements of Section 6.24 with this design's numbers; the three sections after it check the parts fitted against them.

Item	Requirement
Primary drain-source voltage	$\geq 485 \text{ V}$, so a 600 V class part
Primary body diode	fast recovery: a capacitive excursion turns a device on onto the other's conducting body diode (Section 2.8)
Primary $R_{DS(on)}$ per device	$\leq 27.7 \text{ m}\Omega$ hot to meet the loss budget
Primary $C_{o(tr)}$	small enough that the tank can swing the node inside 220 ns less the driver mismatch
Gate drivers	one half-bridge driver per leg, floating side above the bus peak; bootstrap diode rated like the primary switches
Secondary drain-source voltage	$\geq 80 \text{ V}$ including the centre-tap doubling
Secondary rectifier current	28.50 A rms per leg, 99.7 A peak
Secondary package	check the lead and clip rating, not only the die
V_{CC} pass transistor (NPN)	$\beta \geq 50$ at 144 mA; dissipates 2.12 W at OVP1; blocks 30.8 V plus the winding spike
V_{CC} Zener	15 V $\pm 5 \%$, rated above 393 mW
V_{CC} rectifier and bypass diodes	reverse rating above 30.8 V plus the winding spike

Table 23: Semiconductor requirements for the worked design.

7.17 The primary switches: STO60N045DM9

One ST STO60N045DM9 per position, four in the full bridge: 600 V, 45 m Ω maximum at 25 °C and $V_{GS} = 10 \text{ V}$, a fast-recovery body diode, and a TO-LL package with a separate driver-source pin, which keeps the gate loop out of the power source lead [STO]. Three of its datasheet lines enter the design.

$R_{DS(on)}$ hot. The normalised curve, read from the vector drawing of the datasheet, gives $k_T = 1.9$ at 125 °C, and the standing device spends 9.27 W (Section 7.15). That is a heatsink question: junction to case is 0.4 °C/W, but

Single-Stage PFC LLC Converter Design

the same part on 40 × 40 mm of 2 oz copper is quoted at 22 °C/W, a 204 °C rise at this loss. With the junction at 125 °C and 25 °C ambient, as the workbook takes it, the case-to-ambient path may have 10.4 °C/W, and every 10 °C of real ambient takes 1.08 °C/W off it: the exposed pad has to reach a heatsink.

Output charge. $C_{o(tr)} = 897 \text{ pF}$ is the constant capacitance that charges in the same time as C_{oss} from 0 to 400 V, so it stands for a charge of 359 nC per device. The ZVS corner swings only 245 V, but C_{oss} is at least 86 pF above it, so the full charge over-states the corner by a few per cent and is taken as it is. Two devices swing together, so the midpoint sees

$$C_{HB} = \frac{2 \cdot 897 \text{ pF} \cdot 400}{\sqrt{2} \cdot 173.24} + 100 \text{ pF} = 3029 \text{ pF}$$

in place of the 800 pF assumed before the part was chosen. The dead-time limit on Q moves to $Q_{ZVS2} = 4.22$, still above the gain-peak limit 1.975, so the tank does not change; the swing takes $T_T = 170 \text{ ns}$ of the 220 ns dead time.

Gate charge. The datasheet gives 90 nC from 0 to 10 V and its curve 105 nC at 12 V; between them it is the straight line of Equation 47, $C_g = 7.5 \text{ nF}$. At the highest regulated rail and at V_{CCon} , where start-up begins, the charges $Q_{g,run}$ and $Q_{g,SU}$ are

$$Q_{g,run} = 90 + 7.5 (14.35 - 10) = 122.6 \text{ nC}, \quad Q_{g,SU} = 90 + 7.5 (17 - 10) = 142.5 \text{ nC}$$

Both figures are measured at $V_{DD} = 400 \text{ V}$, hard switched. Under ZVS the Miller share (33 nC) is not drawn, so they over-state the run current; they are kept because the first pulses of a start are hard switched.

7.18 The gate drivers: two L6498LD

The L6790A drives no gate. Each leg gets one L6498LD, the SO-14 version of the ST L6498 high-voltage half-bridge gate driver [L6498], fed from the regulated V_{CC} (Figure 49).

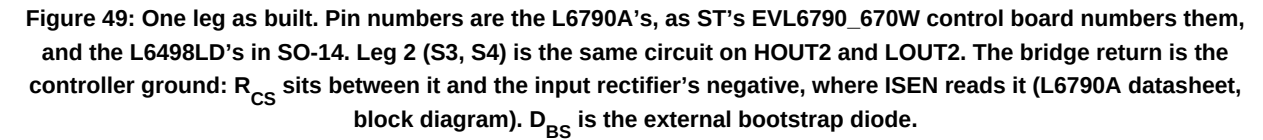


Table 24: L6498LD connections, one per leg (x = 1, 2).

Single-Stage PFC LLC Converter Design

Logic. The controller outputs reach at least 4 V high and at most 0.4 V low; the driver reads high above 2.5 V and low below 0.95 V ($k = 1.60$ and 2.38). Both inputs high gives both outputs low. The L6498 has no enable input, so DRV_EN stays open. One connection needs a second look: at start-up the L6790A drives 300 μA into LOUT2 and reads it against 2.5 V to tell a fixed half bridge from morphing. The LIN pull-down is at least 58 k Ω , which would take 17.4 V, so LOUT2 still reads open ($k_{\text{LOUT2}} = 6.96$).

Floating side. OUT is rated 480 V dc (600 V for under 1 ms); the bus peaks at 373 V ($k = 1.286$).

Supply. The driver is specified from 10 to 20 V. The regulated band, 12.85 to 14.35 V, is inside it, and so is $V_{\text{CCCon}} = 17$ V, the highest the start-up unit takes the rail ($k = 1.176$). The two drivers draw 1.20 mA quiescent, the floating sections included, since they are fed from V_{CC} through the bootstrap.

Bootstrap. The L6498 replaces the bootstrap diode with an integrated switch of about 175 Ω . At f_{Max} the low side is on for $T_{\text{charge}} = 1/(2f_{\text{Max}}) - t_D = 1531$ ns, and Equation 40 gives

$$V_{\text{drop}} = \frac{122.6 \text{ nC} \cdot 175 \Omega}{1531 \text{ ns}} = 14.0 \text{ V}$$

nearly the whole rail, so an external fast diode from VCC to BOOT does the charging, as the driver datasheet allows. It blocks the bus and is rated like the primary switches. With its drop taken as 1.0 V, $C_{\text{BOOT}} = 470$ nF and the longest high-side pulse $1/(2f_{\text{Min}}) = 10.36$ μs , Equation 41 gives

$$\Delta V_{\text{boot}} = \frac{142.5 \text{ nC} + 120 \mu\text{A} \cdot 10.36 \mu\text{s}}{470 \text{ nF}} = 0.31 \text{ V}, \quad V_{\text{BO}} = 12.85 - 1.0 - 0.31 = 11.54 \text{ V}$$

against a recommended 9.3 V minimum ($k = 1.241$). In half-bridge morphing the idle leg holds its low side on, so its bootstrap stays charged; after a burst pause the controller restarts with a pattern that recharges both.

The floor the drivers put under V_{CC} follows from Equation 42:

$$V_{\text{CC, floor}} = \max(8, 10, 9.3 + 1.0 + 0.31) = 10.61 \text{ V}$$

The high-side supply, not the controller's 8 V lockout, decides how far C_{VCC} may run down at start-up (Section 7.28).

Dissipation. The datasheet gives the output only as short-circuit currents, at least 1.4 A source and 1.55 A sink over temperature at 15 V, so the output is taken as 10.7 and 9.7 Ω . With $R_G = 4.7$ Ω and the MOSFET's own 4.5 Ω , the driver keeps $s_{\text{drv}} = 0.525$ of the gate power, and at f_{Max} and the top of the rail (Equation 43)

$$P_{\text{drv}} = 2 \cdot 122.6 \text{ nC} \cdot 14.35 \cdot 285.6 \text{ kHz} \cdot 0.525 + 600 \mu\text{A} \cdot 14.35 = 0.537 \text{ W}$$

per driver, against 1 W for the SO-14 ($k = 1.864$), a 64 $^\circ\text{C}$ rise at 120 $^\circ\text{C/W}$. f_{Max} is reached only at light load; the bound is deliberately high.

Timing. The driver's delay mismatch is at most 30 ns, so the dead time at the gates covers the swing with $(t_D - \text{MT})/T_T = 1.118$. This ratio falls as L_r does: with C_r , L_m and R_T held, it reaches 1 at $L_r \approx 23.5$ μH , just above the low end of the leakage the field solution allows (Section 7.10). c_{HB} is an upper bound and the controller adapts its dead time between 40 and 420 ns, but if the first samples come in low, t_D is the first thing to raise. The MOSFET's own turn-off and turn-on delays (109 and 38 ns) are quoted for hard switching at 28 A and do not apply under ZVS; measure the gate-to-midpoint timing.

Single-Stage PFC LLC Converter Design

Slope. The driver's OUT may slew at most 50 V/ns. Near the top of the swing C_{oss} is only 86 pF, so the commutation current $I_{Lm,pk} = 11.13$ A gives the slope

$$S_{mid} = \frac{11.13 \text{ A}}{2 \cdot 86 + 100 \text{ pF}} = 40.9 \text{ V/ns}$$

($k = 1.222$). Above resonance the switched current can exceed $I_{Lm,pk}$; a small capacitor across each switch is the remedy if the measurement asks for it, paid for in T_T .

What ST's EVL6790_670W does differently. Three things on its schematics are worth copying where they apply. It switches the drivers' supply from DRV_EN through a P-MOSFET (BSS84, level-shifted by a 2N7002), so the drivers draw nothing in idle. Its half-bridge boards feed the bootstrap through 3.3 Ω in series with the diode (STTH1R06), which limits the first charging current. And it puts 10 k Ω from each gate to its source: the L6498 holds its outputs low only once its own V_{CC} is above 3 V, and the resistor covers the time before that.

7.19 The SR MOSFETs: STL160N10F8

Each centre-tap leg has 2 STMicroelectronics STL160N10F8 in parallel [STL]: 100 V STripFET F8 parts in PowerFLAT 5×6, drain on pins 5–8, gate on 4, source on 1–3. Against what Section 7.16 asks of them:

Quantity	Datasheet	Asked	Margin
V_{DS}	100 V	80 V	1.250
$R_{DS(on)}$ max., 25 °C, $V_{GS} = 10$ V	3.2 m Ω	4.93 m Ω	1.539
k_T , 25 to 125 °C (Fig. 13)	1.5	—	—
Loss per device, per leg budget	0.974 W	3.0 W per leg	1.539
Gate charge at the highest drive	81.6 nC	130.4 nC	1.599

Table 25: STL160N10F8 against the secondary requirements (DS14249 Rev 5).

k_T is read from the vector curve of Fig. 13 (1.504 at 125 °C, typical), so $R_{DS(on)}$ is 4.8 m Ω hot and the four devices lose 3.90 W together. On the datasheet's 2s2p board, 16 °C/W, one device would rise 16 °C; the real board decides.

The gate charge is the one that matters for an SR MOSFET, which turns on with its drain already at the body-diode drop: $Q_{g,sync} = 72$ nC at 10 V, typical. The controller drives up to 11.2 V, and past the plateau the gate-charge curve (Fig. 8, read from its vector) rises $C_{g,SR} = 7.96$ nC per volt, so

$$Q_{g,SR} = 72 + 7.96 \cdot (11.2 - 10) = 81.6 \text{ nC}$$

per device. Two on each gate pin present $C_{iss} = 2 \times 5.4 = 10.8$ nF at 50 V, typical: about the 10 nF into which the TEA2095TE datasheet characterises its driver. The body diode, V_{SD} at most 1.2 V at 60 A and Q_{rr} 160 nC typical, conducts only in the dead times the controller leaves.

7.20 Synchronous rectification: TEA2095TE

One NXP TEA2095TE drives both centre-tap legs, each leg 2 MOSFETs in parallel [TEA] (Figure 50). It needs no design here beyond its connection and its limits; its gate drive, regulation and timing are its own.

Single-Stage PFC LLC Converter Design

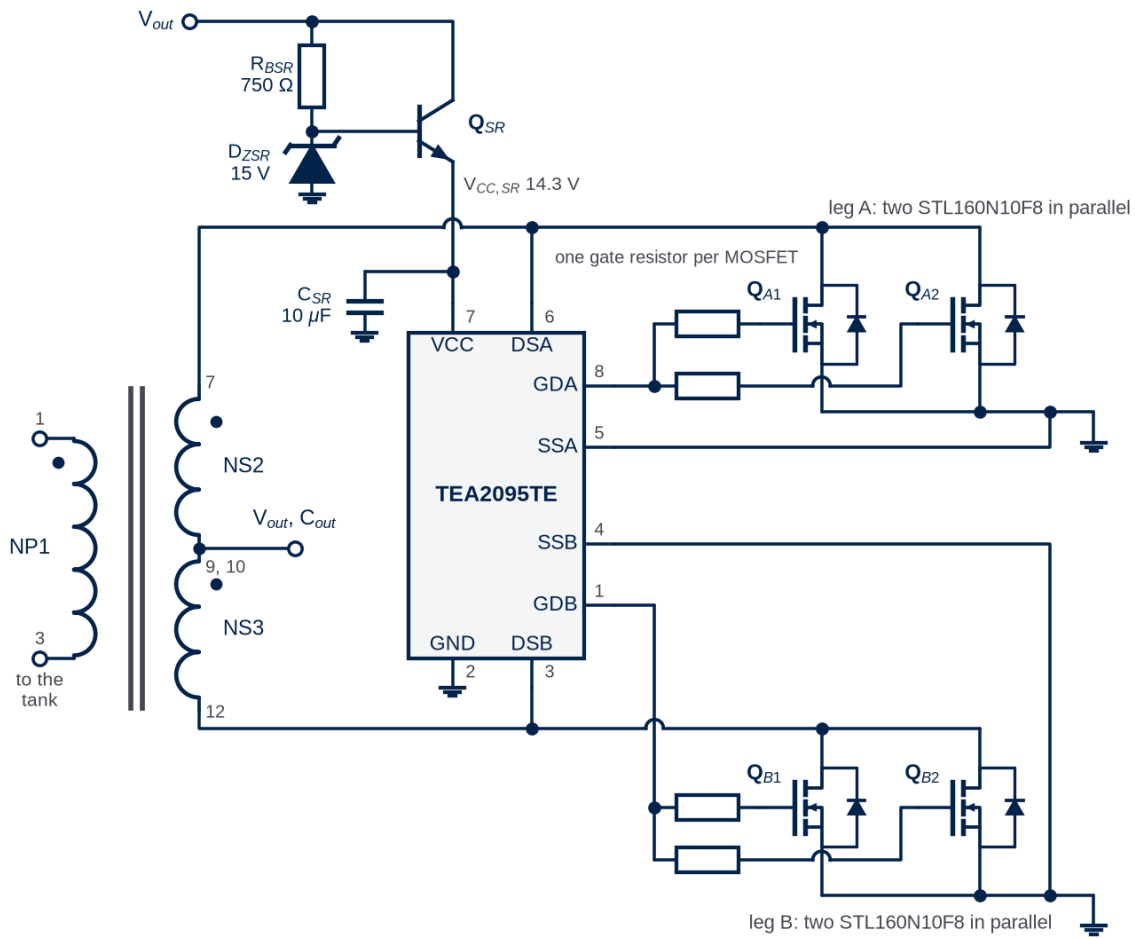


Figure 50: The synchronous rectifier as built: Q_{A1} , Q_{A2} on leg A and Q_{B1} , Q_{B2} on leg B; $V_{CC,SR}$ from the Zener follower R_{BSR} , D_{ZSR} , Q_{SR} with C_{SR} at the pin, sized below. Each MOSFET has its own gate resistor; DSx and SSx are separate sense lines to the drains and the sources of their pair, never along the power ground. Transformer pins as in Figure 46; the centre tap is the output.

Pin	Name	Connection
1, 8	GDB, GDA	gates of the leg B and leg A pairs
2	GND	secondary ground at the SR sources
3, 6	DSB, DSA	drain sense, a separate trace to the drains of each pair
4, 5	SSB, SSA	source sense, a separate trace to the sources of each pair, never along the power ground
7	VCC	the 12 V-class follower below, with C_{SR} at the pin

Table 26: TEA2095TE connections (HSO8).

Its supply. Fed straight from the 25 V output, the controller would drop the difference to its own gate-drive supply, and that drop times the SR gate current is heat in an HSO8. A 12 V-class Zener follower, the circuit of Equation 45 without the bypass diode, takes the drop outside: R_{BSR} from the output to a Zener D_{ZSR} on the base, the collector on the output, the emitter on VCC with $C_{SR} = 10 \mu F$ at the pin. The datasheet gives the gate drive at $V_{CC} = 12 V$,

Single-Stage PFC LLC Converter Design

so the bottom of the band, not its middle, is held there: with a 15 V Zener of $\pm 5\%$ and $V_{BE} = 0.7\text{ V}$,

$$V_{CC,SR} = 15 - 0.7 = 14.30\text{ V}, \quad 13.55 \dots 15.05\text{ V}$$

$k = 1.129$ over 12 V. A 13 V Zener would centre the band near 12 V and put its bottom at 11.65 V, where the gate drive is not specified. The follower is sized for a current, $I_{SR,max} = 150\text{ mA}$, which sets the most gate charge an SR MOSFET may bring. With 4 MOSFETs switched per period and the run frequency bounded at f_{Max} , as for the primary,

$$Q_{g,SR} \leq \frac{150\text{ mA} - 1.05\text{ mA}}{4 \cdot 285.6\text{ kHz}} = 130.4\text{ nC}$$

per MOSFET, against the 81.6 nC of the STL160N10F8 ($k = 1.599$); at f_{Max} the follower then carries

$$I_{SR} = 1.05 + 4 \cdot 81.6\text{ nC} \cdot 285.6\text{ kHz} = 94.2\text{ mA}$$

with $I_{SR,q} = 1.05\text{ mA}$ the controller's own current. The discharge-mode current, 41 mA at most at 12 V and less above, sits well inside ($k = 3.659$). R_{BSR} comes from Equation 48 with the output at the end of hold-up and the same β_{min} and $I_{Z,min}$:

$$R_{BSR} \leq \frac{19 - 15.75}{150/50 + 1} = 812\ \Omega \rightarrow 750\ \Omega$$

Ratings to buy, set at OVP1 (28.93 V): the Zener 308 mW with no load, R_{BSR} 0.287 W, the pass transistor 2.31 W at $I_{SR,max}$ (1.60 W at the nominal output). The transistor share scales with the real gate current, $4 \cdot Q_g \cdot f_{sw}$; it must block the 31.45 V of the OVP2 output. What it buys: at $I_{SR,max}$ the controller's own drop is $(15.05 - 10.4)\text{ V} \times 149\text{ mA} = 0.693\text{ W}$, a 32 °C rise at the 46 °C/W the datasheet quotes on a four-layer board; fed from the output at OVP1 it would be 2.76 W and 127 °C. At the real I_{SR} the two figures are 1.01 W in the pass transistor at the nominal output and 0.433 W in the controller.

Its other limits. V_{CC} is at most 38 V; it would see the 31.45 V OVP2 output only with the pass transistor shorted ($k = 1.208$). Its drain-sense inputs take 120 V, above the 80 V asked of the SR MOSFETs ($k = 1.50$). Its gate drive is held between 10.4 and 11.2 V once its supply is 12 V or more, which clears the 10 V at which the SR $R_{DS(on)}$ is quoted ($k = 1.040$). Below 4.75 V of supply it is off and the body diodes rectify.

Two of its features meet this design. Each gate pin drives two MOSFETs, twice the charge of one, while the datasheet characterises the driver into 10 nF; check the turn-on and turn-off times on the prototype. And after 1.4 s (at least 1.1 s) without rectifier activity it draws 0.4 W / V_{CC} to discharge the output. Through the follower that current comes from the output, so the output gives up $25/14.3 \times 0.4 = 0.70\text{ W}$: after a mains disconnect the 75.2 mF bank, 23.5 J at 25 V, is emptied in about 34 s, but a deep-burst pause longer than 1.1 s at no load would trip it too and add 0.70 W to standby. Measure the longest burst-off time at no load.

ST's EVL6790_670W SR board uses a TEA2096, on the same pinout, with two ISC079N15NM6 per leg: a 0 Ω position for each gate, 220 Ω in series with each drain-sense line, and V_{CC} from an auxiliary rail through 10 Ω with 10 μF at the pin.

7.21 The voltage loop, as built

The method of Sections 6.28 to 6.35, with this converter's numbers, in the order they are found.

Single-Stage PFC LLC Converter Design

Step 1 — the output divider, with $R_I = 100 \text{ k}\Omega$ chosen:

Equation 54, repeated

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$
$$R_O = \frac{V_R R_I}{V_{out} - V_R} = \frac{2.495 \text{ V} \times 100 \text{ k}\Omega}{25.0 \text{ V} - 2.495 \text{ V}} = 11.09 \text{ k}\Omega \rightarrow \mathbf{11 \text{ k}\Omega}$$

$$V_{out} = 2.495 \text{ V} \times \left(1 + \frac{100}{11} \right) = \mathbf{25.18 \text{ V}}$$

Step 2 — the plant. With $K_{pwr} = 2K_{HV}/(K_M K_{FF})$, the slope of Equation 15, $V_{FB} = K_{pwr} R_{CS} P_{in,LLC} = 0.167 \times 24 \text{ m}\Omega \times 670.9 \text{ W} = 2.689 \text{ V}$ (Equation 15). Then

Equation 50, repeated

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$
$$G_o = \frac{657.5 \text{ W}}{25.0 \text{ V} \times 2.689 \text{ V} \times 75.2 \text{ mF}} = \mathbf{130.1 \text{ rad/s}}, \quad f_{cto} = \frac{130.1}{2\pi} = 20.7 \text{ Hz}$$

Step 3 — the gain allowed at $2f_l$, for 5 % third harmonic:

Equation 51, repeated

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_l C_{out}}$$
$$\Delta V_{loop} = \frac{657.5 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} = \mathbf{1.184 \text{ V}}$$

Equation 53, repeated

$$G_{EA}(2f_l) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$
$$G_{EA}(2f_l) \leq \frac{4 \times 2.689 \text{ V} \times 0.05}{1.184 \text{ V}} = \mathbf{0.4541}$$

Step 4 — the K factor, for a target phase margin of 50° ($\tan 50^\circ = 1.1918$) and $\alpha_v = 1.2$:

Single-Stage PFC LLC Converter Design

Equation 59, repeated

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$
$$K_v = \frac{1}{2 \times 1.2} \left[(1 + 1.2^2) \times 1.1918 + \sqrt{(1 + 1.2^2 \times 1.1918)^2 + 4 \times 1.2^2} \right] = \mathbf{2.7218}$$

Step 5 — the compensator gain constant:

Equation 60, repeated

$$EA_o = \frac{2\pi (2f_i) G_{EA} (2f_i)}{K_v^2}$$
$$EA_o = \frac{2\pi \times 94 \text{ Hz} \times 0.4541}{2.7218^2} = \mathbf{36.20 \text{ rad/s}}$$

Step 6 — zero and pole, with $\Gamma_v = 0.744 \times 264.0 / 173.24 = 1.1338$:

Equation 61, repeated

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v EA_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$
$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{130.06 \times 2.7218 \times 36.20}{1.1338}} = \mathbf{16.92 \text{ Hz}}$$
$$f_p = 2.7218 \times 16.92 = \mathbf{46.06 \text{ Hz}}, \quad f_z = \frac{16.92}{2.7218} = \mathbf{6.217 \text{ Hz}}$$

Step 7 — the bias parts, with the TL431 minimum current 0.8 mA, $V_Z = 12 \text{ V}$, $V_{Fo} = 1.15 \text{ V}$, $I_{FB, \text{steady}} = 150 \text{ }\mu\text{A}$, $I_{FB, \text{max}} = 300 \text{ }\mu\text{A}$, $\text{CTR}_s = 0.37$, $\text{CTR}_m = 0.53$:

Equation 55, repeated

$$R_P \leq \frac{V_{Fo}}{I_{min}}$$
$$R_P \leq \frac{1.15 \text{ V}}{0.8 \text{ mA}} = 1.437 \text{ k}\Omega \rightarrow \mathbf{1.3 \text{ k}\Omega} \text{ (rounded down)}$$

Single-Stage PFC LLC Converter Design

Equation 58, repeated

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,steady}/CTR_s}$$
$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,max}/CTR_m}$$
$$R_{B,max} = \frac{12 - (2.495 + 1.15) \text{ V}}{1.15/1.3 \text{ mA} + 150 \mu\text{A}/0.37} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 0.405 \text{ mA}} = \mathbf{6.48 \text{ k}\Omega}$$
$$R_{B,min} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 300 \mu\text{A}/0.53} = \mathbf{5.76 \text{ k}\Omega} \rightarrow R_B = \mathbf{6.2 \text{ k}\Omega}$$

Step 8 — the compensation parts, with $R_{FB} = 35 \text{ k}\Omega$, $C_{opto} = 3 \text{ nF}$ and the high-frequency pole placed at 1000 Hz:

Equation 62, repeated

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_I R_B E A_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$
$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$
$$C_{Fo} = \frac{6.217}{46.06} \cdot \frac{0.37 \times 35 \text{ k}\Omega}{100 \text{ k}\Omega \times 6.2 \text{ k}\Omega \times 36.20} = 77.9 \text{ nF} \rightarrow \mathbf{82 \text{ nF}}$$
$$C_F = 82 \text{ nF} \times \left(\frac{46.06}{6.217} - 1 \right) = 526 \text{ nF} \rightarrow \mathbf{560 \text{ nF}}$$
$$R_F = \frac{1}{2\pi \times 6.217 \text{ Hz} \times 560 \text{ nF}} = 45.7 \text{ k}\Omega \rightarrow \mathbf{47 \text{ k}\Omega}$$
$$C_{fx} = \frac{1}{2\pi \times 1000 \text{ Hz} \times 35 \text{ k}\Omega} - 3 \text{ nF} = 1.547 \text{ nF} \rightarrow \mathbf{1.55 \text{ nF}}$$

Step 9 — what the standard parts give:

Single-Stage PFC LLC Converter Design

Equation 57, repeated

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \quad [\text{rad/s}], \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

$$EA_o = \frac{0.37 \times 35 \text{ k}\Omega}{(560 + 82) \text{ nF} \times 100 \text{ k}\Omega \times 6.2 \text{ k}\Omega} = \mathbf{32.53 \text{ rad/s}}$$

$$f_z = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 560 \text{ nF}} = \mathbf{6.047 \text{ Hz}}$$

$$C_{ser} = \frac{560 \times 82}{560 + 82} \text{ nF} = 71.5 \text{ nF}, \quad f_p = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 71.5 \text{ nF}} = \mathbf{47.34 \text{ Hz}}$$

$$f_{px} = \frac{1}{2\pi \times 35 \text{ k}\Omega \times (3 + 1.55) \text{ nF}} = \mathbf{999 \text{ Hz}}$$

Step 10 — crossover and phase margin. $G_o EA_o = 4231$, so the iteration starts at $\omega_c = 65.0 \text{ rad/s}$ and settles at

Equation 64, repeated

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

$$\omega_c = \mathbf{110.4 \text{ rad/s}}, \quad f_c = \frac{110.4}{2\pi} = \mathbf{17.57 \text{ Hz}}, \quad |T(\omega_c)| = 1.0003 \quad (\text{check : } 1)$$

Equation 65, repeated

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

$$\Phi_M = \arctan \frac{110.4}{38.0} - \arctan \frac{110.4}{297.5} - \arctan \frac{110.4}{6279} = 71.0^\circ - 20.4^\circ - 1.0^\circ = \mathbf{49.6^\circ}$$

Step 11 — the gain margin:

Equation 66, repeated

$$f_{180} = \sqrt{f_p f_{px} - f_z (f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})|$$

$$f_{180} = \sqrt{47.34 \times 999 - 6.047 \times (47.34 + 999)} \text{ Hz} = \mathbf{202 \text{ Hz}}$$

$$|T(f_{180})| = 0.0196, \quad GM = -20 \log_{10} 0.0196 = \mathbf{34.2 \text{ dB}}$$

Single-Stage PFC LLC Converter Design

Step 12 — the third harmonic of the built loop, at $2f_l = 94$ Hz ($\omega = 590.6$ rad/s):

$$G_{EA}(2f_l) = \frac{32.53}{590.6} \cdot \frac{\sqrt{1 + (590.6/38.0)^2}}{\sqrt{1 + (590.6/297.5)^2} \sqrt{1 + (590.6/6279)^2}} = \mathbf{0.3843} \quad (\leq 0.4541)$$

Equation 52, repeated

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

$$D_3 = \frac{0.3843 \times 1.184 \text{ V}}{4 \times 2.689 \text{ V}} = \mathbf{4.23 \%} \quad (\text{budget } 5 \%)$$

Step 13 — the feedback ripple at the burst point, with burst entry at 75 W:

Equation 67, repeated

$$\Delta V_{FB} = \frac{P_{in, BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

$$\Delta V_{FB} = \frac{75 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} \times 0.3843 = \mathbf{51.9 \text{ mV}}$$

Equation 68, repeated

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

$$\Delta R_{BM} = \frac{51.9 \text{ mV}}{2 \times 0.01 \text{ V/k}\Omega} = \mathbf{2.60 \text{ k}\Omega}, \quad R_{BM} = 30 - 2.60 = \mathbf{27.4 \text{ k}\Omega}$$

The fitted 30 k Ω is kept for now: the burst entry it is set from is an assumption (Table 6), so R_{BM} is traded against this ripple once the standby power has been measured (Appendix 10.4).

Single-Stage PFC LLC Converter Design

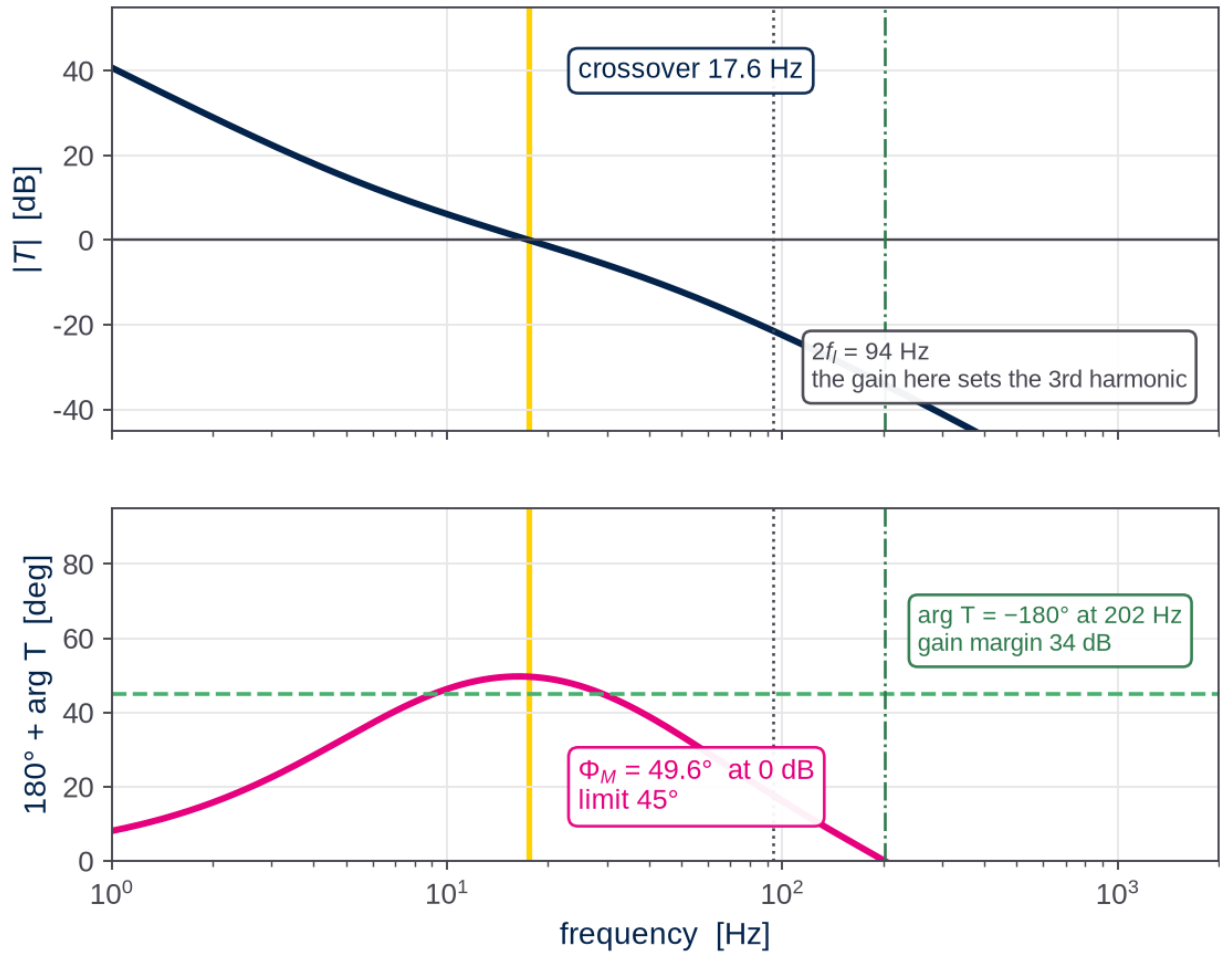


Figure 51: $|T|$ and $180^\circ + \arg T$ of the loop as built. The second trace is the phase margin only at the crossover; the gain margin is read at f_{180° .

Quantity	Value	Against
Divider R_I / R_O	100 / 11 k Ω	regulates to 25.18 V
Bias R_P, R_B	1.3 k Ω , 6.2 k Ω	$R_P \leq 1.44$ k Ω ; R_B in 5.76 to 6.48 k Ω
Compensation C_{F0}, C_F, R_F, C_{fx}	82 nF, 560 nF, 47 k Ω , 1.55 nF	zero 6.05 Hz, pole 47.3 Hz, pole 999 Hz
Crossover f_c	17.57 Hz	the 15 to 20 Hz this topology lands in
Phase margin	49.6°	target 50°, floor 45°
Gain margin	34.2 dB at 202 Hz	6 dB floor, 10 dB comfortable
Third harmonic on the input current	4.23 %	budget 5 %
Feedback ripple at the burst point	52 mV	clears the burst threshold if R_{BM} moves from the fitted 30 k Ω to 27.4 k Ω

Table 27: The voltage loop, as built.

7.22 The parts around the controller

The results; the following sections derive each value in the order it has to be done.

Single-Stage PFC LLC Converter Design

Part	Value	Result
C_T / R_T	470 pF / 21.5 kΩ	f_{Min} 48.3 kHz, f_{Max} 285.6 kHz
R_{CS}	24.0 mΩ (120 mΩ × 5 in parallel)	OCP1 at 1.265 × the composite peak
R_{CFG}	30 kΩ	V_{BO} 120 V peak (84.8 Vac rms), morphing enabled
R_{BM}	30 kΩ	burst-mode entry point
ZCD divider	220 kΩ / 19 kΩ	OVP1 28.93 V, OVP2 31.45 V
C_{in}	2200 nF film	about 3.2 nF/W — there is no bulk capacitor
V_{CC} regulator	Zener 15 V, R_{BZ} 560 Ω, C_{VCC} 1000 μF	V_{CC} 13.60 V (12.85 to 14.35 V) from N_{aux} 2 T
Compensation	82 nF / 560 nF / 47 kΩ / 1.55 nF	f_c 17.57 Hz, Φ_M 49.6°

Table 28: Controller network for the worked design.

7.23 The oscillator: C_T first, then R_T

The VCO charges C_T from V_{ref}/R_T plus the error-amplifier current I_{EA} up to $V_{\text{ref}} = 1.5$ V, then adds a fixed idle time:

Equation 70

$$\frac{T_{\text{sw}}}{2} = \frac{C_T V_{\text{ref}}}{\frac{V_{\text{ref}}}{R_T} + I_{\text{EA}}} + T_{\text{idle}}$$

I_{EA} is the only control input. More current, higher frequency, less power. C_T is chosen first because it sets the span, R_T after it because it sets the floor:

Equation 71

$$C_{T,\text{max}} = \frac{I_{\text{EA,max}}}{2V_{\text{ref}}} \cdot \frac{(1 - 2T_{\text{idle}}f_{\text{sw,des}})(1 - 2T_{\text{idle}}f_{\text{sw,min}})}{f_{\text{sw,des}} - f_{\text{sw,min}}}, \quad C_{T,\text{min}} = \frac{1}{R_{T,\text{max}}} \left(\frac{1}{2f_{\text{sw,min}}} - T_{\text{idle}} \right)$$

Equation 72

$$R_{T,\text{ceil}} = \frac{1}{C_T} \left(\frac{1}{2f_{\text{sw,min}}} - T_{\text{idle}} \right)$$

with $f_{\text{sw,min}} = f_o = 44.8$ kHz; $f_{\text{sw,des}} = 177.7$ kHz, the frequency the oscillator span is designed to: 1.5 times the highest frequency the converter runs at, 118.5 kHz (the full-bridge edge at the line peak), and no more than the 225 kHz specification; $I_{\text{EA,max}} = 400$ μA, $V_{\text{ref}} = 1.5$ V and $T_{\text{idle}} = 249$ ns:

$$C_{T,\text{max}} = \frac{400 \times 10^{-6}}{2 \cdot 1.5} \cdot \frac{(1 - 2 \cdot 249 \times 10^{-9} \cdot 177.7 \times 10^3)(1 - 2 \cdot 249 \times 10^{-9} \cdot 44.8 \times 10^3)}{(177.7 - 44.8) \times 10^3} = 894 \text{ pF}$$

and the floor condition, f_{Min} brought down to f_o with R_T at its 30 kΩ maximum, gives $C_{T,\text{min}} = 363$ pF. The datasheet allows 270 to 1000 pF, so the window is 363 to 894 pF, and the part is taken from its middle: **$C_T = 470$ pF, C0G**. Then

Single-Stage PFC LLC Converter Design

$$R_{T,ceil} = \frac{1}{470 \times 10^{-12}} \left(\frac{1}{2 \cdot 44.8 \times 10^3} - 249 \times 10^{-9} \right) = 23.20 \text{ k}\Omega \rightarrow 21.5 \text{ k}\Omega$$

$R_{T,ceil}$ is a maximum. f_{Min} falls as R_T grows, so rounding up drops the clamp below f_o . Round it **down**.

The selected pair produces

Equation 73

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2 \left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle} \right)}$$

$f_{Min} = 1/[2(470 \text{ pF} \times 21.5 \text{ k}\Omega + 0.25 \text{ }\mu\text{s})] = \mathbf{48.29 \text{ kHz}}$, which must clear f_o .

Quantity	Value	Limit	Margin
f_{Min}	48.3 kHz	above $f_o = 44.8 \text{ kHz}$	1.077
f_{Max}	285.6 kHz	above the highest operating $f_{sw} = 118.5 \text{ kHz}$ (FB edge, line peak)	2.410
f_{SU}	476.6 kHz	below the 675 kHz silicon ceiling	1.416
$R_T C_T$	10.11 μs	2.5 to 12 μs	4.042 / 1.188
R_T	21.5 k Ω	5 to 30 k Ω	4.300 / 1.395
C_T	470 pF	270 to 1000 pF	1.741 / 2.128

Table 29: Oscillator: what the two parts produce, and the limits each result has to clear.

T_{idle} is the weakest number in this chapter. The draft datasheet implies 700 ns in its frequency expressions and 350 ns in its $C_{T,max}$ expression, and its Table 2 back-solves to about 250 ns; this design uses 249 ns. At 700 ns f_{Min} would be 46.3 kHz, only just above $f_o = 44.8 \text{ kHz}$, and f_{Max} 227.1 kHz against the 118.5 kHz the FB edge needs. Measure it first.

7.24 The current-sense resistor, which sets three things at once

Two conditions, the smaller wins:

Equation 74

$$R_{CS,max1} = \frac{16.8 \text{ }\Omega \cdot \text{W}}{P_{in}} \quad R_{CS,max2} = \frac{0.55 \text{ V}}{I_{Lr,pk}}$$

The first is the maximum-power law (Section 4.6), the second the OCP1 trip point. The first is written with P_{in} , as the datasheet writes it, not with the $P_{in,LLC}$ the pin senses; that is 2 % conservative:

Single-Stage PFC LLC Converter Design

$$R_{CS,max1} = \frac{16.8}{684.8} = 0.02453 \, \Omega = 24.53 \, \text{m}\Omega, \quad R_{CS,max2} = \frac{0.55}{18.12} = 0.03036 \, \Omega = 30.36 \, \text{m}\Omega$$

so the power law limits. Dissipation at the worst switching cycle fixes how many resistors share it:

$$P_{CS} = I_{pri,rms}^2 R_{CS} = (13.50)^2 \cdot 24.0 \times 10^{-3} = 4.38 \, \text{W} \Rightarrow N = 5 \text{ in parallel}$$

5 × 120 mΩ in parallel give **R_{CS} = 24.0 mΩ**.

R_{CS} sits in the bridge return, so R_{CS,max2} uses the composite peak. I_{trafo,pk} = 16.58 A instead of I_{Lr,pk} = 18.12 A would loosen the over-current protection by about 9 %.

Result	Value	Against
Maximum input power	700.0 W	P _{in} = 684.8 W
OCP1 trip	22.92 A	composite peak 18.12 A, margin 1.265
OCP2 trip	31.25 A	switching stops at once; restart after 50 μs at f _{SU}
Sense dissipation, worst switching cycle	4.38 W total, 0.875 W each	1 W parts, margin 1.142

Table 30: What the selected R_{CS} then fixes.

7.25 Burst mode: one resistor, read once at power-up

Equation 75

$$R_{BM} = 16.7 \frac{\text{k}\Omega}{\text{V}^2} R_{CS} P_{in,BM}, \quad V_{BM,eq} = 0.01 \frac{\text{V}}{\text{k}\Omega} R_{BM} + 0.5 \, \text{V}$$

Burst starts at P_{in,BM} = 11 % × 684.8 W = 75.3 W:

$$R_{BM} = 16.7 \cdot 24.0 \times 10^{-3} \cdot 75.3 = 30.1 \, \text{k}\Omega \rightarrow 30 \, \text{k}\Omega, \quad V_{BM,eq} = 0.01 \times 30 + 0.5 = 0.800 \, \text{V}$$

Valid range 15 to 140 kΩ; tied to ground, burst mode is off. Check this threshold against the feedback ripple (Section 6.35) or the converter chatters in and out of burst.

7.26 Brown-out and bridge configuration: the CFG pin

One resistor, two jobs, both read at power-up:

Equation 76

$$V_{BO,pk} = R_{CFG} \times 4 \frac{\text{V}}{\text{k}\Omega}, \quad V_{BO,rms} = \frac{V_{BO,pk}}{\sqrt{2}}$$

Brown-out is compared against the *peak* of the mains, so the √2 is easy to drop. Brown-out just below the lowest input voltage:

Single-Stage PFC LLC Converter Design

$$R_{CFG,max} = \frac{\sqrt{2} \cdot 90}{4} = 31.82 \text{ k}\Omega \rightarrow 30 \text{ k}\Omega$$

That is a maximum, so it rounds down; the selected part sets

$$V_{BO,rms} = \frac{30 \cdot 4}{\sqrt{2}} = 84.85 \text{ Vac}$$

against a 90 Vac minimum, margin 1.061. Rounding up would stop the converter starting at the lowest input voltage.

R _{CFG}	LOUT2	Configuration
15 kΩ	open	morphing, fixed brown-out: off below 60 V peak, on above 70 V peak
15 to 47 kΩ	open	morphing, adjustable brown-out — this design
47 to 100 kΩ	open	fixed full bridge
15 kΩ	to GND	fixed half bridge, split C _r , fixed brown-out
15 to 100 kΩ	to GND	fixed half bridge, split C _r , adjustable brown-out

Table 31: What R_{CFG} and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

For universal input the window is 15 kΩ to 31.8 kΩ; the 47 kΩ morphing limit never applies. Hang nothing on LOUT2: a pull-down under about 8 kΩ reads as fixed half bridge.

7.27 Output sensing and over-voltage: the ZCD divider

The divider ratio alone sets both over-voltage thresholds; the absolute values only set the bias current:

Equation 77

$$R_{ZCD,L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD,H} = R_{ZCD,L} \left(\frac{n_{aux}}{n_{sec}} \frac{V_{OVP1,out}}{2.3 \text{ V}} - 1 \right)$$

Equation 78

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left(\frac{R_{ZCD,H}}{R_{ZCD,L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1}$$

OVP1 10 % above the output is 27.50 V. The lower resistor comes from the bias current, 120 μA, the upper from the ratio, with $n_{aux}/n_{sec} = 1.0$:

$$R_{ZCD,L} = \frac{2.3}{120 \times 10^{-6}} = 19.17 \text{ k}\Omega \rightarrow 19 \text{ k}\Omega$$

Single-Stage PFC LLC Converter Design

$$R_{ZCD,H} = 19 \left(\frac{1.0 \cdot 27.5}{2.3} - 1 \right) = 208.2 \text{ k}\Omega \rightarrow 220 \text{ k}\Omega$$

The upper resistor is the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest. The fitted pair gives

$$V_{OVP1} = \frac{2.3}{1.0} \left(\frac{220}{19} + 1 \right) = 28.93 \text{ V}, \quad V_{OVP2} = \frac{2.5}{2.3} \cdot 28.93 = 31.45 \text{ V}$$

Start-up hands over to the loop when the ZCD pin reaches 1.36 V, the datasheet's start-up end threshold. Through the same divider that is 1.36/2.3 of the OVP1 output voltage: 17.11 V of output.

Swapping the two resistors makes OVP1 trip below a volt of output, so the converter never starts. Reversed winding polarity makes the bridge hard switch from the first pulse.

Wired straight to V_{CC} , this winding would put 31.45 V on the pin at OVP2, over its 25 V rating; the ST tool's ratio test reads 0.836. The regulator of the next section is why that does not matter here.

7.28 V_{CC} from the auxiliary winding, as built

There is no external rail. The auxiliary winding is 2 turns of triple-insulated wire over the secondary, $n_{aux} = 1.0$, so it sits at the output voltage: 25.0 V on the winding and 24.3 V on its capacitor after the diode. With a 15 V Zener of $\pm 5\%$ and $V_{BE} = V_D = 0.7 \text{ V}$ assumed, Equation 45 gives

$$V_{CC,reg} = 15 - 0.7 - 0.7 = 13.60 \text{ V}, \quad 12.85...14.35 \text{ V}$$

The bottom of the band clears $V_{CC,HVSUon} = 12 \text{ V}$ by 1.071; the top stays under the 25 V operating limit by 1.742. The load is the controller, 3 mA, the two drivers, 1.20 mA, and 4 switch positions at $Q_{g,run} = 122.6 \text{ nC}$ (Section 7.17), bounded at f_{Max} (Equation 46):

$$I_{VCC} = 3 \text{ mA} + 1.20 \text{ mA} + 4 \cdot 122.6 \text{ nC} \cdot 285.6 \text{ kHz} = 144.3 \text{ mA}$$

The feed resistor comes from Equation 48 at the end of hold-up, with $\beta_{min} = 50$ and $I_{Z,min} = 1 \text{ mA}$ set as requirements on the parts:

$$R_{BZ} \leq \frac{1.0 \cdot 19 - 0.7 - 15.75}{144.3/50 + 1} = 0.656 \text{ k}\Omega \rightarrow 560 \text{ }\Omega$$

What the two parts burn is set at OVP1, where the capacitor stays highest, 28.23 V: the Zener, with no load to share the feed current, 393 mW; the pass transistor, at I_{VCC} , 2.12 W (1.44 W at the nominal output, the figure that goes into the loss account). Both are ratings to buy, not margins. The transistor must also block 30.75 V at OVP2 plus any spike the winding carries, which only a measurement gives.

Start-up. In full bridge the oscillator runs at $f_{SU} = 476.6 \text{ kHz}$ for its first 1 ms only and then stays under f_{Max} . With the gate charge at V_{CCon} the drivers take, and lose B per volt as C_{VCC} runs down,

Single-Stage PFC LLC Converter Design

$$I_{VCC,SU} = 3 + 1.20 + 4 \cdot 142.5 \text{ nC} \cdot 285.6 \text{ kHz} = 167.0 \text{ mA}$$

$$B = 4 \cdot 7.5 \text{ nF} \cdot 285.6 \text{ kHz} = 8.57 \text{ mA/V}$$

so 124.1 mA at $V_{CC,HVSUon}$ and 112.2 mA at the floor $V_{CC,floor} = 10.61 \text{ V}$, which the high-side driver supply sets (Section 7.18). The first millisecond at f_{SU} costs $\Delta Q_{OSC} = 1 \text{ ms} \cdot 4 \cdot 142.5 \text{ nC} \cdot (476.6 - 285.6) \text{ kHz} = 108.9 \text{ } \mu\text{C}$. The winding holds the pin at the floor once the output reaches $(V_{CC,floor} + 3V_D + I_{VCC,SU} R_{BZ}/\beta_{min})/n_{aux} = 14.58 \text{ V}$. Equation 49 then gives

$$t_{hand} = \frac{75.2 \cdot 14.58}{26.3} = 41.7 \text{ ms}$$

$$C_{VCC} \geq \frac{8.57 \cdot (41.7 + 108.9/167.0)}{\ln \frac{167.0}{124.1} + \ln \frac{124.1 - 13}{112.2 - 13}} = 884 \text{ } \mu\text{F} \rightarrow 1000 \text{ } \mu\text{F}$$

Units: mA/V times ms, over a pure number, is μF ; μC over mA is ms.

The hand-over falls well inside the 120 ms the start-up unit stays available. The price is the delay before the first pulse, 1.89 s at 230 Vac, during which the start-up unit carries the whole charge; check its temperature on the prototype.

t_{hand} assumes the set holds its load off until the rail is up, as a TV sequences its panel. Any load during start-up stretches it, and C_{VCC} with it.

Two wiring rules for the start-up path:

- **HVSU connects ahead of the bridge**, on the ac side, through one 1000 V diode per line; behind the bridge X-capacitor discharge and brown-out detection both stop working.
- **100 nF at the V_{CC} pin**, beside C_{VCC} .

Single-Stage PFC LLC Converter Design

7.29 Controller pin rules

Pin	Rule
ISEN	No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.
CFG, BM	No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.
DRV_EN, RT	No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault. DRV_EN stays open here: the L6498 has no enable input.
LOUT2	No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge; the driver input's own 58 k Ω is far enough above that.
HVSU	Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working.
ZCD	Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.
HOUTx, LOUTx	Logic-level outputs, not gate drivers: each pair drives one L6498LD (Section 7.18).

Table 32: Pin rules that must not be broken.

7.30 Protections, and where each threshold is set

Each threshold is fixed by a resistor already chosen, so a change made for one reason moves a threshold set for another.

Protection	Set by	This design
Brown-out, on the rectified mains	R_{CFG} , read at power-up	V_{BO} 120 V peak = 84.8 Vac rms, clear of the 90 Vac minimum by 1.061
First-level over-current, OCP1: raises the switching frequency	R_{CS} — the same resistor as the maximum-power law	22.92 A against a composite peak of 18.12 A, a margin of 1.265
Second-level over-current, OCP2	R_{CS} , fixed ratio to OCP1	31.25 A
Output over-voltage, OVP1	the ZCD divider on the auxiliary winding	28.93 V, above the ripple crest and below OVP2
Second-level over-voltage, OVP2	the same divider	31.45 V
Capacitive-mode protection	internal, on the ISEN pin: ACP-soft raises the frequency, ACP-hard stops for 50 μ s and restarts at the maximum frequency	the last line of defence only — the frequency floor f_{Min} is the first
Burst mode at light load	R_{BM} , read at power-up	30 k Ω ; the $2f_l$ feedback ripple is 52 mV, so see Section 6.35
Loss of ZVS at the worst corner	not a protection — a design margin	swept, not estimated; Section 6.5

Table 33: Protections and the component that sets each one.

Single-Stage PFC LLC Converter Design

7.31 System design rules

The same material as a checklist.

- **Design the tank at the equivalent range**, 173.2 to 332.3 Vac, not at the mains range.
- **Expect λ near 0.5**. A two-stage inductance ratio cannot reach the gain at the lowest input voltage.
- **Keep f_{Min} above f_o** ; the anti-capacitive protection is the last line, not the first.
- **Size the output bank from ripple and hold-up**, take the larger, then check the rms ripple current separately.
- **Crossover in the tens of hertz**; handle the surviving feedback ripple with R_{BM} , not a faster loop.
- **Measure both morphing edges** at full and light load.
- **Check the turns ratio as $nV_{o,eff}$** , never as a bare ratio.
- **Judge over-current on the composite tank peak**.
- **Include the $2f_l$ component** in the capacitor ripple current; judge a centre tap at the output node.
- **Carry $\sqrt{d}$ into the secondary rms**; dropping it overstates the loss by about 27 % here.
- **Put L_{open} and L_{short} on the transformer drawing** beside the turns.
- **ESR at each current component's own frequency**: switching and $2f_l$.
- **Recheck ZVS after raising L_m** .
- **Verify ZVS by sweeping the selected tank**; the closed form has no fixed error sign.
- **Record calculated and selected values separately**, and make every downstream check read the selected one.
- **Estimate the leakage before the insulation is placed**. If a gap carries the reinforced insulation it cannot be narrower than the creepage, and Equation 26 then puts a floor under L_{short} . Where that floor is above L_r , put the insulation in the wire, and check the section winding against L_r again before the tank is fixed.
- **Foil only in a concentric winding**. Between sections the leakage field crosses the window and the face of a foil.
- **An auxiliary winding that feeds V_{CC}** : take the lowest whole-turn ratio that still regulates at the end of hold-up, and size C_{VCC} for the start-up hand-over, not for the steady state.

7.32 What to measure first on hardware

In order; the first four decide whether the design is sound.

- **$f_{sw}(\theta)$ over a line half cycle**. It settles T_{idle} , the unknown behind both oscillator clamps.
- **Cold start into the full 75.2 mF bank**. The start-up window is finite and this risk is new to the architecture.
- **ZVS at the half-bridge edge** ($245 V_{pk}$, full load), which is also the worst point of the sweep: calculated 636 ns against 220 ns.
- **Switching frequency at the full-bridge edge** ($235 V_{pk}$, full load): calculated 118.5 kHz; it must not hit the VCO ceiling.
- **Input current and THD at 90 Vac, full load**: the zero-crossing dead zone.
- **Output ripple against what the load tolerates**.
- **Ripple current and temperature rise in the bank**: 30.54 A rms calculated, 0.19 A per part.
- **Device temperatures, in half-bridge morphing** above $245 V_{pk}$, where the standing device dissipates most. This decides the accepted loss-budget shortfall.
- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the ripple trough**: output still above 19 V after 12 ms.

Single-Stage PFC LLC Converter Design

- **A step across the morphing band**, both directions.
- **V_{CC} and the auxiliary voltage against the output**, at no load, at full load and through a cold start: whether the winding follows the output, and when it takes over from C_{VCC} .
- **Gate and midpoint timing** at both legs: the dead time left at the gates, V_{BO} through a cold start (it must stay above 9.3 V), and the midpoint dv/dt against 50 V/ns.
- **The SR gate waveforms** with two MOSFETs per gate pin, the longest burst-off time at no load, and $V_{CC,SR}$ with the temperatures of Q_{SR} and the TEA2095TE at full load.
- **The transformer's working voltage and an electric-strength test** on the first samples, before they go to the certifying body.

7.33 The whole circuit, with this design's values

Figures 52 and 53 put the parts of this chapter together on two sheets, which meet at the transformer and at the optocoupler U4. A net that continues elsewhere on its sheet carries a flag with the name of the net. Every value printed is the value selected in the sheet. The parts that are named here for the first time are those of the V_{CC} regulator of Section 7.28: D_{HV} , the two start-up diodes from the ac lines to HVSU; D_{aux} and C_{aux} , the rectifier and capacitor on the auxiliary winding; Q_{VCC} , the pass transistor, with D_Z on its base and the bypass diode D_{byp} on its emitter; and C_{HF} , the ceramic beside the output bank.

What this note does not size is drawn as a block or without a value: the EMI filter; the input bridge BR1, the start-up, bootstrap and auxiliary diodes, Q_{VCC} and Q_{SR} , which Section 7.16 gives as requirements; C_{aux} ; the SR gate resistors; and the 12 V rail V_Z that feeds the optocoupler LED, which the loop design takes as given.

Single-Stage PFC LLC Converter Design

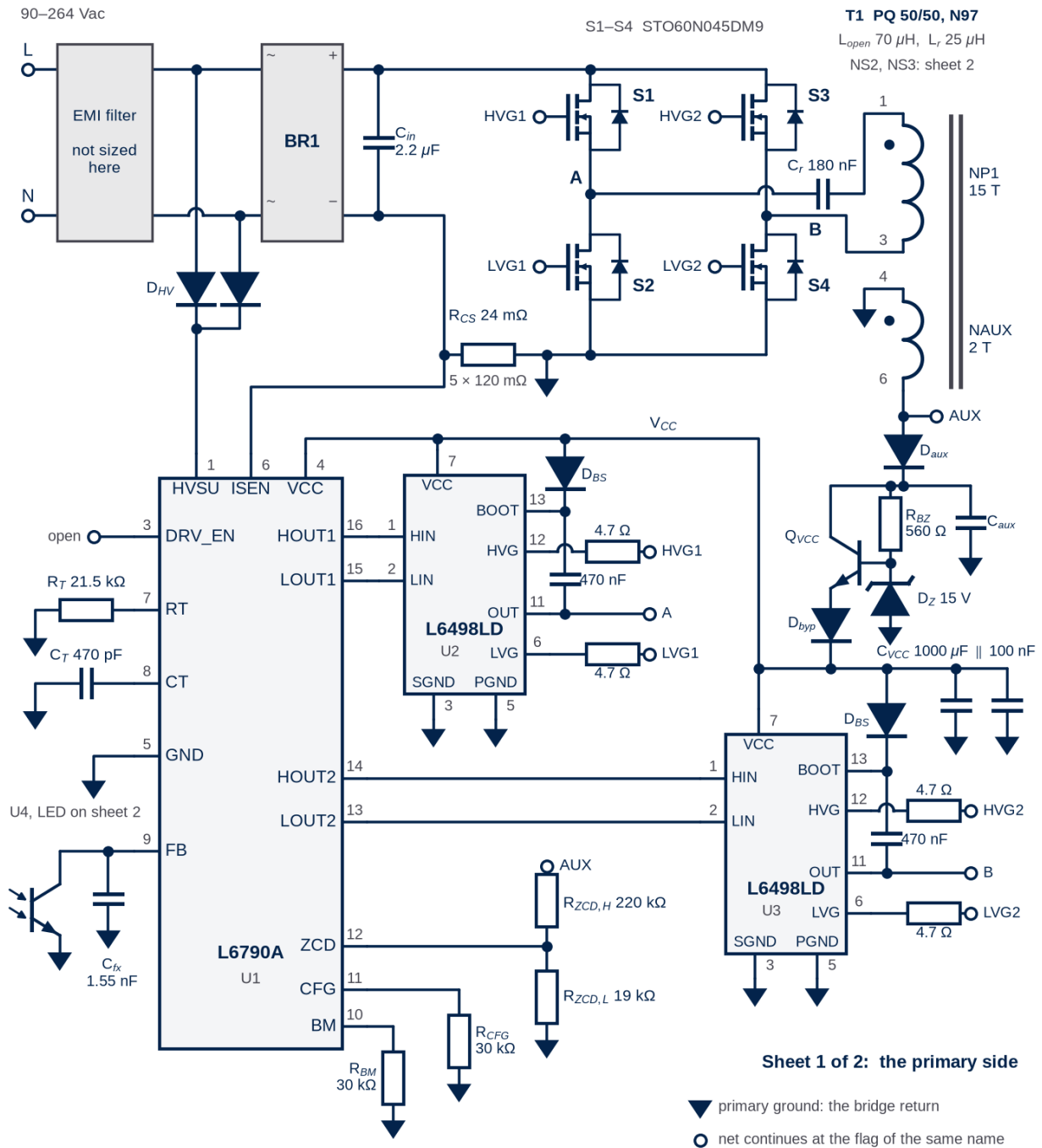


Figure 52: The whole circuit, sheet 1 of 2: the primary side. Pin numbers are the L6790A's as ST's EVL6790_670W control board numbers them, the L6498LD's in SO-14 and the transformer's as in Figure 46. The bridge return is the primary ground, with R_{cs} between it and the rectifier's negative, where ISEN reads it.

Single-Stage PFC LLC Converter Design

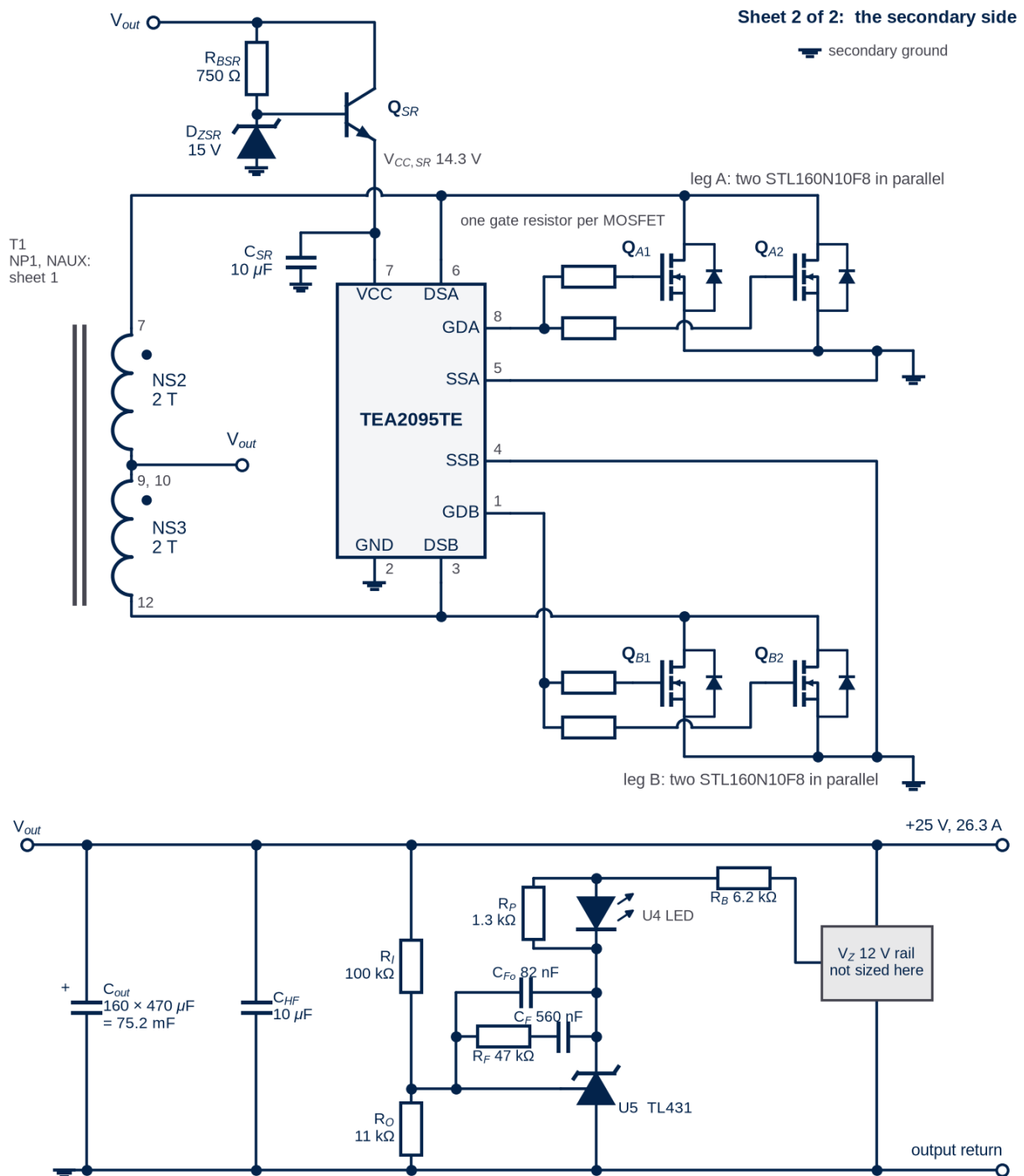


Figure 53: The whole circuit, sheet 2 of 2: the secondary side. The SR stage of Figure 50 above, the output bank and the voltage loop below; the V_{out} flags are one net. The transistor of $U4$ is on sheet 1.

8 List of symbols

The symbols used in this note. The controller's own electrical parameters are in its datasheet.

Single-Stage PFC LLC Converter Design

Symbol	Meaning
The tank and its gain	
C_r, L_r, L_m	resonant capacitor, series inductance and magnetising inductance of the tank model
$Z_0, Z_{0,\text{design}}$	characteristic impedance $\sqrt{L_r/C_r}$; the value the tank is designed to, $R_{ac} Q_{ZVS}$
Z_{in}	input impedance the bridge drives; $\arg Z_{in} = 0$ is the capacitive boundary
f_r, f_o	series resonance, and the lower resonance with L_m included
f_{sw}, f_n	switching frequency, and the same normalised to f_r
$f_{sw,\text{max}}, f_{sw,\text{min}}$	the specified maximum switching frequency, an input to λ ; and the lowest frequency the oscillator must reach, f_o in this design
$f_{sw,\text{des}}$	the frequency the oscillator span is designed to: 1.5 times the highest switching frequency the converter runs at, capped by the specified maximum
T_{sw}, T_r	switching period $1/f_{sw}$, and the resonant period $1/f_r$
λ, m	L_r/L_m , and $(L_r+L_m)/L_r = 1 + 1/\lambda$
$\lambda_{act}, \lambda_{req}$	the λ of the selected parts, and the λ the design asked for
$\lambda_1, \lambda_2, \lambda_{TD}, \lambda_3$	the four candidate λ : minimum gain, the same with the frequency ceiling, the dead-time condition, and the minimum-frequency condition; the largest binds
Q, Q_{pk}, Q_{ZVS}	quality factor Z_0/R_{ac} ; its value at the line peak; the cap the ZVS condition puts on it
R_{ac}	the rectifier and load as one resistance at the fundamental
V_{RI}, V_{RI}^F, i_{RI}	the square voltage at the rectifier input, referred to the primary; its fundamental; and the sine current the tank delivers into the rectifier
M	tank gain, $n V_{o,\text{eff}}$ over the drive
M_{req}, M_{pk}	the gain the operating point demands, and its value at the line peak
M_{min}, M_{max}	in the classic LLC procedure: the gain range the tank must cover, from the input range and the output tolerance
M_{low}, M_{high}	the required gain at the low and the high equivalent corner
M_{OL}, M_{∞}	the gain curve at no load ($Q = 0$, output open), and the floor $1/(1+\lambda)$ it flattens onto
M_Z	the gain on the capacitive/inductive boundary ($\arg Z_{in} = 0$), traced as Q varies; a little to the right of the gain peaks
x, u, q	the substitutions that turn the gain equation into a cubic: $1/f_n^2$, $\sin^2\theta$, and $Q_{pk}^2 u^2$
d	conduction ratio f_{sw}/f_r of the secondary, 1 above resonance
n, n_T	equivalent-model turns ratio, and the physical (wound) turns ratio
N_{rect}	devices in the secondary conduction path: 1 centre tap, 2 full bridge
V_{drive}	the amplitude the bridge applies to the tank: the rectified mains in full bridge, half of it in half bridge
V_{refl}	reflected output voltage $n V_{o,\text{eff}}$
$V_{o,\text{eff}}$	$V_{out} + N_{rect} V_f$
V_f	forward drop of one secondary rectifier

Single-Stage PFC LLC Converter Design

Symbol	Meaning
The mains, and power factor	
θ	line phase angle
$f_l, f_{l,min}, \omega_l$	line frequency, its lowest specified value, and $2\pi f_l$
v_{ac}, V_{ac}, I_{ac}	the instantaneous mains voltage, and the rms voltage and current drawn
$I_{1,rms}$	the rms of the input current's fundamental alone
$p_{in}(t), P$	instantaneous and average input power
$v_{in}(\theta), i_{in}(\theta)$	the rectified mains voltage and the current drawn from it, over the line half cycle
PF, THD	power factor, and total harmonic distortion of the input current
ϕ_1, D	displacement angle of the fundamental, and the distortion factor $I_{1,rms}/I_{ac}$
D_3	third harmonic on the input current, as a fraction of the fundamental
V_{in}, V_{pk}	the rail the bridge works from, and the peak of the mains
V_{bus}	the regulated dc bus between the two stages of a two-stage converter
$V_{ac,eq}, V_{ac,eq,low}, V_{ac,eq,high}$	equivalent input voltage the tank sees after morphing, and its value at the two morphing thresholds
$V_{eq,min}, V_{eq,max}$	the lowest equivalent input (the HB edge) and the mains maximum, as the ST tool's margin factor Γ_v uses them
Power and loss	
P_{in}, P_{out}	input power, output power
P_{LLC}, P_{EMI}, P_{BR}	the three budgeted losses: the LLC stage, the EMI filter, the input bridge
$P_{in,LLC}$	power into the tank, P_{out}/η_{HB}
η_{HB}	assumed efficiency of the LLC stage
$P_{mos,dc}$	conduction loss of the primary device that stands on in half-bridge morphing
$P_{SR}, P_{SR,leg}$	secondary rectifier loss, per device and per rectifier leg
P_{CS}	dissipation in the current-sense resistors
P_{budget}	the loss a device position is allowed, before parts are chosen
E, C, V, V_{min}	the energy a capacitance C gives up between two voltages, in the hold-up expression
Currents and times of one switching cycle	
$i_{Lr}, I_{Lr,pk}$	tank current, and its peak: reflected load plus magnetising
$i_{trafo}, I_{trafo,pk}$	the reflected load current alone, and its peak
$i_{Lm}, I_{Lm,pk}$	magnetising current of the tank model, and its peak
$i_{\mu}, i_{\mu,pk}$	magnetising current of the physical transformer, and its peak
i_p, i_s	the instantaneous primary and secondary winding currents
$I_{sec,pk}$	peak secondary current per rectifier leg

Single-Stage PFC LLC Converter Design

Symbol	Meaning
$I_{pri,rms}$	primary rms over the worst switching cycle
$I_{lc}, I_{pri}, I_{rect}$	a line-cycle rms: the rms of each switching period, averaged in I^2 over the line half cycle; for the primary and for one rectifier leg
I_{Cout}, I_{out}	ripple current in the output bank, and the load current
$t_D, T_{ZC}, T_{ZC,min}$	bridge dead time; the time the tank current takes to reach zero after the gates turn off, and its smallest value over the operating space
T_T	the swing time: how long the bridge node takes to cross the rail in the dead time
v_d, v_d^F, V_{ds}	bridge mid-point voltage, its fundamental, and a device's drain-source voltage
v_A, v_B	the two bridge mid-points A and B against 0; $v_d = v_A - v_B$
i_{S1}	drain current of S1, positive drain to source
$S_1, S_2, S_3, S_4, D_1, D_2$	the four bridge switches, and the two secondary rectifiers
i_{D1}, i_{D2}, i_D	the current in each secondary rectifier, forward; i_D is either of them
The transformer and its core	
$B(t), B_{pk}, B_{max}$	flux density, its peak, and the ceiling the design keeps under
N	turns of the winding a voltage is applied to, in Faraday's law
A_e, A_{min}	core effective area, and its narrowest section
A_N	winding window area of the coil former
A_L, g, μ_0	inductance factor of the gapped core, the total centre-leg gap, and the permeability of free space
N_p, N_s, N_x	primary turns, secondary turns per winding, number of units in the assembly
N_{s1}, N_{s2}	the upper and lower halves of the centre-tapped secondary (NS2 and NS3 in the design example)
$i_{NS}, i_{NS2}, i_{NS3}, V_{NS}$	in the design example: the current of a secondary winding, named by its winding, and the voltage across one
$N_{aux}, n_{aux}/n_{sec}$	auxiliary winding turns, and its turns ratio to one secondary
L_{open}, L_{short}	inductance across the primary with every other winding open; and with the secondary shorted, on a centre tap one half at a time, the auxiliary open
I_{dc}	the dc current the DC-overlap test overlaps on the LCR signal
$I_{p,pk}, L_p$	in the flyback comparison only: the primary peak current and the primary inductance of a flyback
Φ, Φ_p, Φ_s	in the flyback comparison: the core flux, and the flux the primary and the secondary ampere-turns would set up alone
B_s	saturation flux density of the core material, from the material data at temperature
L_μ, L_{L1}, L_{L2}	physical magnetising inductance and the two leakage inductances
L_1, L_2	the open-circuit inductance measured at the primary and at one secondary
$L, \Delta L$	an inductance and the change of it a tolerance allows

Single-Stage PFC LLC Converter Design

Symbol	Meaning
I_{eq}, I_{sat}	open-circuit current that reproduces the operating flux; the DC-overlap test current
The copper	
A_{cu}, J, I_{rms}	copper cross-section one winding needs, the current density it is sized to, and the rms current of that winding; $I_{rms}(\theta)$ is its rms over one switching period at line phase θ
$N_w, A_{cu,w}$	turns and copper area of winding w , in the sum over all the windings
k_u	window utilisation: the share of A_N that ends up copper
δ, ρ, f	skin depth, the resistivity of copper at the working temperature, and the frequency the depth is evaluated at
σ	conductivity of copper, $1/\rho$, in the strand eddy loss
d_s, a_s, n_s	Litz strand diameter, the copper cross-section of one strand, and the number of strands a winding needs
d_{litz}, k_{litz}	outside diameter of the served bundle, and the share of that area which is copper
$t_f, w_f, w_{f,max}, n_f$	foil thickness, the width one turn needs, the widest single strip assumed, and the number of strips in parallel
N_A, N_B	the turns of the two parts of a split primary
w_A, w_S, w_B	axial widths of primary part A, the secondary and primary part B
g_1, g_2	the gaps between the winding sections
h_w, l_N	depth of the winding window, across which the leakage field runs, and the mean turn length
h_r, k_w	radial room between the coil-former tube and the rim of its flanges (or the outer legs, whichever is closer), and the pitch from layer to layer over the bundle diameter (the looseness of a real winding)
h_A, h_S	radial build of the primary section and of the secondary section with NAUX
k_{ax}	pitch along a layer over the bundle diameter, spread so that the two sections and the partition fill the winding width
J_p, J_s	current density in the primary and in each secondary winding, as built
Semiconductors	
$R_{DS(on)}, R_{DS(on),25}, k_T$	on-resistance, its 25 °C headline value, and the multiplier from 25 °C to $T_{j,max}$
$T_j, T_{j,max}, T_a$	junction temperature, its rating, and ambient temperature
$C_{oss}, C_{o(tr)}, Q_{oss}$	the small-signal and the time-related output capacitance of a MOSFET, and the charge the time-related one stands for
The controller and its network	
V_{FB}, V_{FB}	feedback voltage above its 0.5 V offset at rated power, and its small-signal part
$V_{os}, K_{HV}, K_M, K_{FF}$	the offset and the three internal gains of the feedback law, quoted from the datasheet block diagram
K_{pwr}	gain of the power law, $V_{FB} = K_{pwr} R_{CS} P_{in,LLC}$

Single-Stage PFC LLC Converter Design

Symbol	Meaning
$R_{CS}, R_{CS,max1}, R_{CS,max2}$	current-sense resistor, and the two conditions that bound it: the maximum-power law and the OCP1 trip
I_{OCP1}, I_{OCP2}	over-current thresholds set by R_{CS}
C_T, R_T	oscillator timing capacitor and resistor
$C_{T,min}, C_{T,max}, R_{T,ceil}, R_{T,max}$	the window those two have to be chosen inside
$V_{ref}, I_{EA}, I_{EA,max}$	the oscillator reference, the error-amplifier current that is the only control input, and its maximum
f_{Min}, f_{Max}, f_{SU}	oscillator floor, ceiling and start-up frequency set by R_T, C_T and T_{idle}
T_{idle}	oscillator idle time
$R_{BM}, r_{BM}, V_{BM,eq}, P_{in,BM}$	burst-mode resistor, the fraction of the rated input power P_{in} burst starts at, the equivalent FB threshold, and the input power that corresponds to
$R_{CFG}, R_{CFG,max}$	configuration resistor, and the largest value the lowest input voltage allows
$V_{BO}, V_{BO,pk}, V_{BO,rms}$	brown-out threshold, as a mains peak and as an rms voltage
$R_{ZCD,H}, R_{ZCD,L}, I_{bias}$	the ZCD divider, and the current it is designed to draw
$V_{OVP1}, V_{OVP2}, V_{OVP1,out}$	the two over-voltage thresholds, and the output voltage OVP1 is aimed at
$V_{CC}, V_{CCon}, V_{CCoff}$	the IC supply, and the two thresholds of its under-voltage lockout
$V_{CC,reg}, V_{CC,reg,min}, C_{VCC}, C_{VCC,reg}$	the voltage the regulator holds the V_{CC} pin at and the bottom of its band; the capacitor on the pin and the value the start-up needs
$V_{CC,HVSUon}$	the V_{CC} level below which the start-up unit turns its charge current on
$V_{DZ}, V_{DZ,max}, V_{BE}, V_D$	regulator Zener voltage and its top of tolerance, the pass transistor base-emitter drop, and a silicon diode forward drop
$R_{BZ}, R_{BZ,max}, \beta_{min}, I_{Z,min}$	Zener feed resistor and the largest that still regulates, the least current gain of the pass transistor, and the least Zener bias
$I_{VCC}, I_{VCC,SU}, I_{CC}, I_{drv,q}$	current the V_{CC} rail supplies in run and at the start of switching, the controller's own share, and the quiescent current of the gate drivers
$N_{sw}, Q_g, Q_{g,10}, C_g$	switch positions driven; the gate charge of one primary MOSFET at the drive voltage, at 10 V, and the slope of its curve past the Miller plateau
$B, \Delta Q_{OSC}, V_{CC,floor}, I_{VCC}(V)$	drive current lost per volt of V_{CC} ; extra charge of the start-up oscillator interval; the lowest V_{CC} the controller and drivers are specified at; the supply current at rail voltage V
$C_{BOOT}, V_{BO}, \Delta V_{boot}, V_{drop}, R_{BS}$	bootstrap capacitor, the high-side driver supply and its per-cycle ripple, the drop across an integrated bootstrap switch of resistance R_{BS}
$T_{charge}, T_{on,max}, I_{QCC}, I_{QBO}$	low-side on-time that recharges the bootstrap, the longest high-side on-time, and the quiescent currents of the low and floating driver sections
$R_G, R_{g,int}, R_{so}, R_{si}, s_{drv}, P_{drv}$	external and internal gate resistance, driver source and sink resistance, the driver's share of the gate power, and its dissipation
MT	delay mismatch of the gate driver between its channels
$D_{BS}, Q_{A1}, Q_{A2}, Q_{B1}, Q_{B2}$	the external bootstrap diode; the SR MOSFETs, two per centre-tap leg

Single-Stage PFC LLC Converter Design

Symbol	Meaning
$D_{HV}, D_{aux}, C_{aux}, Q_{VCC}, D_Z, D_{byp}$	the start-up diodes to HVSU; the rectifier and capacitor on the auxiliary winding; the V_{CC} pass transistor, its base Zener and the bypass diode on its emitter
C_{HF}	the ceramic capacitor beside the output bank
$V_{CC,SR}, R_{BSR}, D_{ZSR}, Q_{SR}, C_{SR}$	the SR controller supply, and the feed resistor, Zener, pass transistor and pin capacitor of the follower that makes it
$I_{SR,max}, I_{SR}, I_{SR,q}$	the current the SR supply follower is sized for, the current it carries with the chosen MOSFETs, and the SR controller's own current
C_{iss}, V_{SD}, Q_{rr}	a MOSFET's input capacitance, and its body diode's forward voltage and reverse-recovery charge
$Q_{g,SR}, Q_{g,sync}, C_{g,SR}$	gate charge of one SR MOSFET at the highest drive, the datasheet's synchronous-rectifier gate charge at 10 V, and the slope of its gate-charge curve past the plateau
$Q_{g,run}, Q_{g,SU}$	the gate charge at the highest regulated V_{CC} , and at V_{CCon} where start-up begins
$V_{BO,min}, V_{CC,drv,min}, P_{drv,max}, R_{drv}$	the lowest recommended floating and low-side driver supplies, the dissipation the driver package allows, and its output resistance
$S_{mid}, S_{OUT,max}$	slope of the bridge midpoint during the swing, and the slew rate the driver's OUT pin allows
C_{HB}, Q_{ZVS2}	the capacitance the bridge midpoint swings, two devices and the layout; the dead-time limit on the quality factor it sets
$V_{GS}, V_{DD}, V_{G,SR}$	gate-source voltage; the drain supply of a datasheet test circuit; the gate drive of the SR controller
k_{LOUT2}	how far the driver input pull-down keeps LOUT2 above the level that reads as a fixed half bridge
$I_{HVSU}, t_{hand}, V_{out,UV}$	start-up unit charge current; time from the first pulse until the auxiliary winding holds V_{CC} , and the output voltage at which it does
The voltage loop	
$G_{plant}(s), G_{EA}(s), T(s)$	plant v_{out}/v_{FB} , compensator $-v_{FB}/v_{out}$ (the feedback sign taken out), and loop gain $T = G_{plant}G_{EA}$
G_o, EA_o	gain constant of the plant integrator and of the compensator, in rad/s
$\omega_z, \omega_p, \omega_{px}, \omega_c$	compensator zero, pole, high-frequency pole, and the crossover, in rad/s
$f_z, f_p, f_{px}, f_{180}$	the same zero and poles in hertz, and the frequency where $\arg T = -180^\circ$
f_c, Φ_M, GM	loop crossover frequency, phase margin, gain margin
f_{cto}	where the bare plant integrator would cross 0 dB, $G_o/2\pi$
f_{pHF}, f_{MB}	the high-frequency pole chosen for C_{fx} , and the centre frequency the K-factor method places the zero and the pole around
K_v, Γ_v, α_v	K factor of the Type II compensator, and the input-voltage margin factor and weighting constant it is built from
Z_f	feedback impedance of the compensator, C_{Fo} in parallel with $R_F + C_F$
v_K	small-signal voltage at the TL431 cathode
R_i, R_o	output divider of the compensator

Single-Stage PFC LLC Converter Design

Symbol	Meaning
C_{F0}, C_F, R_F, C_{fx}	compensator parts: the two capacitors and the resistor of the TL431 network, and the FB-pin capacitor
C_{opto}, C_{ser}	the optocoupler's own output capacitance, and $C_F + C_{F0}$ in series
$R_B, R_{B,max}, R_{B,min}, R_P, R_{FB}$	optocoupler LED resistor and the window it has to sit in, the TL431 bias resistor, and the pull-up inside the FB pin
$CTR, CTR_s, CTR_m, i_{LED}$	optocoupler current transfer ratio, its value at the steady-state and at the maximum LED current, and the LED current itself
i_{FB}	small-signal current the optocoupler sinks from the FB pin, $CTR i_{LED}$
$I_{FB,steady}, I_{FB,max}, I_{min}$	FB-pin current at steady state and at maximum, and the least current the TL431 needs to stay in regulation
V_R, V_Z, V_{F0}	TL431 reference, the regulated rail that feeds the LED, and the LED forward drop
v_{out}, i_{out}, v_C	small-signal output voltage and the current the converter delivers into the bank; the voltage on a capacitor in a waveform
$\Delta V_{loop}, \Delta V_{FB}$	$2f_l$ output ripple seen by the loop, and the ripple it leaves on the FB pin
The output, and hold-up	
C_{out}, C_{in}	output capacitor bank, input film capacitor
V_{out}, I_{out}, R_L	output voltage and load current; the load as a resistance
$\Delta v, \Delta v_{pp}$	allowed and achieved $2f_l$ output ripple
$T_{hold}, t_{hold}, V_{o,min}$	required and achieved hold-up time, and the lowest output allowed at its end
ESR	equivalent series resistance of the bank, taken at the switching frequency
Verification margins	
k, X, X_{act}, X_{req}	a verification margin X_{act}/X_{req} , where X is whatever quantity the row is about; every k in this note is ≥ 1 when the design passes
k_{floor}, k_{ceil}	how far f_{Min} clears f_o , and how far f_{Max} clears the highest operating frequency
k_{Ploss}	the loss budget of a device position over the loss computed for it
k_{RBZ}	$R_{BZ,max}/R_{BZ}$: how far the Zener feed resistor sits below the largest that still regulates at the end of hold-up
w_k, θ_k	the weights and the nodes of the five-point Simpson rule the line-cycle rms values are integrated with

Table 34: Symbols.

9 References

- STMicroelectronics, *L6790A LLC-PFC controller*, preliminary datasheet, 30 April 2026. **Draft.**
- STMicroelectronics, *EVL6790_670W* evaluation board schematics.
- Infineon Technologies, *LLC Converter Design Note*, AN 2013-03, V1.0, March 2013.
- Infineon Technologies, *Resonant LLC converter: operation and design*, AN 2012-09, V1.0, September 2012.

Single-Stage PFC LLC Converter Design

- onsemi (formerly Fairchild), *Half-bridge LLC resonant converter design using FSFR-series Fairchild power switch*, AN-4151.
- Monolithic Power Systems, *Understanding LLC operation*, part 2.
- Toshiba Electronic Devices & Storage, *Power factor correction (PFC) circuits*, application note.
- Toshiba Electronic Devices & Storage, *Resonant circuits and soft switching*, application note, 12 November 2019.
- ROHM Semiconductor, TechWeb, *LLC operating regions* (region framing of Figure 12).
- W. Wenbo et al., *A single-stage 1.65 kW ac-dc LLC converter*, IEEE ECCE.
- onsemi, *The TL431 in the control of switching power supplies*, TND381-D (compensator design with a TL431 and an optocoupler).
- [PI] Power Integrations, *If your power converter is not safe, you may have an expensive recall*, PSMA IS293, 2020 (IEC 62368-1 creepage and clearance tables, altitude factors).
- [ATIS] M. Maytum, *IEC 62368-1 and pluggable mains powered equipment surge protection*, ATIS PEG, 2019 (mains transients, Table 14 clearances).
- [TUV] TUV Rheinland, CB test report 60431427 001 to IEC 62368-1, Class II AV/ICT adapter, evaluated for 5000 m.
- [UL] UL, CB test report E135803-A6002-CB-1 to IEC 62368-1, Advanced Energy GB130Q, evaluated for 5000 m.
- [TI] Texas Instruments, *Demystifying clearance and creepage distance for high-voltage end equipment*, SLUP419, 2024.
- [GB] GB 4943.1-2022, *Audio/video, information and communication technology equipment — Part 1: Safety requirements* (foreword and scope).
- [TIW] Furukawa Electric, *TEX-E triple-insulated wire, safety approvals*.
- [STO] STMicroelectronics, *STO60N045DM9, N-channel 600 V, 35 mΩ typ., 56 A MDmesh DM9 Power MOSFET in a TO-LL package*, DS14711 Rev 4, June 2026.
- [L6498] STMicroelectronics, *L6498, high voltage high and low-side 2 A gate driver*, DocID030318 Rev 3, September 2017.
- [STL] STMicroelectronics, *STL160N10F8, N-channel 100 V, 3.2 mΩ max., 158 A STRipFET F8 Power MOSFET in a PowerFLAT 5x6 package*, DS14249 Rev 5, July 2024.
- [TEA] NXP Semiconductors, *TEA2095TE, GreenChip dual synchronous rectifier controller*, product data sheet Rev. 1.3, 20 October 2025.

10 Errata and open items

Everything here is worked around rather than solved, and most of it only the released datasheet or a prototype can close. Re-read it before a production release.

Single-Stage PFC LLC Converter Design

10.1 Datasheet: the draft is not self-consistent

Item	What the draft says	What is used here, and why
Oscillator idle time T_{idle}	Section 5.3.2 implies 700 ns in the frequency expressions and 350 ns in the $C_{T,max}$ expression; Table 2 (recommended operating range) back-solves to about 250 ns	249 ns. The two clamps bracket the operating range at 250 ns; at 700 ns they still do, but the floor margin falls to 1.032, so this is a measurement, not a rounding
Brown-out expression	$V_{BO} = \min(60 \text{ V}, R_{CFG} \cdot 4 \text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table	Read as $R_{CFG} \cdot 4 \text{ V/k}\Omega$ with 15 k Ω as the floor. That is the only reading consistent with the table
Brown-out units	the threshold is a <i>peak</i> voltage; board notes and design tools quote rms	Converted explicitly wherever it is compared ($\sqrt{2} = 1.414$)
Thermal resistance	TBD	No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead
ZCD absolute maximum	lower limit given as TBD	The divider is sized from the OVP thresholds only
Driver naming in Section 5.3.1	LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table	The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge
Start-up unit after its timeout	the charge current turns on below $V_{CC,HVSUon} = 12 \text{ V}$ and the generator shuts down 120 ms after V_{CCon} ; whether the charge current comes back below 12 V after that is not stated	$V_{CC,reg}$ is kept above 12 V at the bottom of its band, so the run supply never depends on the answer

Table 35: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.

10.2 Traps in the obvious way of building a design sheet

Places where the obvious way to set up a design sheet gives a plausible wrong number that nothing flags.

Where	The trap	Consequence if missed
Equivalent input range	taking the maximum from the ac maximum instead of the full-bridge morphing edge	The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets λ
λ from the minimum-gain condition	the same corner again, one step earlier	The required λ comes out far too small — a factor of over a hundred in one case — and L_m far too large
Line-cycle rms currents	evaluating one phase, usually $\theta = \pi/4$, and calling it the line-cycle value	The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough
Secondary rms above resonance	squaring a full half sine when the current is truncated	Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time
Primary device rating	rating on the reflected load current $I_{trafo,pk}$	About 9 % under-rated — the switch carries the <i>composite</i> tank current

Single-Stage PFC LLC Converter Design

Where	The trap	Consequence if missed
Loss and thermal resistance	computing loss from the 25 °C $R_{DS(on)}$ and then asking for a heatsink that holds 125 °C	The required thermal resistance comes out only about half as demanding as it really is
ZVS check	the closed-form shortcut, at the design Q and with the design λ	An error of tens of per cent whose sign nothing fixes; on this tank 7 % conservative. Optimistic is the dangerous direction
Selected versus calculated	a check that reads the calculated value while the board carries the selected one	Every downstream margin is reported for a design that was not built

Table 36: Traps that produce a plausible wrong number.

10.3 Constants used here without a derivation

- The $\pi^2/8$ in the dead-time form of the λ condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **K_v expression** (Equation 59) is the ST tool's as written. With $\alpha_v = 1$ it gives 2.675 for a 50° target, not Venable's $\tan(45^\circ + \Phi_M/2) = 2.748$; the tool does not say what α_v weights. The phase margin is therefore taken from the fitted parts (Equation 65), not from K_v .
- The **16.8 $\Omega \cdot W$** maximum-power constant follows from the feedback span and the multiplier gain (2.8 V / 0.167); it is listed because those gains are draft values.
- The **gate driver output resistance**, taken as 15 V over the short-circuit current at the full-temperature minimum. The L6498 datasheet gives no output resistance; a real output is not a resistor, and this choice puts the larger share of the gate power in the driver.

Single-Stage PFC LLC Converter Design

10.4 Open items in this design

Item	State	What closes it
T_{idle}	three values, 250, 350 and 700 ns	Measure $f_{sw}(\theta)$ on the first board
Primary conduction loss	0.324 against the budget — not met, and accepted	No single 600 V device meets a 3 W budget in the standing position. On board copper alone it would rise 204 °C, so it needs a heatsink of 10.4 °C/W or better from case to ambient at 25 °C (Section 7.17); the rest is a thermal measurement
Secondary loss budget	1.539 — essentially exhausted	One more device in parallel per leg recovers it; the decision waits on the measured temperature
Transformer inductance tolerance	the tank tolerates a 13.8 % fall in open-circuit inductance, which covers the usual ± 10 %	R_T stays at 21.5 k Ω until the first board. A higher R_T narrows the zero-crossing dead zone but spends this margin; choose after measuring $f_{sw}(\theta)$ and the input-current THD at 90 and 230 Vac
Output bank volume and height	75.2 mF, 160 parts	A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage
Standby power	burst entry assumed at 75 W	Measure, then trade R_{BM} against the feedback ripple
Efficiency assumption	$\eta_{HB} = 98$ % assumed	Optimistic. Taking 95 % instead moves R_{CS} and R_{ac} by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement
SR MOSFET gate charge	typical values only: $Q_{g, sync}$ and the Fig. 8 slope of the STL160N10F8	Measure the SR supply current at full load; the follower allows 130.4 nC per device. Two per gate pin set the SR switching times
Bootstrap diode	$V_F = 1.0$ V assumed, part not chosen	Choose a fast diode rated like the primary switches and put its V_F at the charging current into the sheet; $V_{CC, floor}$ and C_{VCC} follow
Dead time at the gates	$(t_D - MT)/T_T = 1.118$; 1 at $L_r \approx 23.5$ μ H	Measure L_r on the first samples and the gate-to-midpoint timing on the prototype; raise t_D if L_r comes in low
Midpoint slope above resonance	40.9 V/ns at $I_{Lm, pk}$; the switched current can be higher	Measure dv/dt at the OUT pins against the driver's 50 V/ns
SR discharge function and burst	trips after 1.1 to 1.7 s without rectifier activity	Measure the longest burst-off time at no load; a longer pause adds 0.4 W to standby
Start-up hand-over	t_{hand} 41.7 ms assumes no load until the rail is up	Measure V_{CC} through a cold start with the real load sequencing, and the start-up unit temperature
Transformer working voltage	187 V rms estimated; creepage taken at the 320 V row	The certifying body measures it on the first samples
Insulation values not read from the standard	from seminars and certification reports (Section 7.12)	Confirm with the certifying body under the edition the certificate will cite: electric strength, the high-frequency clearance table, the production test
TIW approval	500 kHz, Class B (130 °C); f_{SU} 477 kHz	Confirm the approval for the wire sizes used, and hold the winding under 130 °C

Single-Stage PFC LLC Converter Design

Item	State	What closes it
Leakage of each half	NS2 alone 24.50, NS3 alone 25.14 μH with the 3.00 mm partition and $k_w = 1.10$, a field-solution estimate; 23.35–26.40 at the extremes of the back-plate share	Measure both halves on the first samples and judge the tank and R_T with them. A thinner partition lowers them; it is not made thicker than 3.0 mm
Leakage, both halves shorted	23.09 μH estimated, 23.20 wound tight	Recorded on the samples; it is not the value the tank sees
Winding pitch k_w	1.10, assumed	Measure the build of the first sample; at the tolerance limits the winding has 0.23 mm to the leg arc
Partition in the coil former	3.0 mm added to the catalogue B65982E (one section, 12 pins)	The bobbin maker confirms that the partition can be moulded into the former, and the distances around the pins
Secondary pins	28.5 A rms line-cycle on one $\varnothing 1.2$ pin	The bobbin maker gives the pin rating; otherwise the secondary ends go straight to the board
Centre-leg gap	one gap of about 1.78 mm (field solution; 1.34 mm without fringing)	The first layer of NP1 is 1.60 mm from it: measure the winding temperature near it
Core loss	TDK N97 at 125 mT and f_r : not computed; the catalogue gives 3.50 W at most at 100 kHz, 100 mT, 100 °C	Read the TDK N97 material curve at the core temperature (the loss rises below 100 °C); measure it
Ac loss of the Litz	2.3 to 3.3 W from the leakage field on 2.3 W dc, first order, $\varnothing 0.10$ mm strands	Measure the winding temperature; a finer strand (loss as d_s^{-2}) is the remedy, its fill from the vendor
Wire sizes	NP1 triple-insulated Litz $\varnothing 2.9$ mm ($401 \times \varnothing 0.10$, 0.2 mm of insulation assumed); NS2, NS3 Litz $\varnothing 4.0$ mm ($880 \times \varnothing 0.10$)	The wire vendor confirms the sizes, the insulation and the approval, and the bend of the $\varnothing 4.0$ mm Litz on the $\varnothing 23.2$ mm tube

Table 37: What is not settled, and what would settle it.

Every cross-check here is a consistency check, not a correctness check. Three implementations agreeing means they implement the same equations, not that the equations describe the hardware. The first prototype is the correctness check.